



# DFDOF

## Forensic Baseline Report

### Drone Forensic Decision and Orchestration Framework

|                            |                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------|
| <b>Case Identifier:</b>    | CASE-df005-2018-ios                                                                             |
| <b>Operator:</b>           | Floris                                                                                          |
| <b>Analysis Started:</b>   | 2026-06-15 14:24:15                                                                             |
| <b>Evidence Directory:</b> | C:\Users\Floris\Documents\dji_drones\test_phantom_4\df005\2018\ios                              |
| <b>Output Directory:</b>   | C:\Users\Floris\Documents\dji_drones\test_phantom_4\df005\2018\dfdof_output\CASE-df005-2018-ios |
| <b>Evidence Sources:</b>   | 2                                                                                               |
| <b>Pipeline Version:</b>   | DFDOF v1.0 (proof of concept)                                                                   |

*This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.*

## 1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

| File                | Classification       | Acquisition | Size    | Hash Timestamp      |
|---------------------|----------------------|-------------|---------|---------------------|
| flight_logs-001.zip | drone_flight_storage | logical     | 2.7 GB  | 2026-06-15 14:24:24 |
| ios_backup.zip      | controller_ios       | logical     | 51.7 MB | 2026-06-15 14:24:24 |

### Evidence Source Path

C:\Users\Floris\Documents\ddji\_drones\test\_phantom\_4\df005\2018\ios

### Cryptographic Hashes

| flight_logs-001.zip |                                                                  | drone_flight_storage | logical |
|---------------------|------------------------------------------------------------------|----------------------|---------|
| SHA-256:            | 4acdfd6614c1a6459a1480ba358c7e7d0882f30a063f50c34731bc091c98bd32 |                      |         |
| SHA-1:              | bd7a4ca4abdfbeb995c9577997780f98a4c295c0                         |                      |         |

| ios_backup.zip |                                                                  | controller_ios | logical |
|----------------|------------------------------------------------------------------|----------------|---------|
| SHA-256:       | 6226f3fd5cc190356375b90d66ac36a07c088681e2fdb1e9bf46ae9c002ca3d7 |                |         |
| SHA-1:         | d0a2b0c6f334eea4c384282d11f121ec878532b8                         |                |         |

## Pipeline Execution Summary

### Completed Phases

1. p1\_provenance
2. p2\_image\_parsing
3. p3\_artefact\_extraction
4. p4\_decision\_and\_orchestration
5. p5\_normalisation\_and\_anomaly\_checking
6. p6\_multisource\_correlation
7. p7\_analysis\_and\_validation
8. p8\_automated\_reporting

### Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - images]: no artefacts found
4. [p3 - controller ios - videos]: no artefacts found
5. [p3 - drone sd]: source evidence not found
6. [p4 - controller android]: source evidence not found
7. [p4 - drone sd]: source evidence not found
8. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY000.csv
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY001.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY002.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY003.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY014.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY015.csv
14. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY016.csv
15. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY017.csv
16. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY018.csv
17. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY019.csv
18. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY020.csv
19. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY021.csv
20. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY022.csv
21. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY023.csv
22. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY024.csv
23. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY034.csv
24. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY038.csv
25. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY039.csv

### Tool Execution Status

| Tool            | Invocations | Successes | Status |
|-----------------|-------------|-----------|--------|
| datcon          | 41          | 41        | [OK]   |
| parser_ios      | 1           | 1         | [OK]   |
| txtlogtocsvtool | 1           | 1         | [OK]   |

### Processing Coverage Score

| Metric                           | Result |
|----------------------------------|--------|
| Evidence Sources Detected        | 2 / 4  |
| Artefact Categories With Data    | 5 / 7  |
| Flights With Primary Correlation | 1 / 23 |
| Tools Succeeded                  | 3 / 3  |

### 3. Device and Account Identification

| Field                                      | Value                                            | Confidence          | Sources |
|--------------------------------------------|--------------------------------------------------|---------------------|---------|
| Installed DJI App(s)                       | com.dji.go<br>com.dji.pilot                      | [K] Controller-only | 1       |
| Controller Device                          | iPad mini 4                                      | [K] Controller-only | 1       |
| Drone Serial Number                        | 07JDE5300200GN                                   | [K] Controller-only | 2       |
| Drone Model                                | P4 Professional<br>P4P                           | [!] Inconsistent    | 25      |
| Controller Device                          | iPad5,1__weibosdk__003143000__iphone__os<br>11.4 | [K] Controller-only | 2       |
| DJI Application Version                    | 3.1.38<br>4.2.20                                 | [!] Inconsistent    | 3       |
| Account E-Mail                             | droneforensics@vtoinc.com                        | [K] Controller-only | 2       |
| Product Type                               | 24<br>255                                        | [!] Inconsistent    | 2       |
| Last Country Code                          | US                                               | [K] Controller-only | 2       |
| Last App Launch                            | 2018-06-19T18:57:41Z<br>2018-06-19T19:35:26Z     | [!] Inconsistent    | 2       |
| Last Flight Date                           | 1970-01-01T00:00:00Z                             | [K] Controller-only | 2       |
| Last Known Latitude (Find Aircraft)        | 39.9612565                                       | [K] Controller-only | 1       |
| Last Known Longitude (Find Aircraft)       | -106.21627702                                    | [K] Controller-only | 1       |
| Last Known Altitude (Find Aircraft)        | 0.0                                              | [K] Controller-only | 1       |
| Last Known Heading (Find Aircraft)         | 49.9                                             | [K] Controller-only | 1       |
| Last Known Horiz. Accuracy (Find Aircraft) | 1.0                                              | [K] Controller-only | 1       |



## 4. Flight Analysis - Flight 01

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** a27c4f44c6f780b43cff85f387fead85d272b4b18c53fb56aa769871aaf22303  
**Drone Log Name:** FLY025.csv  
**Start:** 2015-10-20T19:33:08.189000+00:00  
**End:** 2015-10-20T19:34:11.703000+00:00  
**Duration:** 63.5 s

### Flight Metrics

**Peak Height:** 0.7959153047485443 m  
**Distance Travelled:** 1.5850594136754008 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

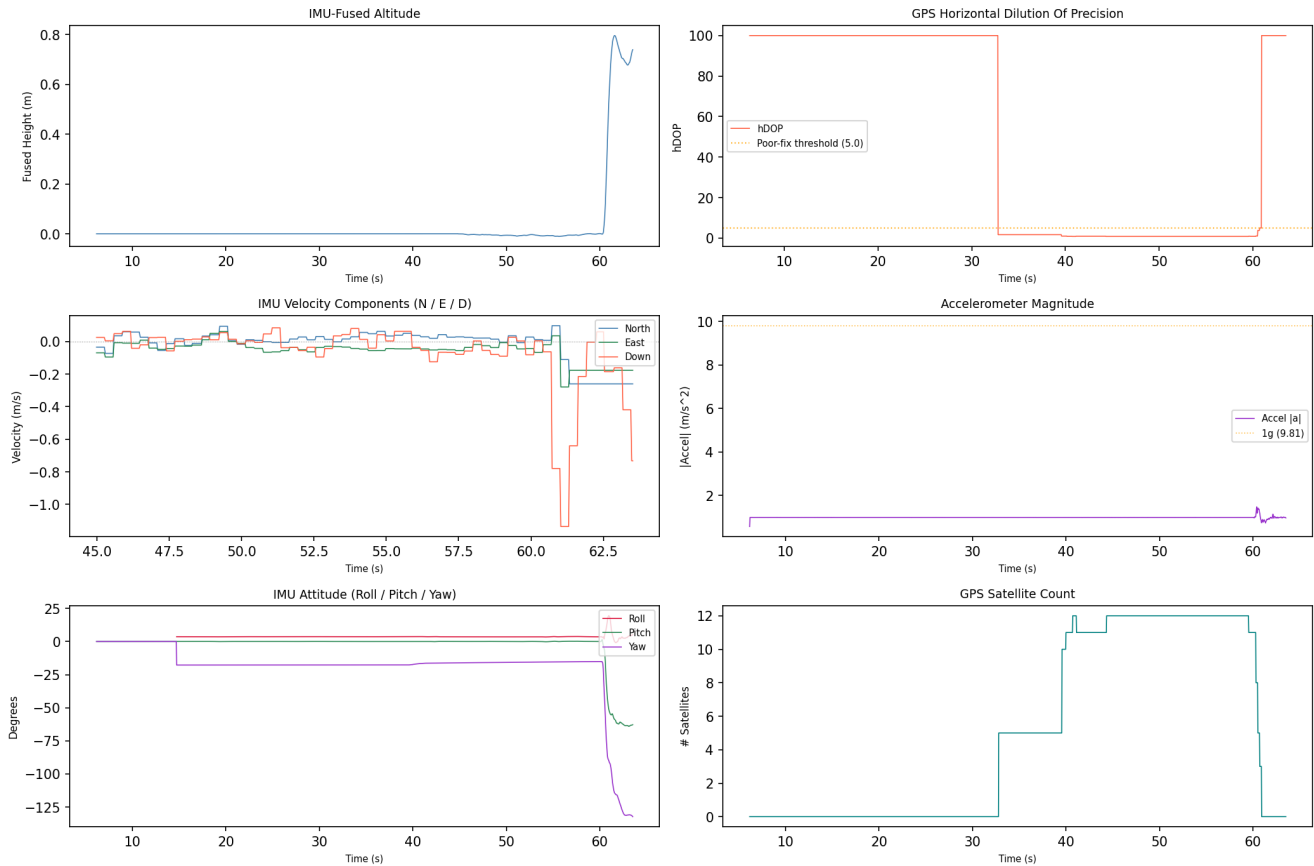
1. Manual
2. Atti
3. GPS\_Atti

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2015-10-20 19:33:41 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-19 19:34:08 | drone_flight_storage:dl | Reached peak height | 0.8 m   |
| 2018-06-19 19:34:10 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

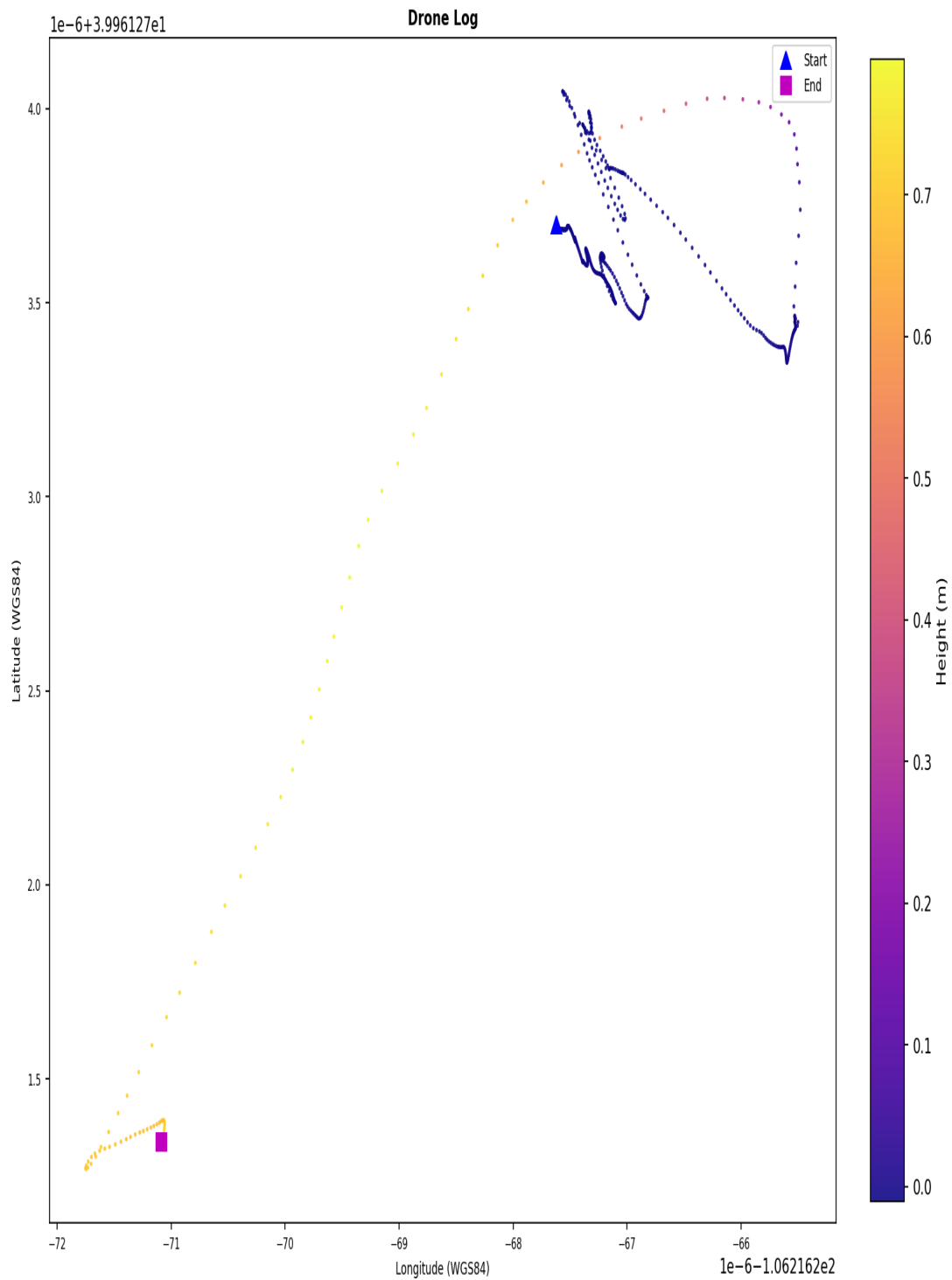
### Drone Log Telemetry - flight\_01



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_01



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY025.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY025.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 02

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** b1de75d6ca539e0f1b32f41cdac3acb1d9142bbb0d85bded399a5677f3d7d07e  
**Drone Log Name:** FLY026.csv  
**Start:** 2015-10-20T19:35:10.496000+00:00  
**End:** 2015-10-20T19:48:58.045000+00:00  
**Duration:** 827.5 s

### Flight Metrics

**Peak Height:** 42.65989867237829 m  
**Distance Travelled:** 3013.7525264403985 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

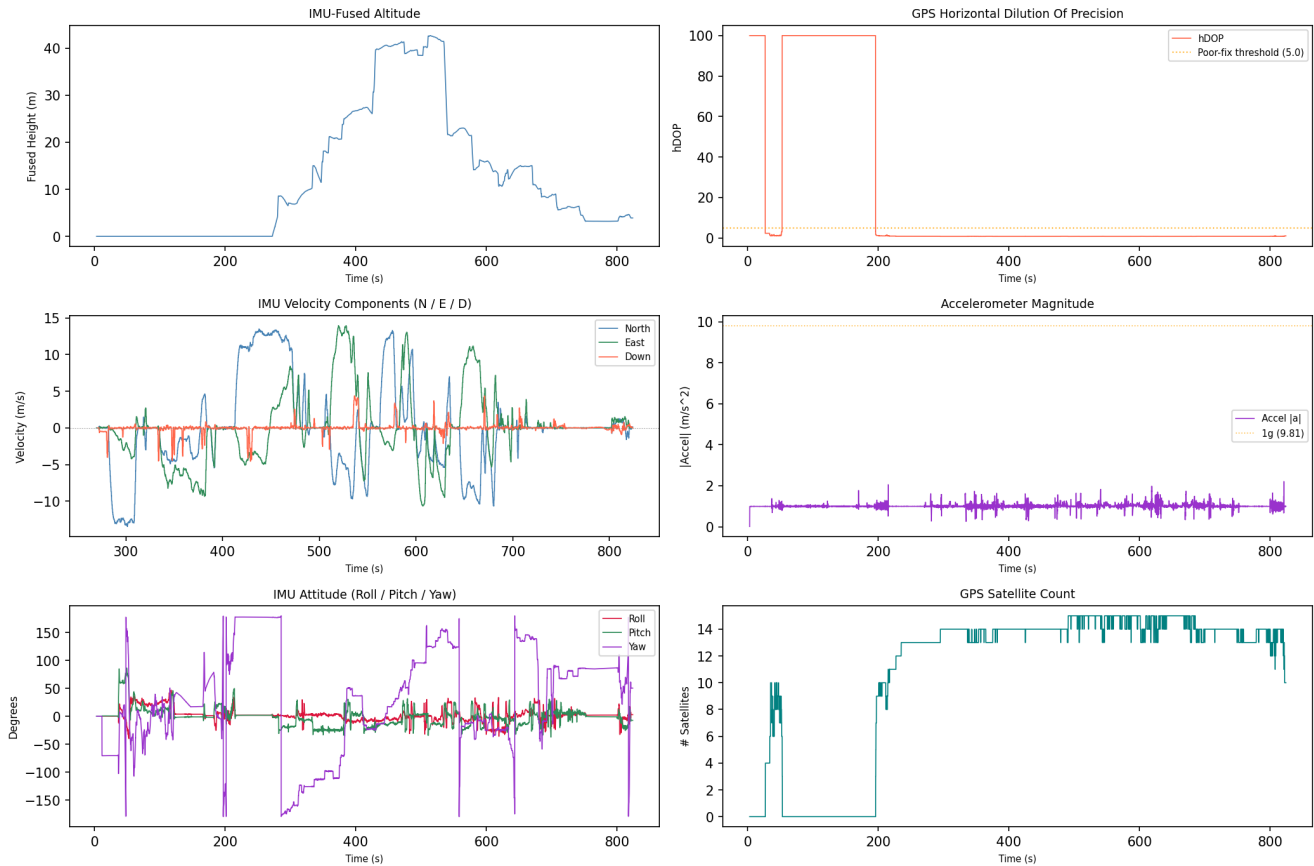
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AutoTakeoff
6. AutoLanding

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2015-10-20 19:35:41 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-19 19:39:42 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-19 19:43:47 | drone_flight_storage:dl | Reached peak height | 42.7 m  |
| 2018-06-19 19:47:46 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-19 19:48:57 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

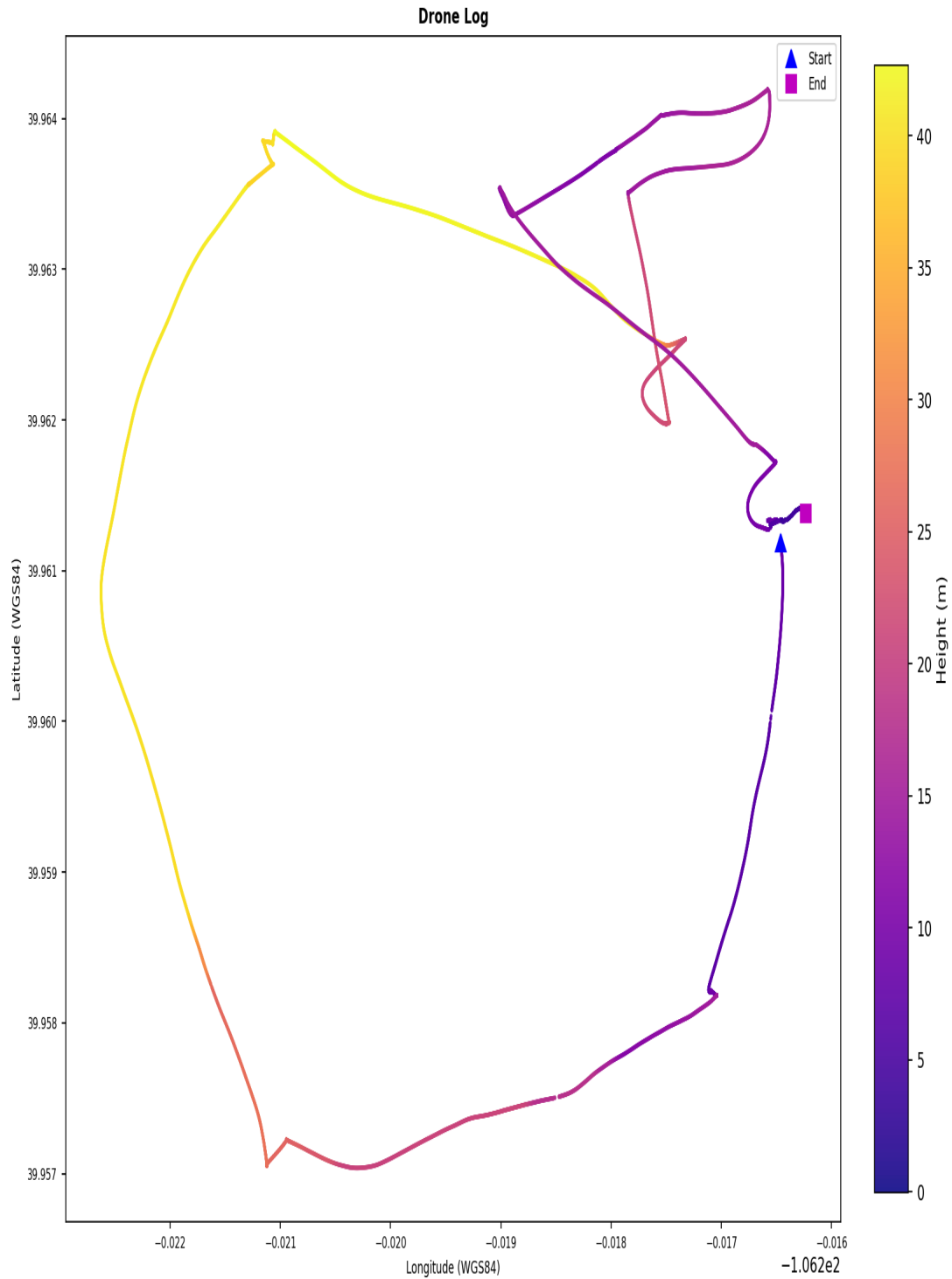
### Drone Log Telemetry - flight\_02



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_02



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY026.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 03

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 66f87be48178896a19a9b0a828b2cad8ea2192bbb47e9b964d2d4fb87ca58dbc  
**Drone Log Name:** FLY032.csv  
**Start:** 2015-10-22T14:43:48.301000+00:00  
**End:** 2015-10-22T14:46:24.952000+00:00  
**Duration:** 156.7 s

### Flight Metrics

**Peak Height:** 0.0013423047548382465 m  
**Distance Travelled:** 4.677371684690006 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

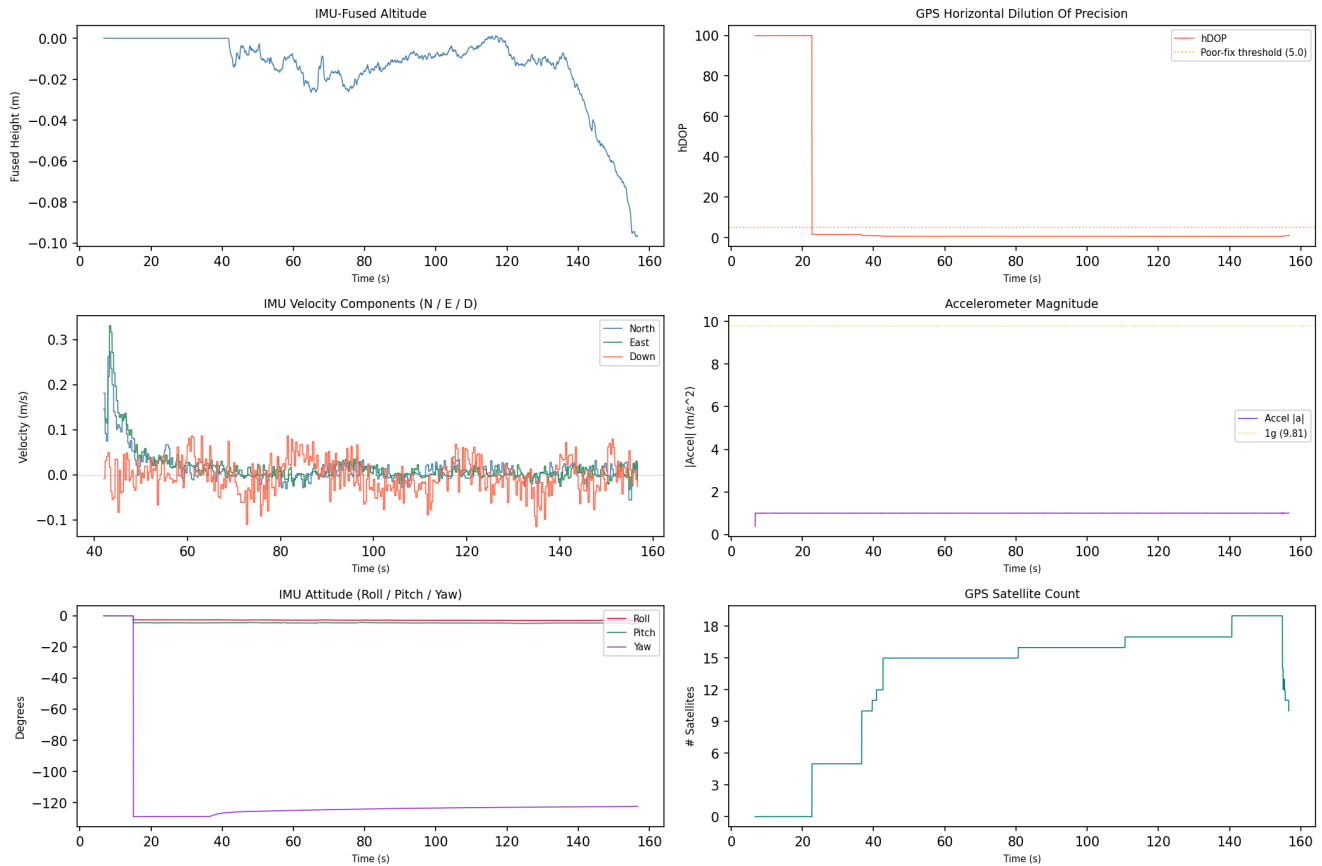
1. Atti
2. GPS\_Atti
3. Manual

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| 0002-11-30 00:00:00 | drone_flight_storage:dl | Motor turned off    | off     |
| 2015-10-22 14:44:11 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-21 14:45:44 | drone_flight_storage:dl | Reached peak height | 0.0 m   |
| 2018-06-21 14:46:24 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

Drone Log Telemetry - flight\_03

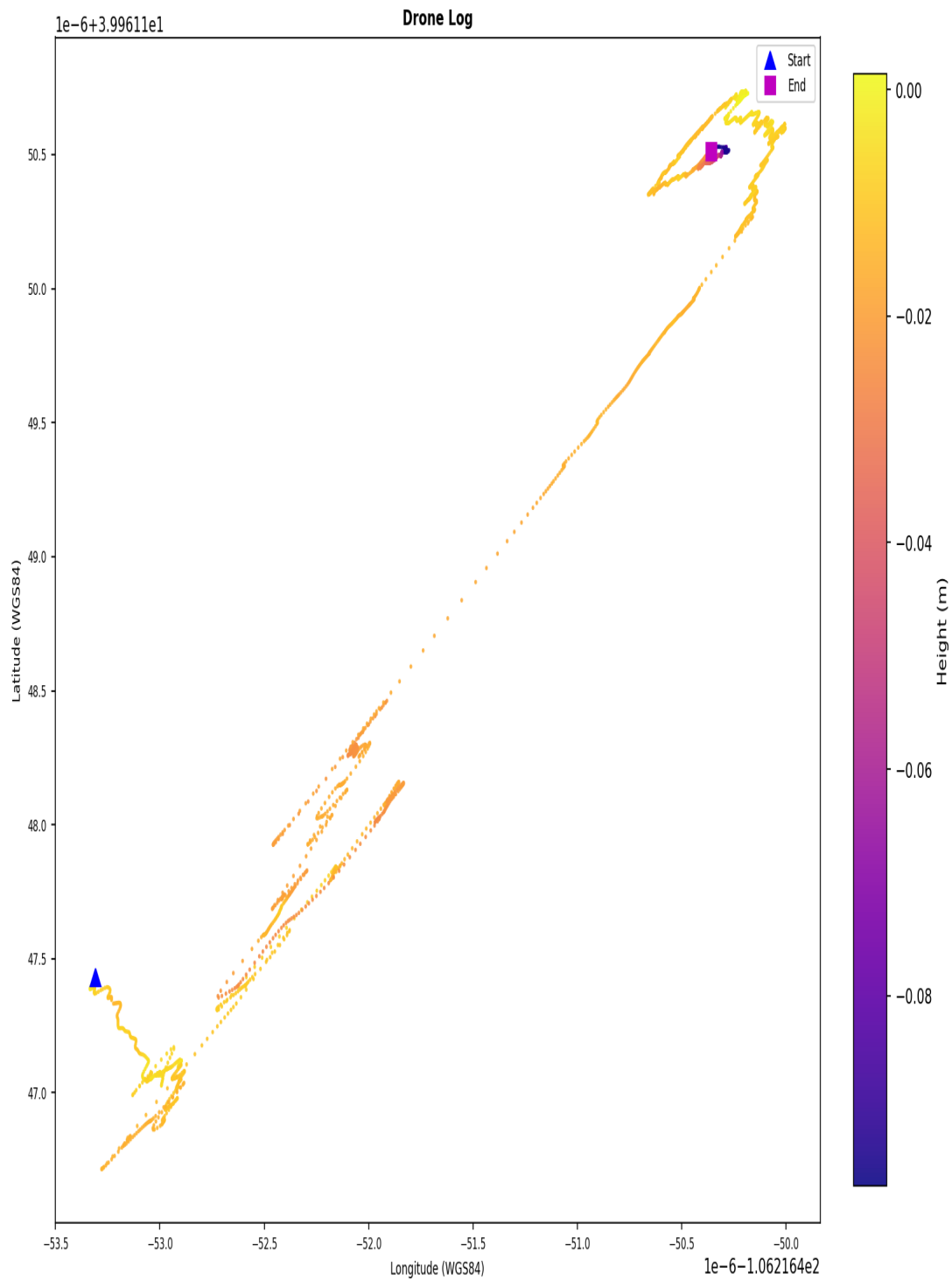


GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.



## Flight Ground Track

### Flight Ground Track - flight\_03



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY032.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY032.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 04

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 01dd846171ae99f8dff56e71d3951b023ce75d7d24daae94658d9172f570ddef  
**Drone Log Name:** FLY011.csv  
**Start:** 2015-10-22T19:25:19.034000+00:00  
**End:** 2015-10-22T19:43:52.553000+00:00  
**Duration:** 1113.5 s

### Flight Metrics

**Peak Height:** 121.05471907021426 m  
**Distance Travelled:** 4389.141749094651 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

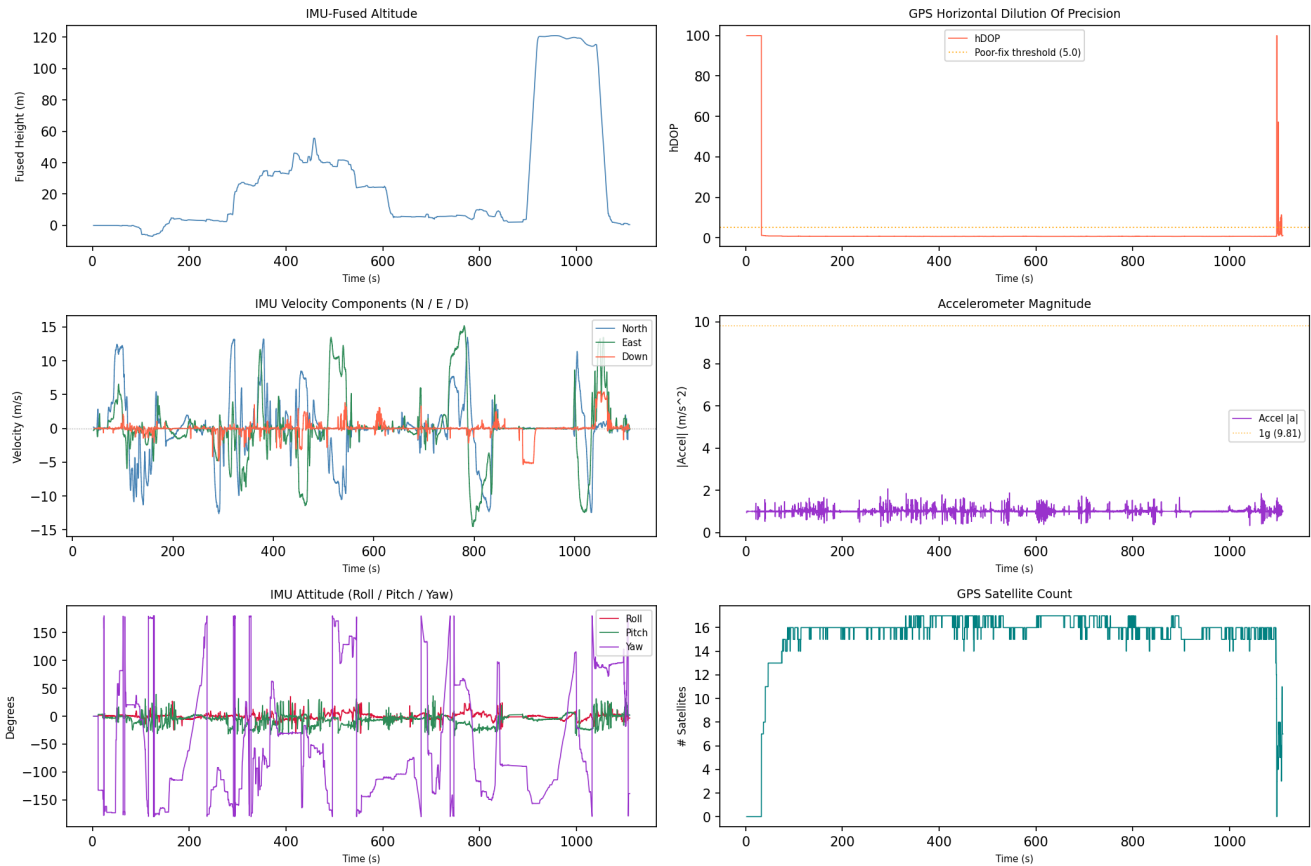
1. Atti
2. GPS\_Atti
3. ASST\_TAKEOFF
4. AssitedTakeoff
5. Manual
6. AutoLanding

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned on     | on      |
| 2015-10-22 19:25:25 | drone_flight_storage:dl | Log started         | P4P     |
| 2015-10-22 19:25:25 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 19:39:42 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 19:40:08 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-08-03 19:41:17 | drone_flight_storage:dl | Reached peak height | 121.1 m |
| 2017-08-03 19:43:35 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 19:43:51 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

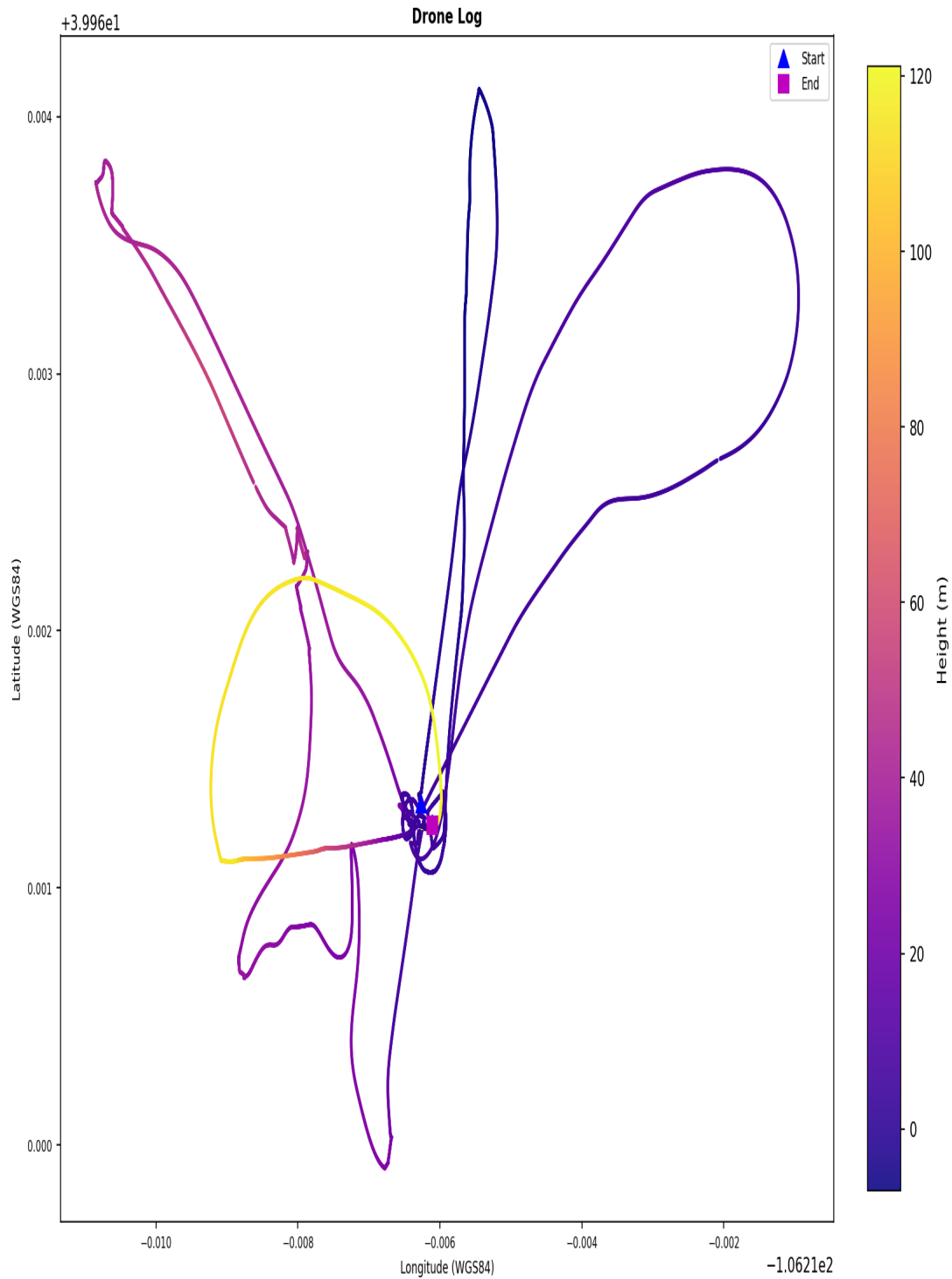
### Drone Log Telemetry - flight\_04



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_04



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY011.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY011.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 05

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** b87040de7f223968e8ec01ad29b57fcae4658a50494d02f6dd74a0029e2890c2  
**Drone Log Name:** FLY004.csv  
**Start:** 2017-06-29T17:09:44.230000+00:00  
**End:** 2017-06-29T17:36:11.938000+00:00  
**Duration:** 1587.7 s

### Flight Metrics

**Peak Height:** 40.489991322089175 m  
**Distance Travelled:** 2783.3967809272376 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

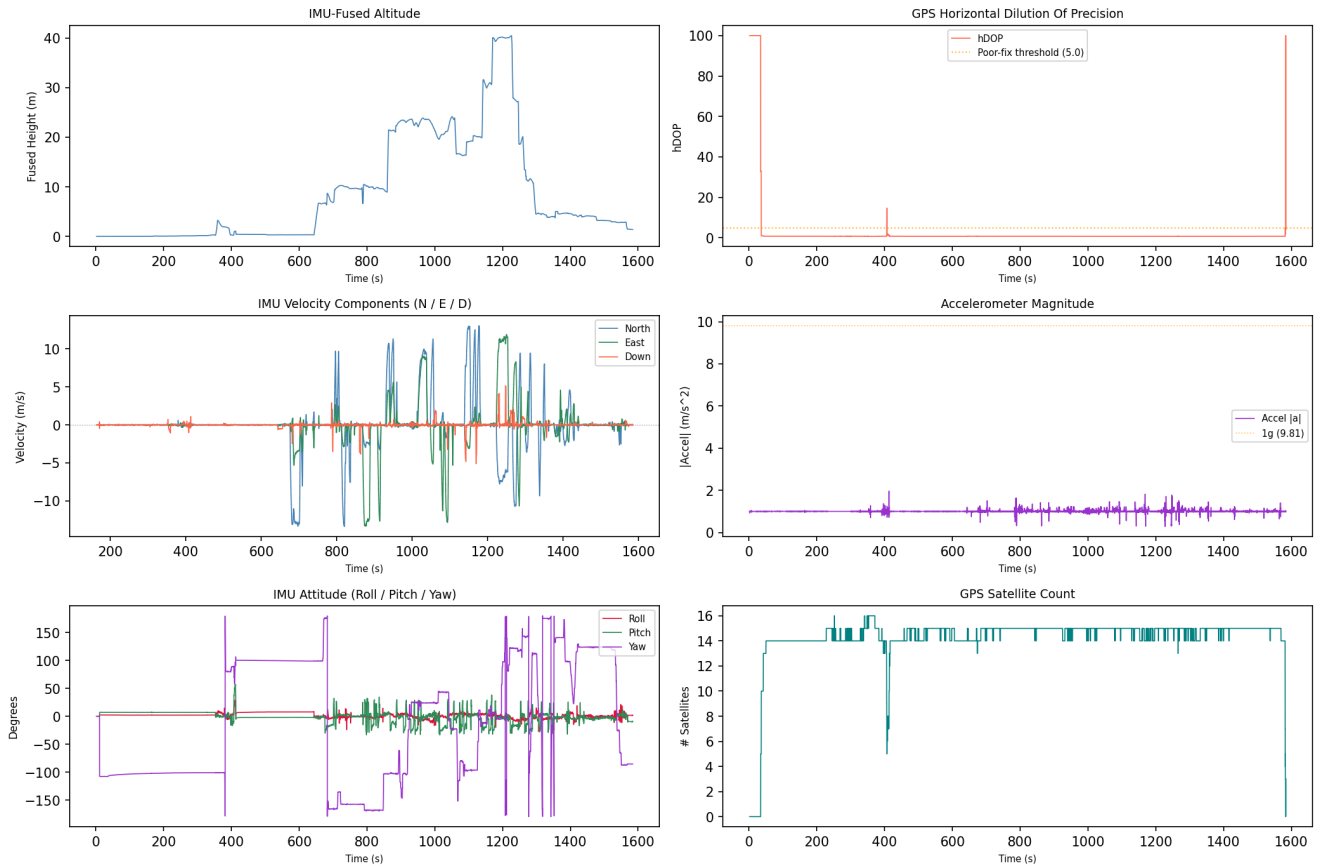
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AssitedTakeoff
6. AutoTakeoff
7. AutoLanding

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-06-29 17:10:24 | drone_flight_storage:dl | Log started         | P4P     |
| 2017-06-29 17:12:31 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-06-29 17:12:38 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-06-29 17:15:14 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-06-29 17:16:30 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-06-29 17:20:29 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-06-29 17:30:13 | drone_flight_storage:dl | Reached peak height | 40.5 m  |
| 2017-06-29 17:35:58 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-06-29 17:36:12 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

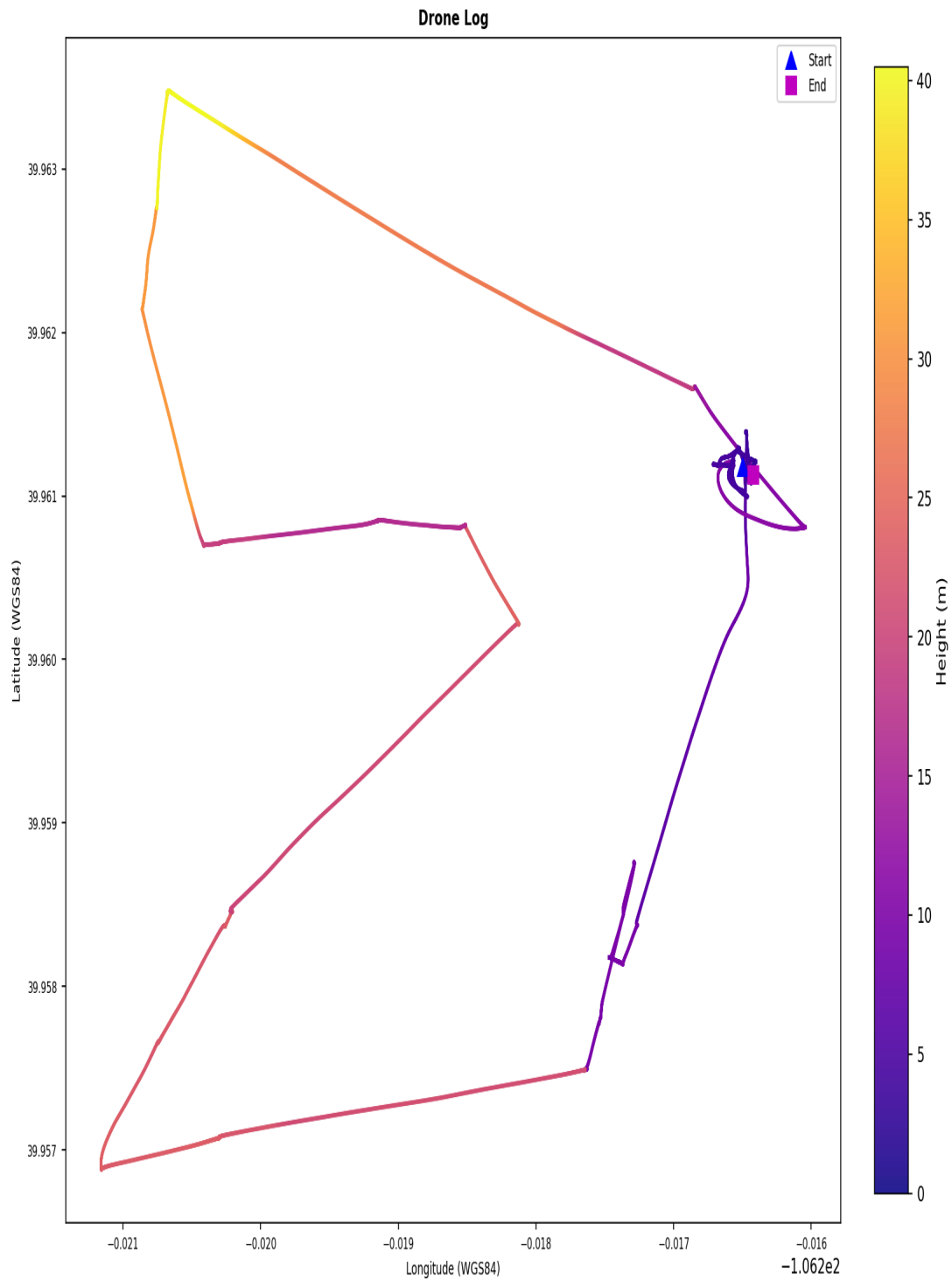
### Drone Log Telemetry - flight\_05



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_05



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY004.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY004.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 06

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** b2692488a43d9b9f7020f910845d6a80d5dd3876914b6f6d33b4780425edf11f  
**Drone Log Name:** FLY005.csv  
**Start:** 2017-06-29T20:51:02.002000+00:00  
**End:** 2017-06-29T21:02:56.039000+00:00  
**Duration:** 714.0 s

### Flight Metrics

**Peak Height:** 38.572346296076404 m  
**Distance Travelled:** 2329.1503786737785 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AutoTakeoff

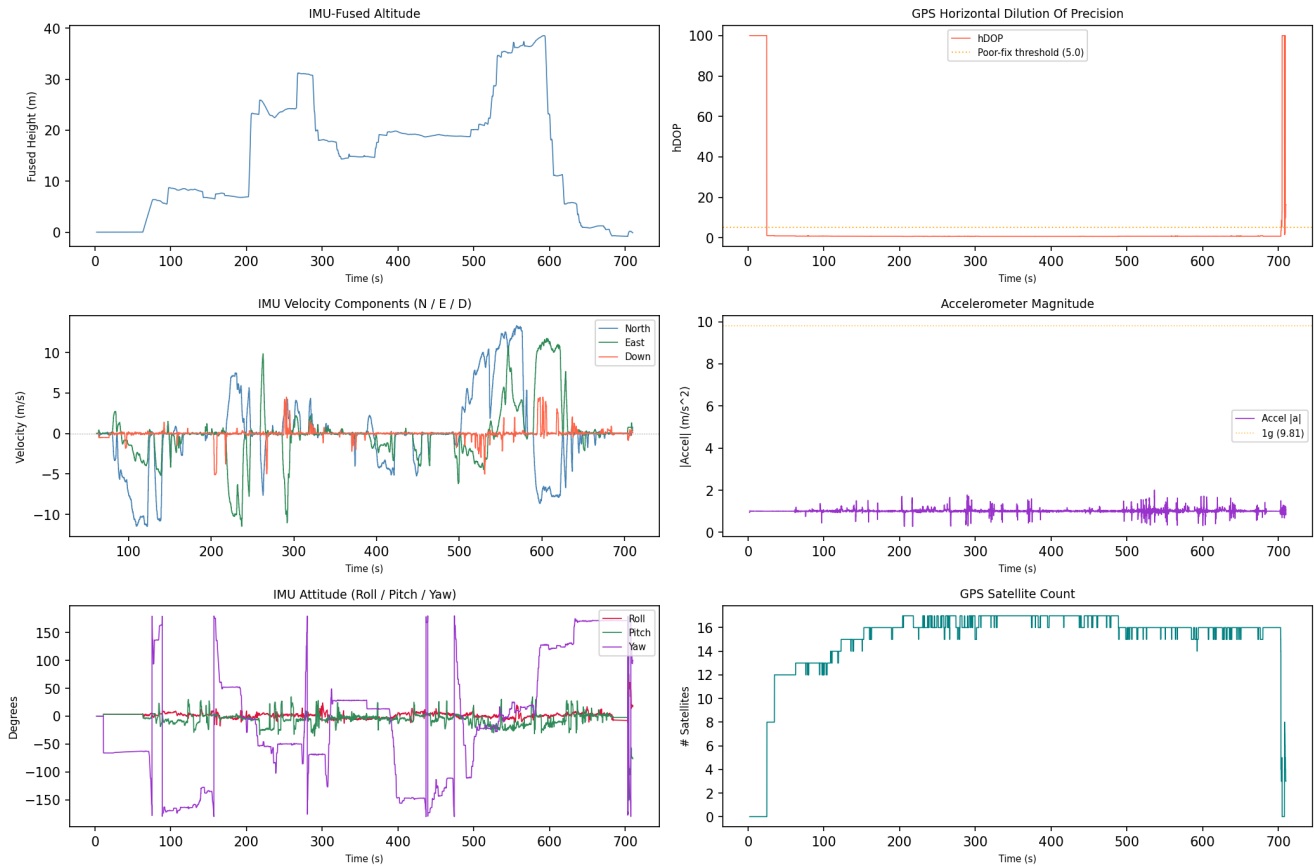
### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-06-29 20:51:31 | drone_flight_storage:dl | Log started         | P4P     |
| 2017-06-29 20:52:06 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-06-29 21:00:57 | drone_flight_storage:dl | Reached peak height | 38.6 m  |
| 2017-06-29 21:02:29 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-06-29 21:02:55 | drone_flight_storage:dl | Log ended           |         |



## Drone Log - Sensor Telemetry

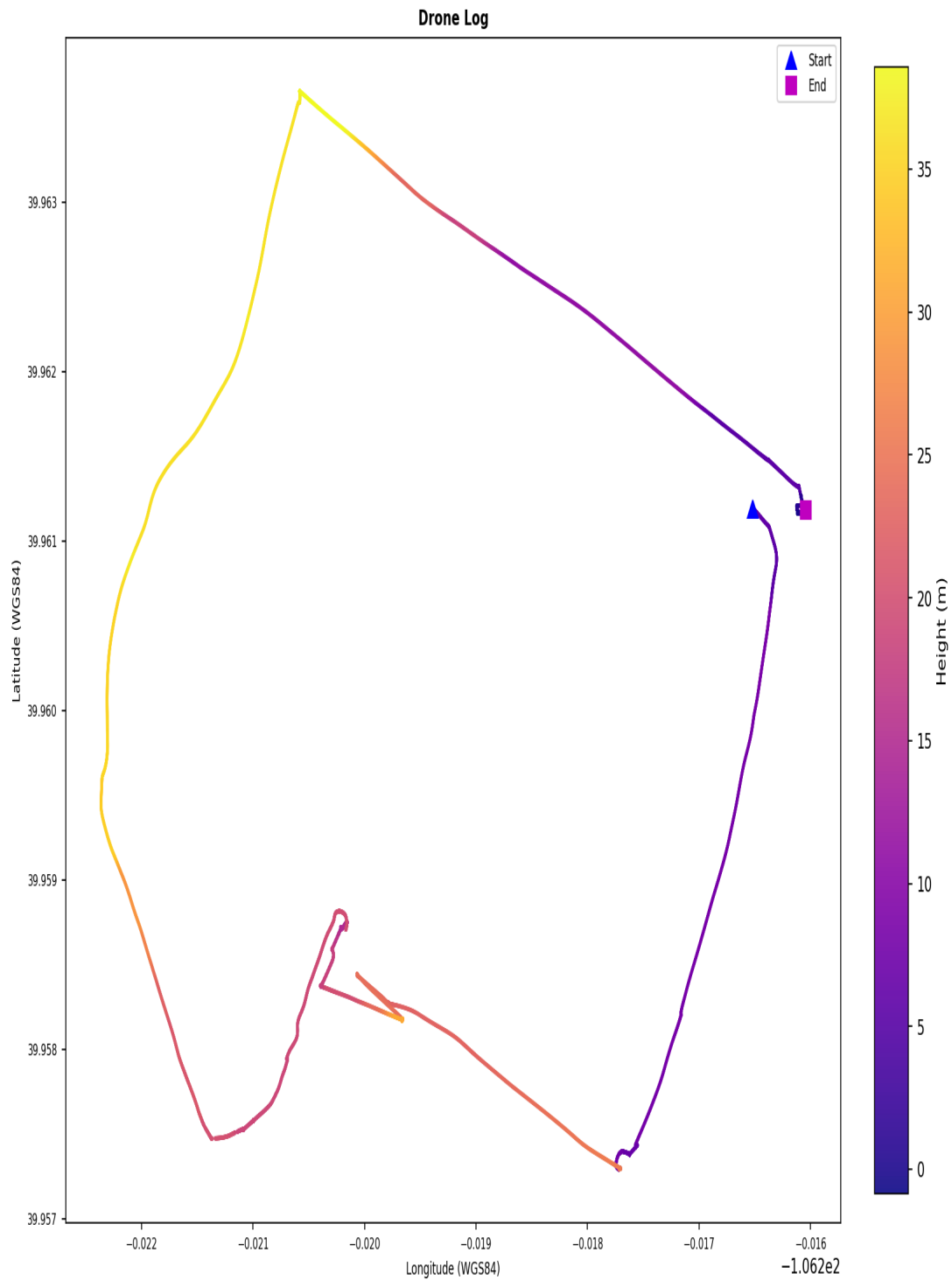
### Drone Log Telemetry - flight\_06



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hdOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hdOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hdOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

## Flight Ground Track - flight\_06



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY005.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY005.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 07

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 4c423ff2eeb138df997756505f5d30ebfecf984db18d2942b3cb91b5edc347c0  
**Drone Log Name:** FLY006.csv  
**Start:** 2017-08-03T17:34:01.033000+00:00  
**End:** 2017-08-03T17:40:10.158000+00:00  
**Duration:** 369.1 s

### Flight Metrics

**Peak Height:** 1.4551563075360554 m  
**Distance Travelled:** 30.648216259408404 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

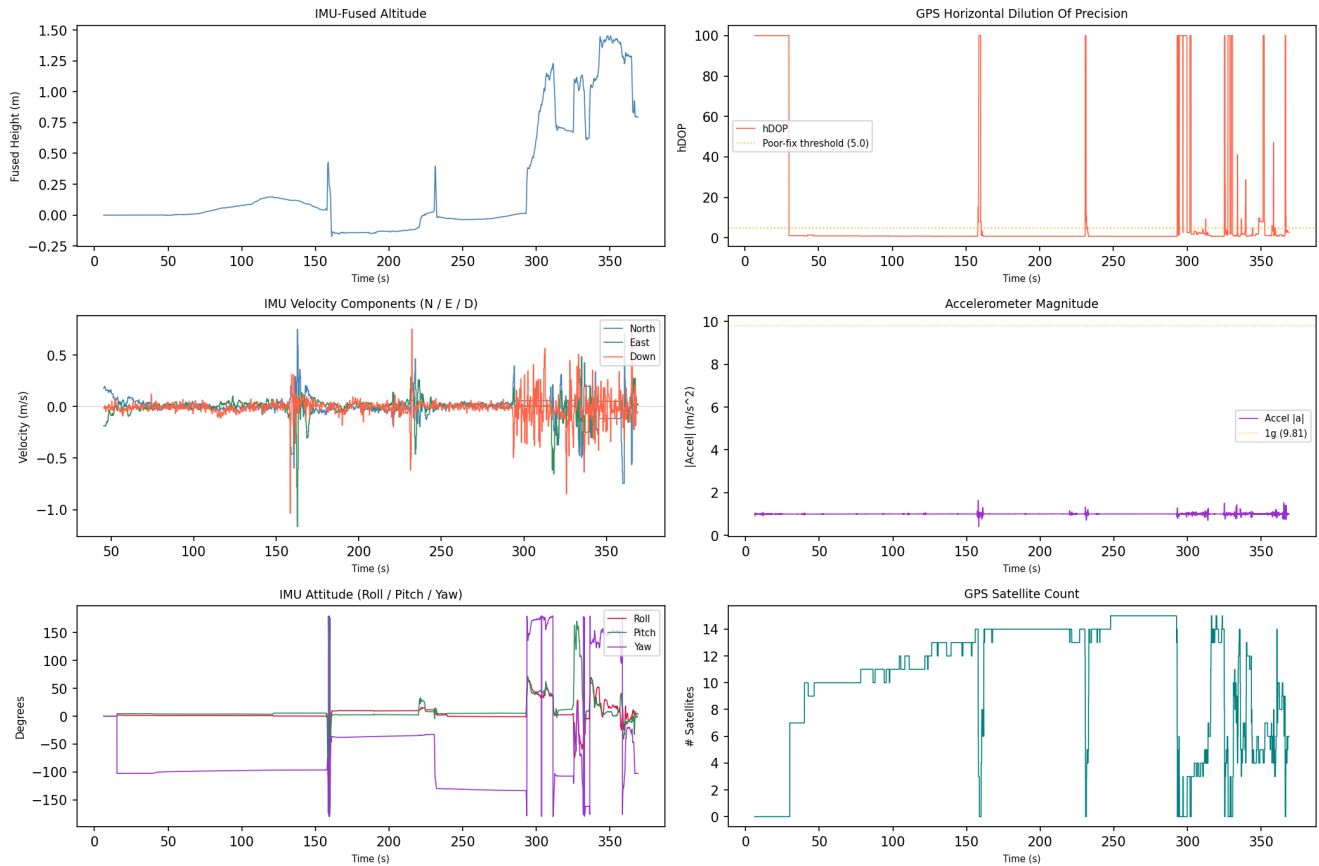
1. Manual
2. Atti
3. GPS\_Atti

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 17:34:31 | drone_flight_storage:dl | Log started         | P4P     |
| 2017-08-03 17:39:48 | drone_flight_storage:dl | Reached peak height | 1.5 m   |
| 2017-08-03 17:40:09 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

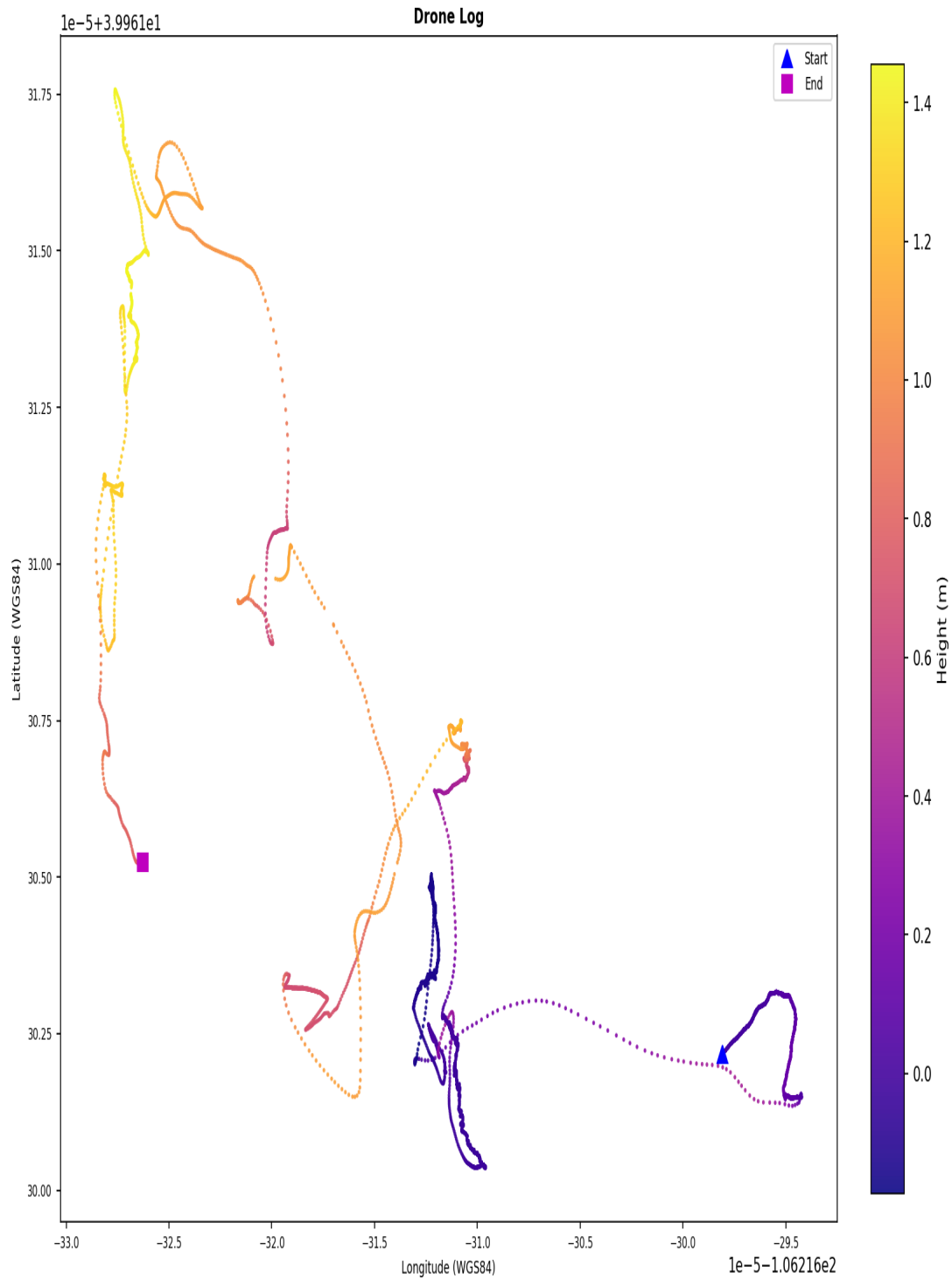
### Drone Log Telemetry - flight\_07



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_07



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY006.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY006.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 08

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 46dff0efe450754b475f8f88c01e824b097e39c8af255d1b1c2c93a29f60485d  
**Drone Log Name:** FLY007.csv  
**Start:** 2017-08-03T17:40:25.051000+00:00  
**End:** 2017-08-03T17:45:14.560000+00:00  
**Duration:** 289.5 s

### Flight Metrics

**Peak Height:** 1.9401717746535234 m  
**Distance Travelled:** 8.805794977746288 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

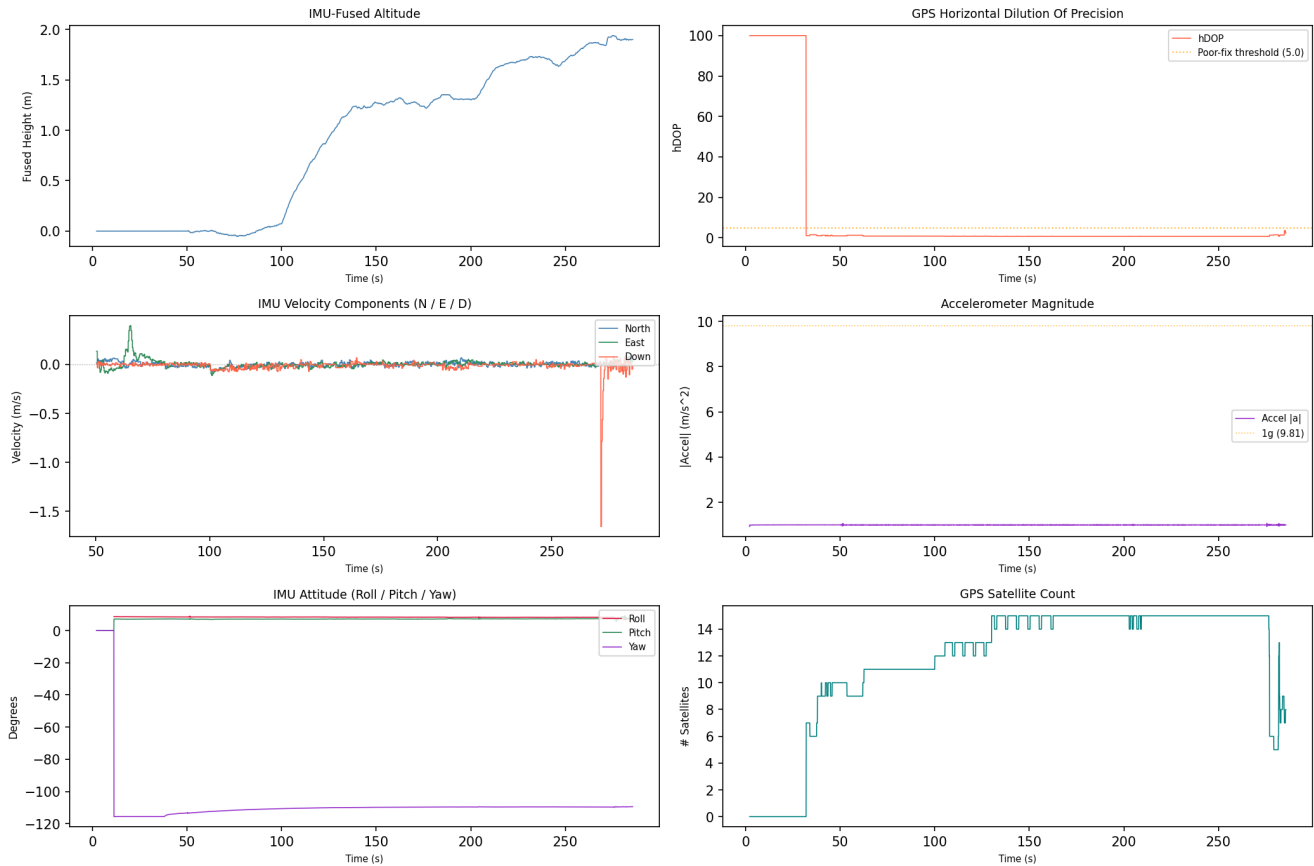
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AssitedTakeoff
6. AutoLanding

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 17:41:01 | drone_flight_storage:dl | Log started         | P4P     |
| 2017-08-03 17:41:18 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-08-03 17:44:59 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 17:45:03 | drone_flight_storage:dl | Reached peak height | 1.9 m   |
| 2017-08-03 17:45:13 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

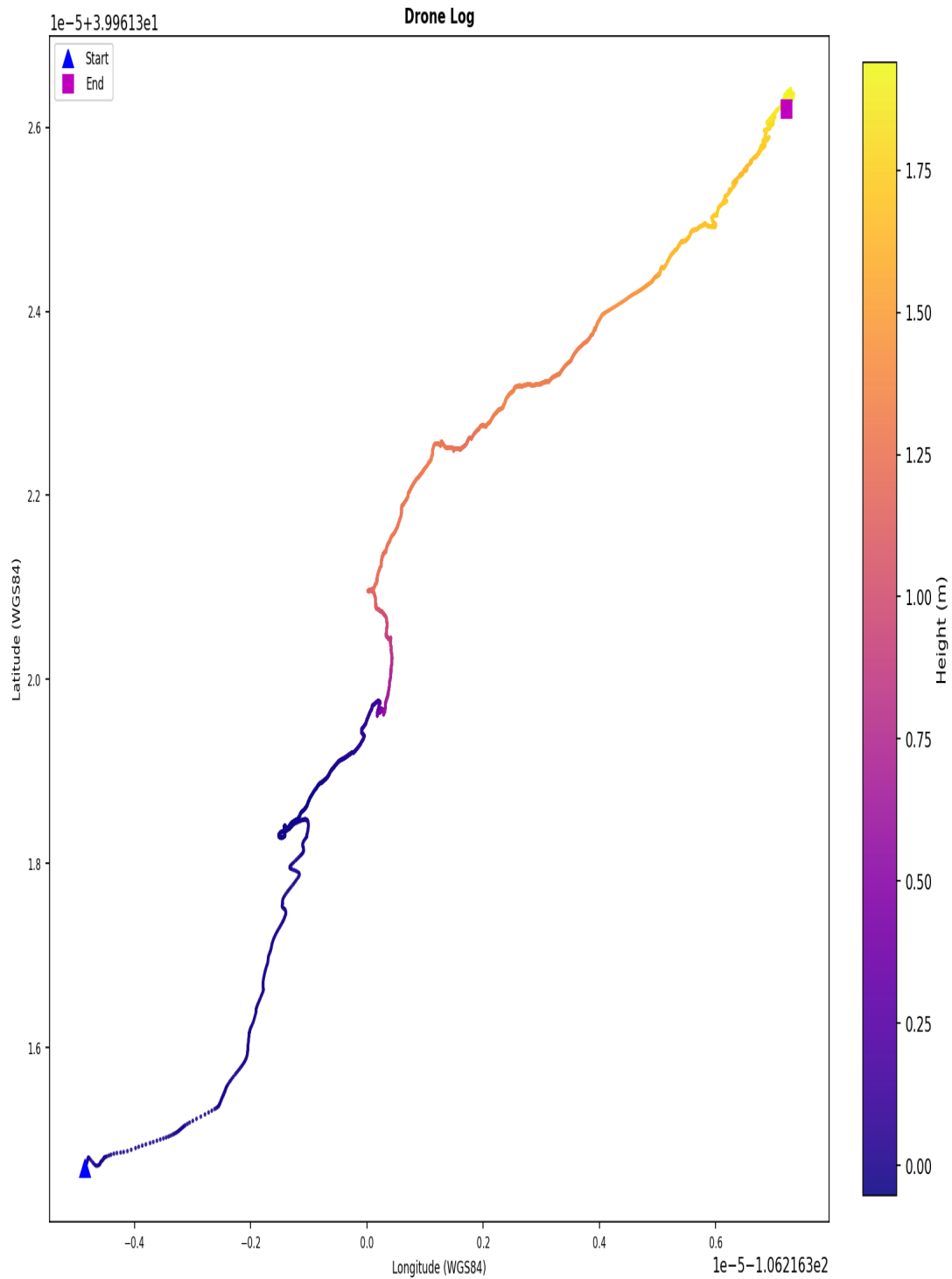
### Drone Log Telemetry - flight\_08



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_08



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY007.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY007.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.



## 4. Flight Analysis - Flight 09

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** d5ae1bfc28173436ac99bb4e8d2590f81851cb594eaa3b5cb5ea4eeffae033d8  
**Drone Log Name:** FLY008.csv  
**Start:** 2017-08-03T17:45:37.726000+00:00  
**End:** 2017-08-03T18:06:39.043000+00:00  
**Duration:** 1261.3 s

### Flight Metrics

**Peak Height:** 86.92763097967683 m  
**Distance Travelled:** 6301.6465540153995 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

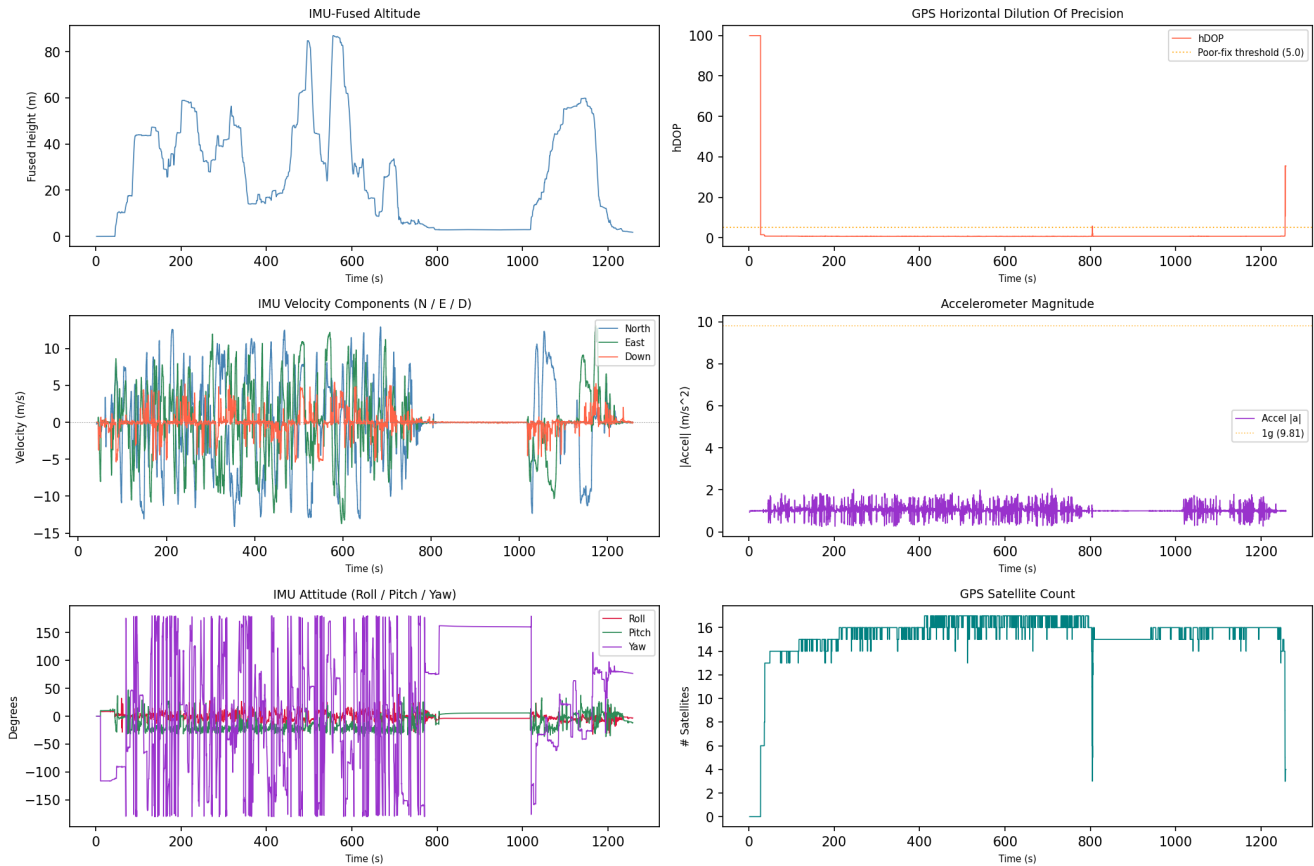
1. Atti
2. GPS\_Atti
3. Manual
4. ASST\_TAKEOFF
5. AssitedTakeoff
6. AutoLanding

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| 0002-11-30 00:00:00 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 17:46:09 | drone_flight_storage:dl | Log started         | P4P     |
| 2017-08-03 17:46:13 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-08-03 17:46:16 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 17:46:19 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-08-03 17:54:57 | drone_flight_storage:dl | Reached peak height | 86.9 m  |
| 2017-08-03 17:58:56 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 18:02:34 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-08-03 18:06:16 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 18:06:38 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

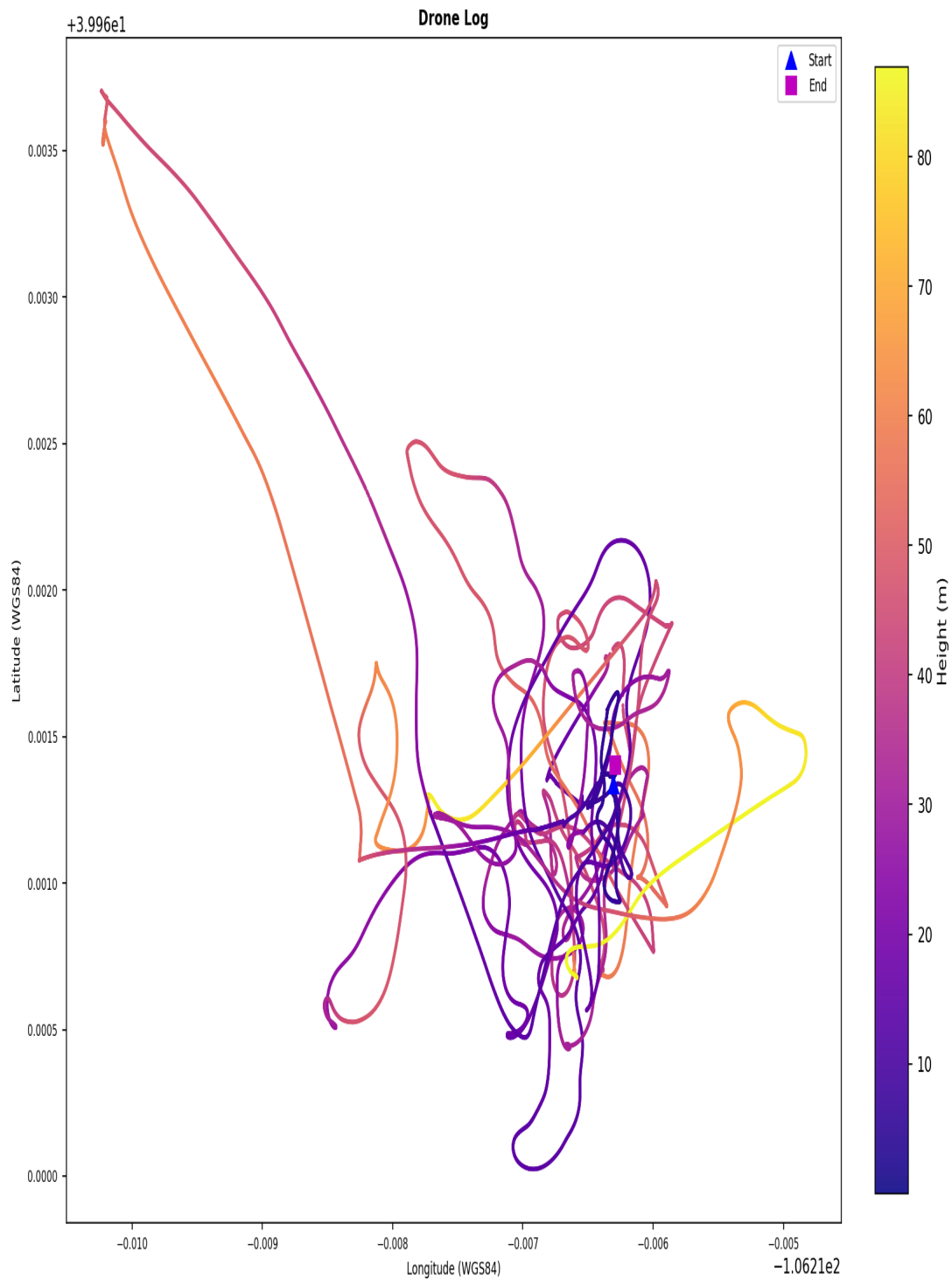
### Drone Log Telemetry - flight\_09



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_09



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY008.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY008.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 10

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** f007238898304209aa7dd66ed09c2faa3652e0da3e301edbe0afa6928fc08342  
**Drone Log Name:** FLY009.csv  
**Start:** 2017-08-03T18:08:06.113000+00:00  
**End:** 2017-08-03T18:14:33.614000+00:00  
**Duration:** 387.5 s

### Flight Metrics

**Peak Height:** 0.1703471956331671 m  
**Distance Travelled:** 7.181130450871256 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

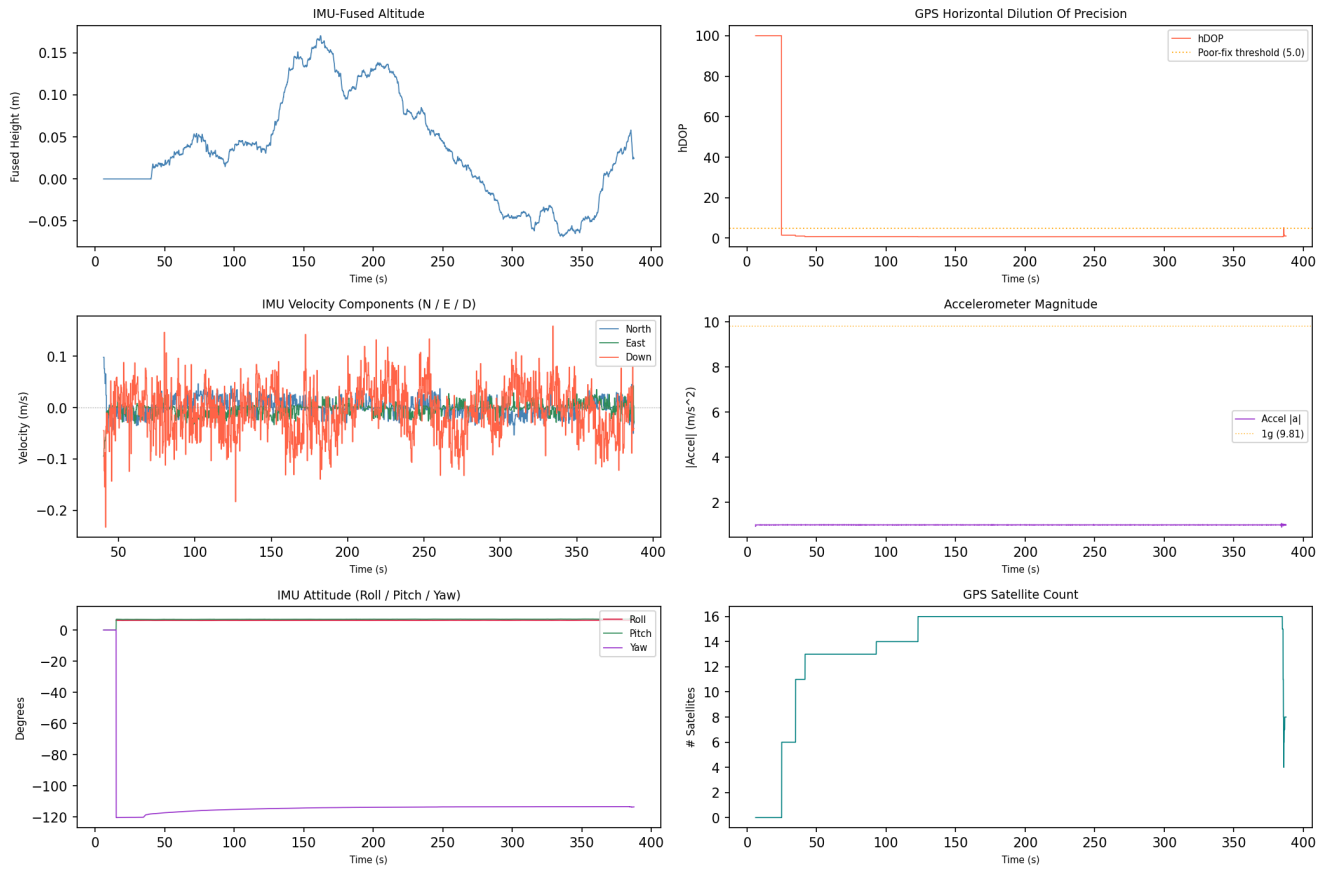
1. Manual
2. Atti
3. GPS\_Atti

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 18:08:31 | drone_flight_storage:dl | Log started         | P4P     |
| 2017-08-03 18:10:47 | drone_flight_storage:dl | Reached peak height | 0.2 m   |
| 2017-08-03 18:14:32 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

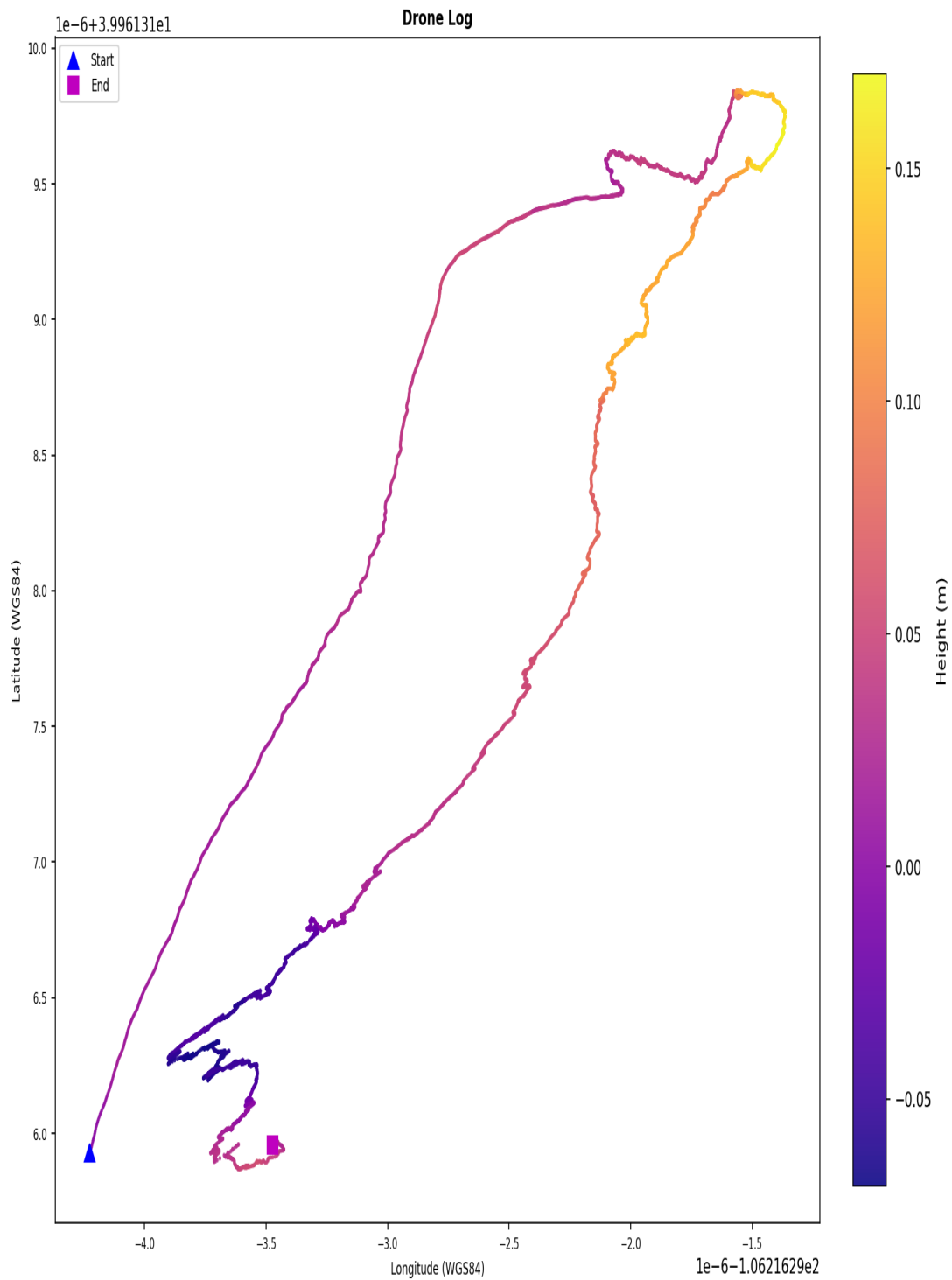
### Drone Log Telemetry - flight\_10



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_10



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY009.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY009.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 11

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** c677f015ebc55a0b16fd5af94ccdbd95c805410f1f0033dd217c3ed450b74e13  
**Drone Log Name:** FLY010.csv  
**Start:** 2017-08-03T19:20:21.896000+00:00  
**End:** 2017-08-03T19:21:59.454000+00:00  
**Duration:** 97.6 s

### Flight Metrics

**Peak Height:** 1.0837763527019018 m  
**Distance Travelled:** 4.416985391548159 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

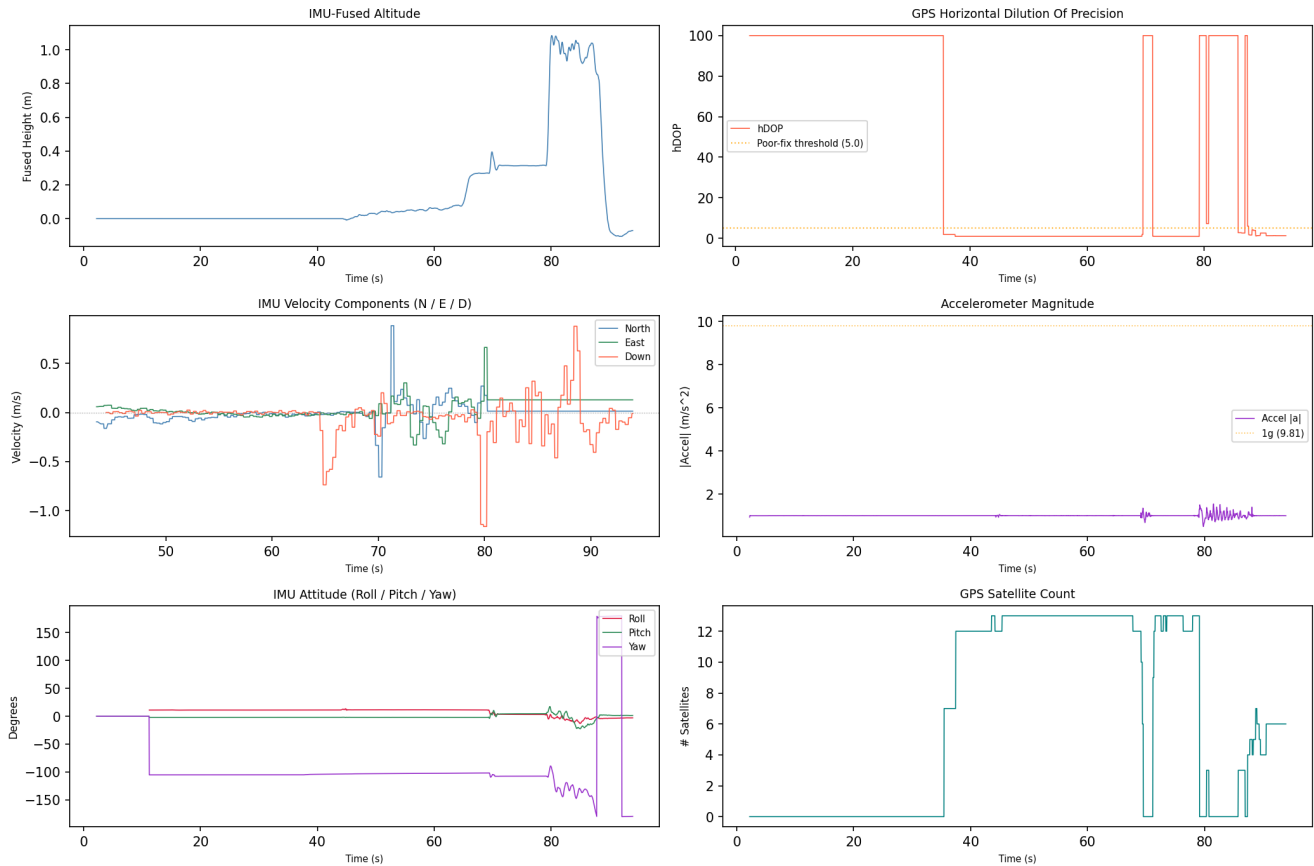
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AssitedTakeoff
6. AutoLanding

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 19:21:01 | drone_flight_storage:dl | Log started         | P4P     |
| 2017-08-03 19:21:07 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-08-03 19:21:29 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 19:21:44 | drone_flight_storage:dl | Reached peak height | 1.1 m   |
| 2017-08-03 19:21:58 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

### Drone Log Telemetry - flight\_11

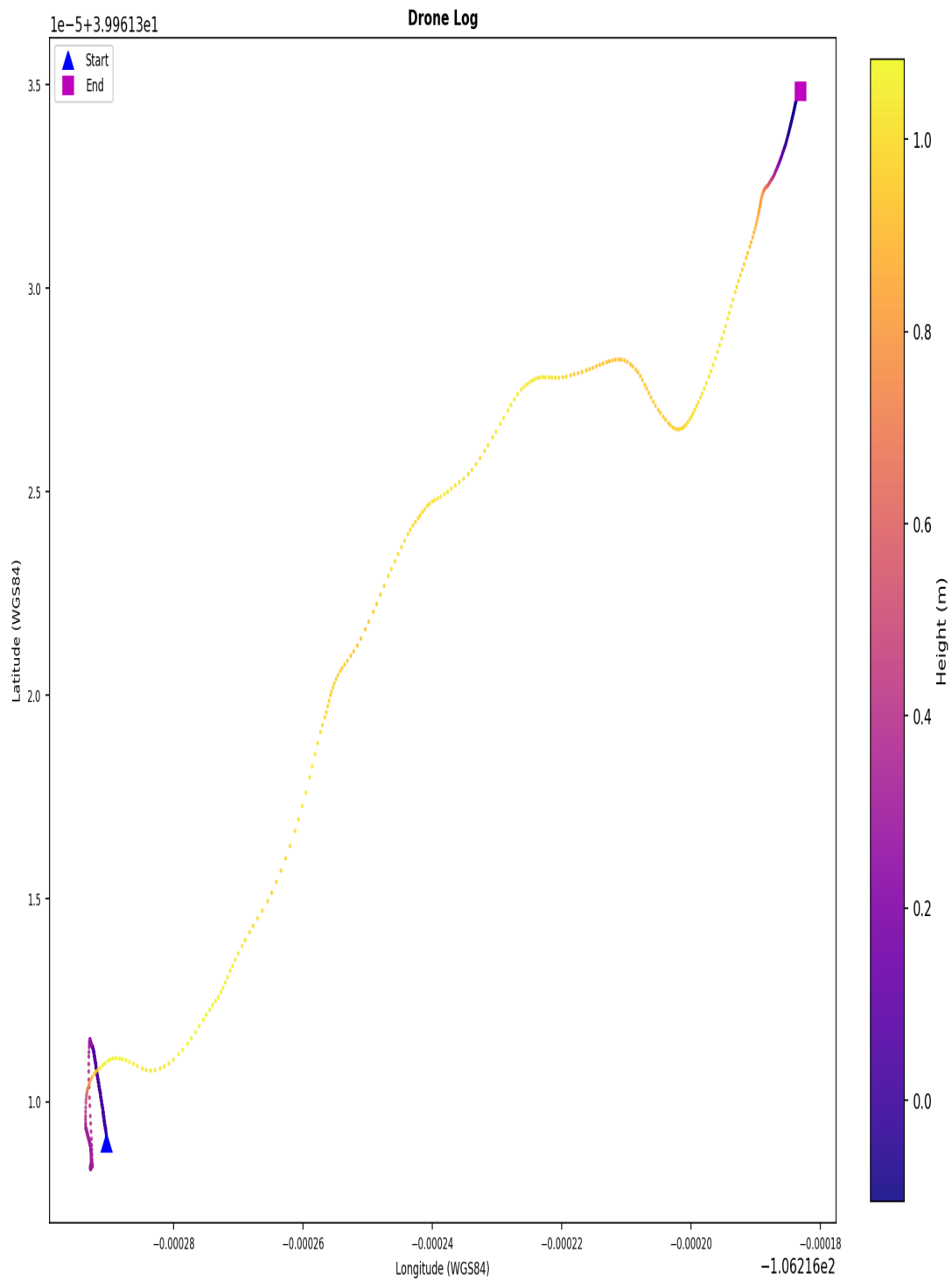


GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.



## Flight Ground Track

### Flight Ground Track - flight\_11



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY010.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY010.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 12

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** fad8c9df4d6b0ad78b6e6bb401e3ecfd9d49af67ba5db15ebfad84e2d3bf093e  
**Drone Log Name:** FLY012.csv  
**Start:** 2017-08-03T20:08:29.033000+00:00  
**End:** 2017-08-03T20:27:32.361000+00:00  
**Duration:** 1143.3 s

### Flight Metrics

**Peak Height:** 108.56425511850321 m  
**Distance Travelled:** 8249.605713470299 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

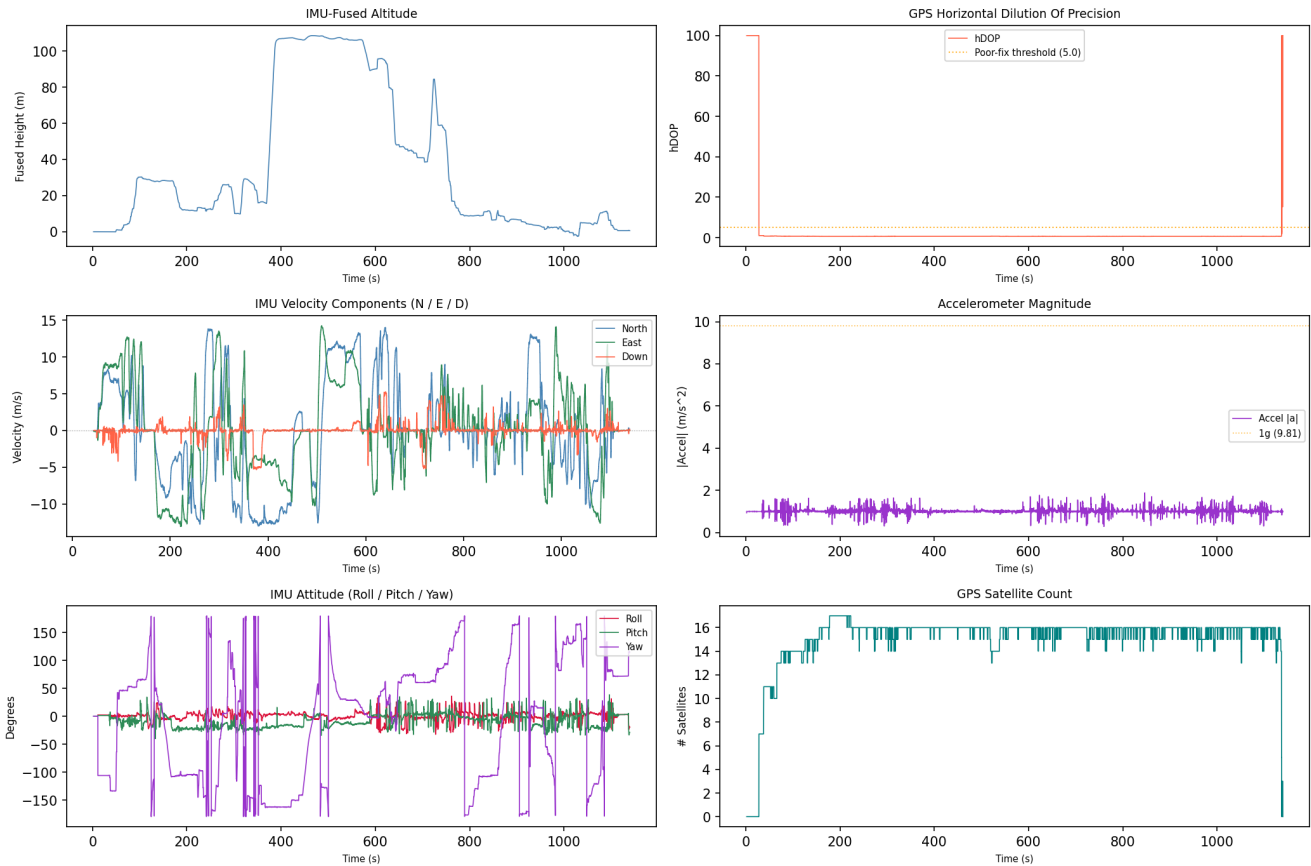
1. Manual
2. Atti
3. ASST\_TAKEOFF
4. AssitedTakeoff
5. GPS\_Atti
6. AutoLanding

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 20:09:01 | drone_flight_storage:dl | Log started         | P4P     |
| 2017-08-03 20:09:05 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-08-03 20:16:17 | drone_flight_storage:dl | Reached peak height | 108.6 m |
| 2017-08-03 20:27:07 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 20:27:31 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

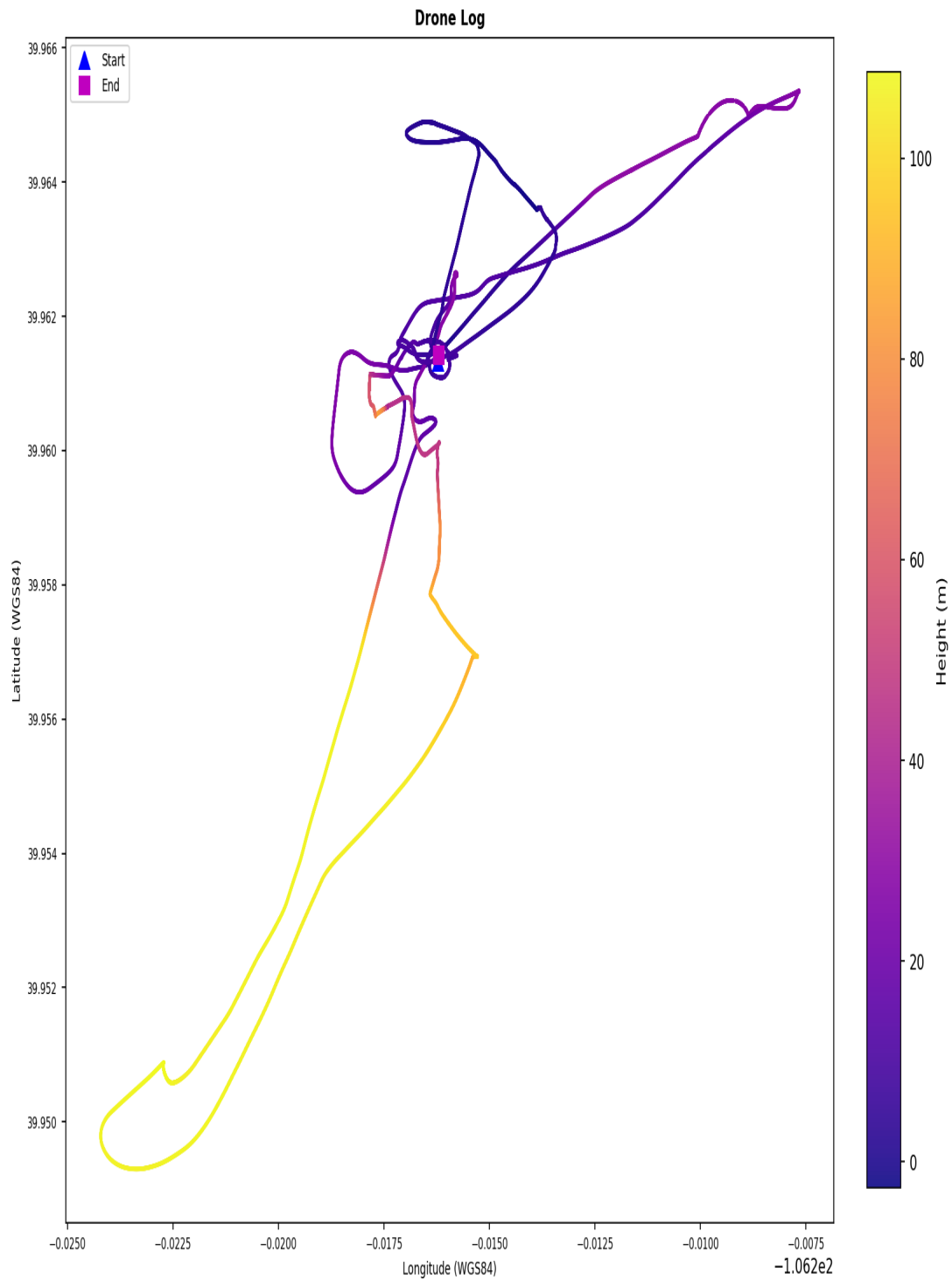
### Drone Log Telemetry - flight\_12



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_12



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY012.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY012.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 13

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 41224a227f46daf1196becc5bfff6e393747fdb684ee520793b24e1df94c748ef  
**Drone Log Name:** FLY013.csv  
**Start:** 2017-08-03T21:25:45.546000+00:00  
**End:** 2017-08-03T21:51:07.434000+00:00  
**Duration:** 1521.9 s

### Flight Metrics

**Peak Height:** 117.6100502158835 m  
**Distance Travelled:** 5017.229998082241 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

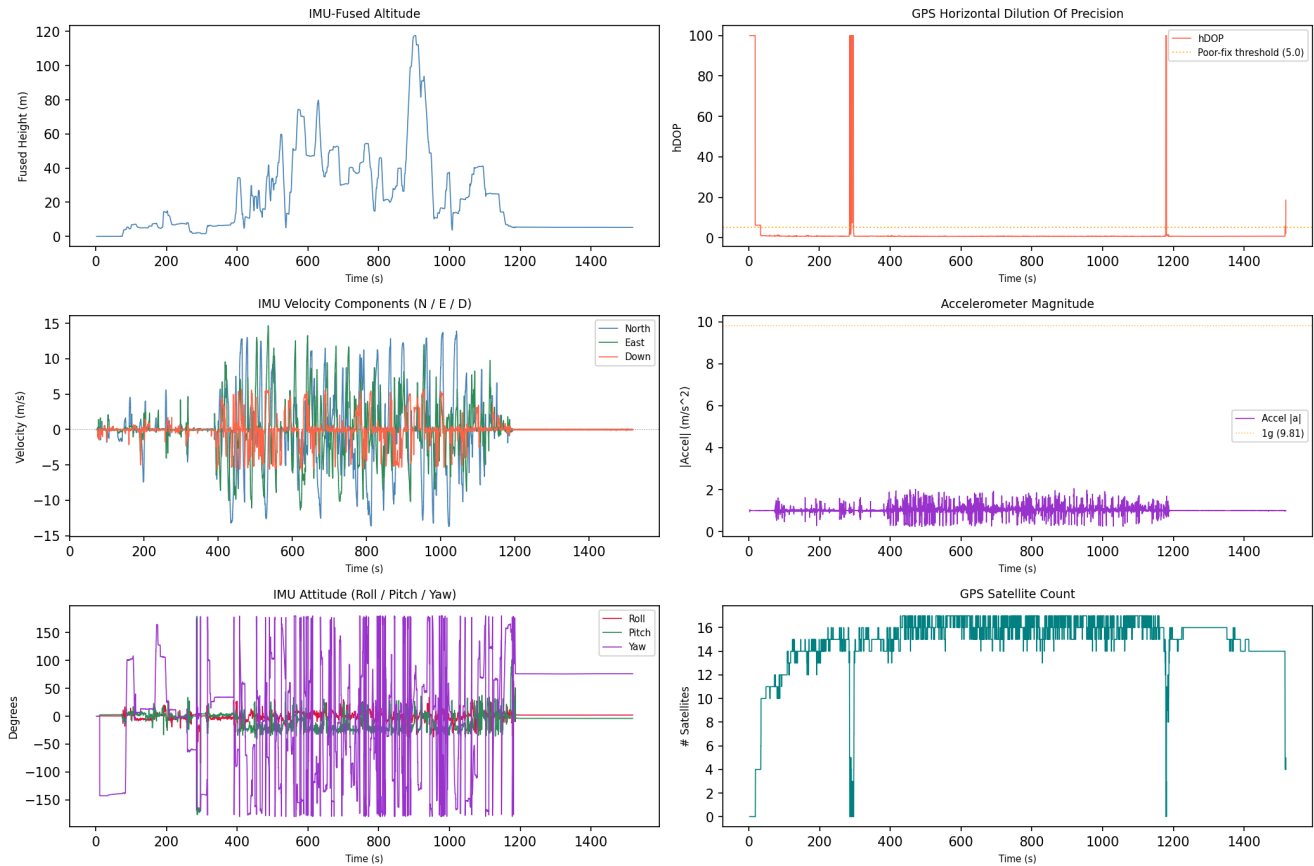
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AutoTakeoff

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 21:26:23 | drone_flight_storage:dl | Log started         | P4P     |
| 2017-08-03 21:27:02 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-08-03 21:30:24 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 21:30:58 | drone_flight_storage:dl | Motor turned on     | on      |
| 2017-08-03 21:40:54 | drone_flight_storage:dl | Reached peak height | 117.6 m |
| 2017-08-03 21:45:23 | drone_flight_storage:dl | Motor turned off    | off     |
| 2017-08-03 21:51:07 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

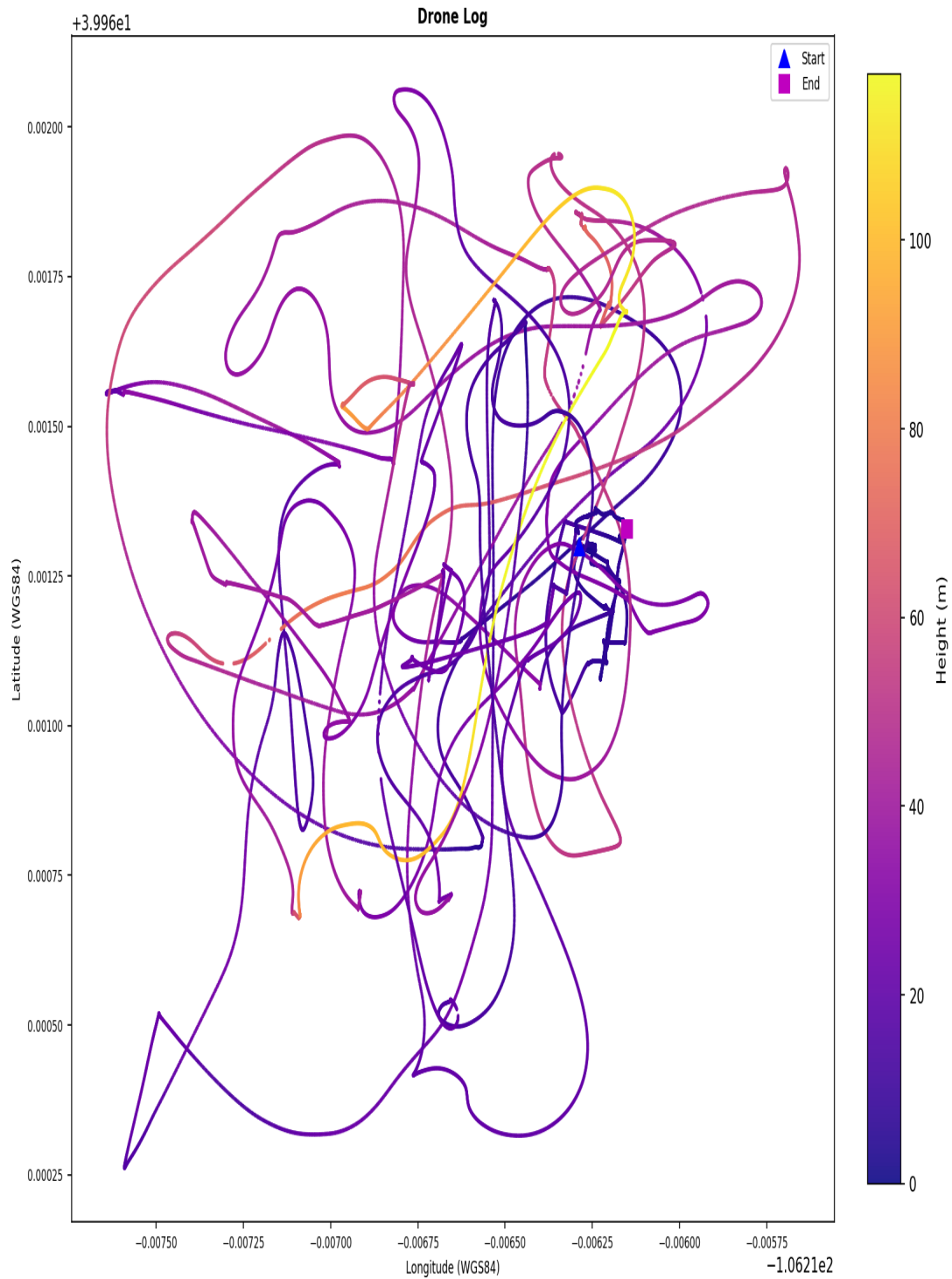
### Drone Log Telemetry - flight\_13



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

## Flight Ground Track - flight\_13



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY013.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY013.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 14

### Correlation and Flight Window

|                        |                                                                  |
|------------------------|------------------------------------------------------------------|
| Status:                | Matched (primary GPS+temporal)                                   |
| Confidence:            | high                                                             |
| GPS Overlap:           | 483.3 s                                                          |
| Median GPS Distance:   | 4.33 m                                                           |
| Overlap:               | 100%                                                             |
| GPS Match Rate:        | 100%                                                             |
| Flight Record SHA-256: | 5bb6399fa343827d7f37cfc4b400d80a4f04e5e49e197463446425580256552a |
| Flight Record Name:    | DJIFlightRecord_2018-06-19_13-39-43_.csv                         |
| Start:                 | 2018-06-19T19:39:43.621000+00:00                                 |
| End:                   | 2018-06-19T19:47:46.970000+00:00                                 |
| Duration:              | 483.3 s                                                          |
| Drone Log SHA-256:     | 350119d390dd0ef0768e37ef5d85971b741b8e14abff0ae12f01fc8c973b3b18 |
| Drone Log Name:        | 2018-06-19_13-38-06_FLY025.csv                                   |
| Start:                 | 2018-06-19T19:35:09.580000+00:00                                 |
| End:                   | 2018-06-19T19:48:57.774000+00:00                                 |
| Duration:              | 828.2 s                                                          |

### Flight Metrics

|                     |           |
|---------------------|-----------|
| Peak Height:        | 44.7 m    |
| Distance Travelled: | 2985.73 m |
| Video Recording:    | No        |
| Photo Capture:      | Yes       |
| Photos Taken:       | 7         |
| Recording Duration: | 0.0 s     |
| Record Complete:    | Yes       |

### Fly Modes Observed (Chronological)

1. GPS\_Atti
2. ASST\_TAKEOFF
3. EngineStart
4. AutoTakeoff
5. AutoLanding

### Key Events

| Timestamp (UTC)     | Source                | Event               | Details                                          |
|---------------------|-----------------------|---------------------|--------------------------------------------------|
| -                   | ctrl_ios:dl           | Motor turned off    | off                                              |
| 2018-06-19 19:38:29 | ctrl_ios:dl           | Log started         | P4P                                              |
| 2018-06-19 19:39:42 | ctrl_ios:dl           | Motor turned on     | on                                               |
| 2018-06-19 19:39:43 | ctrl_ios:fr (v4.2.20) | Log started         | SN: 07JDE5300200GN; App: 4.2.20; P4 Professional |
| 2018-06-19 19:39:43 | ctrl_ios:fr (v4.2.20) | Motor turned on     | on                                               |
| 2018-06-19 19:39:43 | ctrl_ios:fr (v4.2.20) | Record mode changed | No                                               |
| 2018-06-19 19:42:25 | ctrl_ios:fr (v4.2.20) | Reached peak height | 44.7 m                                           |
| 2018-06-19 19:43:47 | ctrl_ios:dl           | Reached peak height | 42.7 m                                           |
| 2018-06-19 19:47:46 | ctrl_ios:dl           | Motor turned off    | off                                              |
| 2018-06-19 19:47:46 | ctrl_ios:fr (v4.2.20) | Log ended           | Dist home: 0.3 m; Height: 0.0 m; Rec: 0.0 s      |
| 2018-06-19 19:47:46 | ctrl_ios:fr (v4.2.20) | Motor turned off    | off                                              |
| 2018-06-19 19:48:57 | ctrl_ios:dl           | Log ended           |                                                  |
| 2018-06-19 19:43:34 | ctrl_ios:fr (v4.2.20) | Photo mode changed  | Single                                           |
| 2018-06-19 19:45:55 | ctrl_ios:fr (v4.2.20) | Photo mode changed  | Single                                           |
| 2018-06-19 19:46:07 | ctrl_ios:fr (v4.2.20) | Photo mode changed  | Single                                           |



| Timestamp (UTC)     | Source                | Event              | Details |
|---------------------|-----------------------|--------------------|---------|
| 2018-06-19 19:46:09 | ctrl_ios:fr (v4.2.20) | Photo mode changed | Single  |
| 2018-06-19 19:47:28 | ctrl_ios:fr (v4.2.20) | Photo mode changed | Single  |

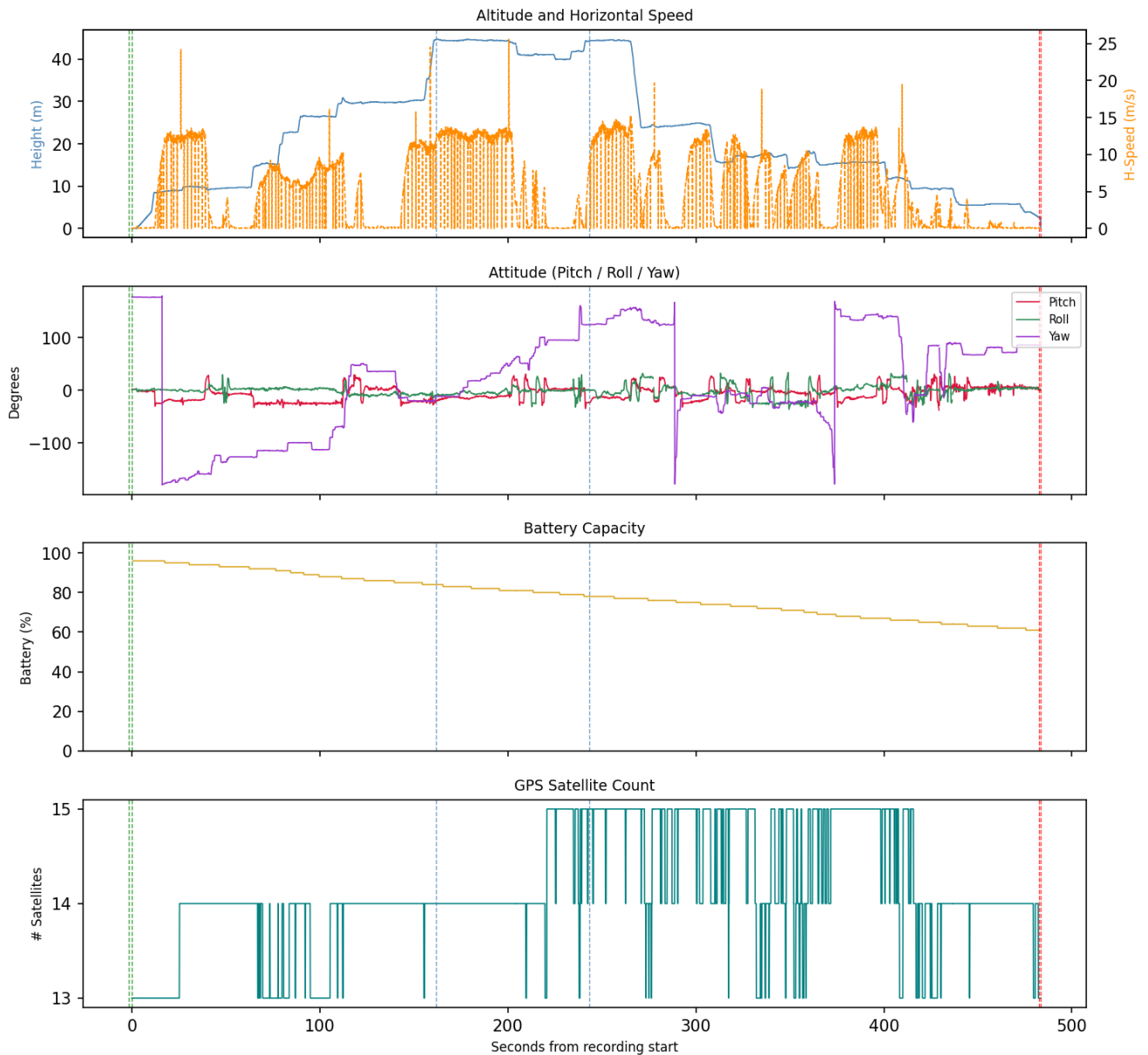
### Correlated Artefacts

Possibly correlated (indirect indicators):

| controller_ios: flight_logs                                         | Log messages match (6 entries, format: json_array_log) |
|---------------------------------------------------------------------|--------------------------------------------------------|
| p5:9d3f351a10b80e9bbc9fe724f7e0e91c07750719efa302e3743d9fdb2e6de33c |                                                        |

## Flight Record - Sensor Telemetry

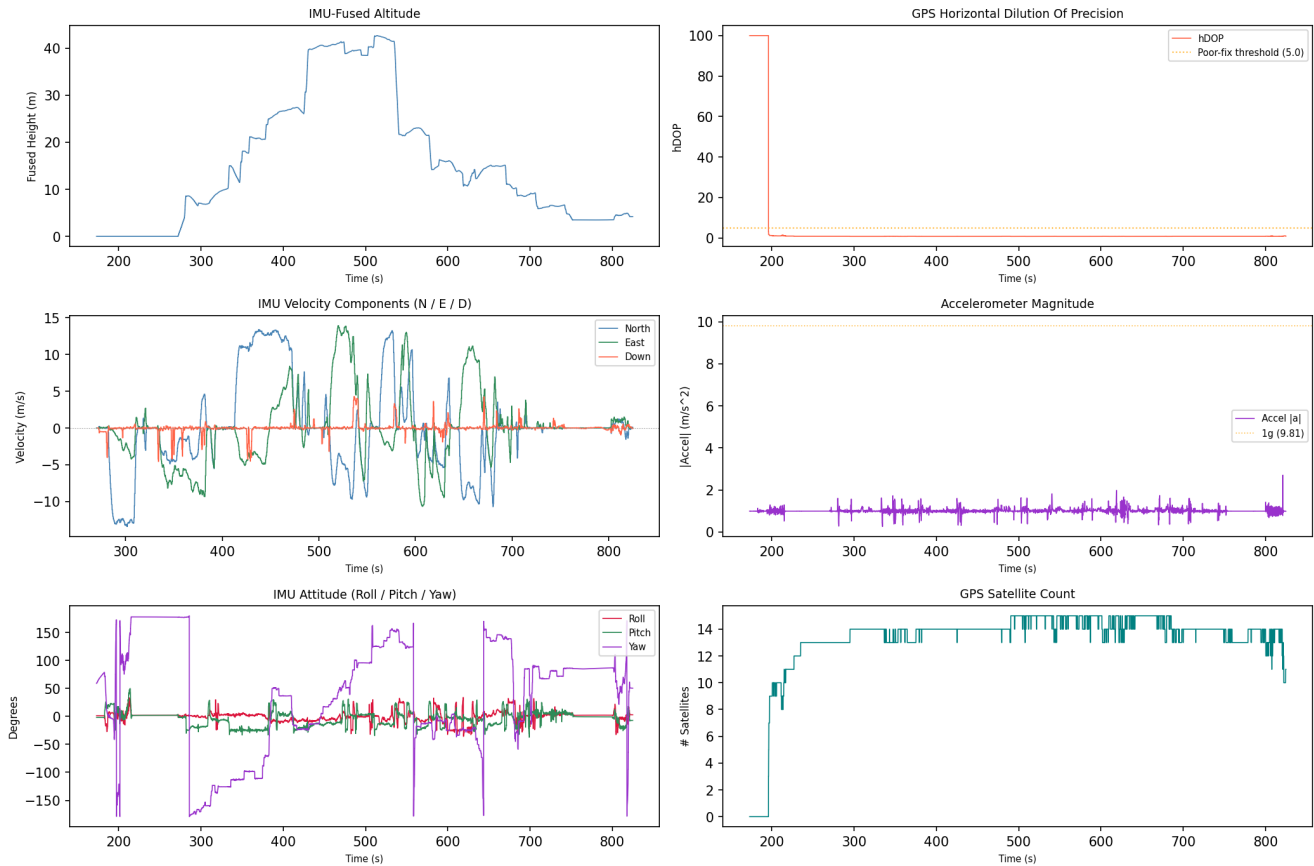
## Flight Record Telemetry - flight\_14



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

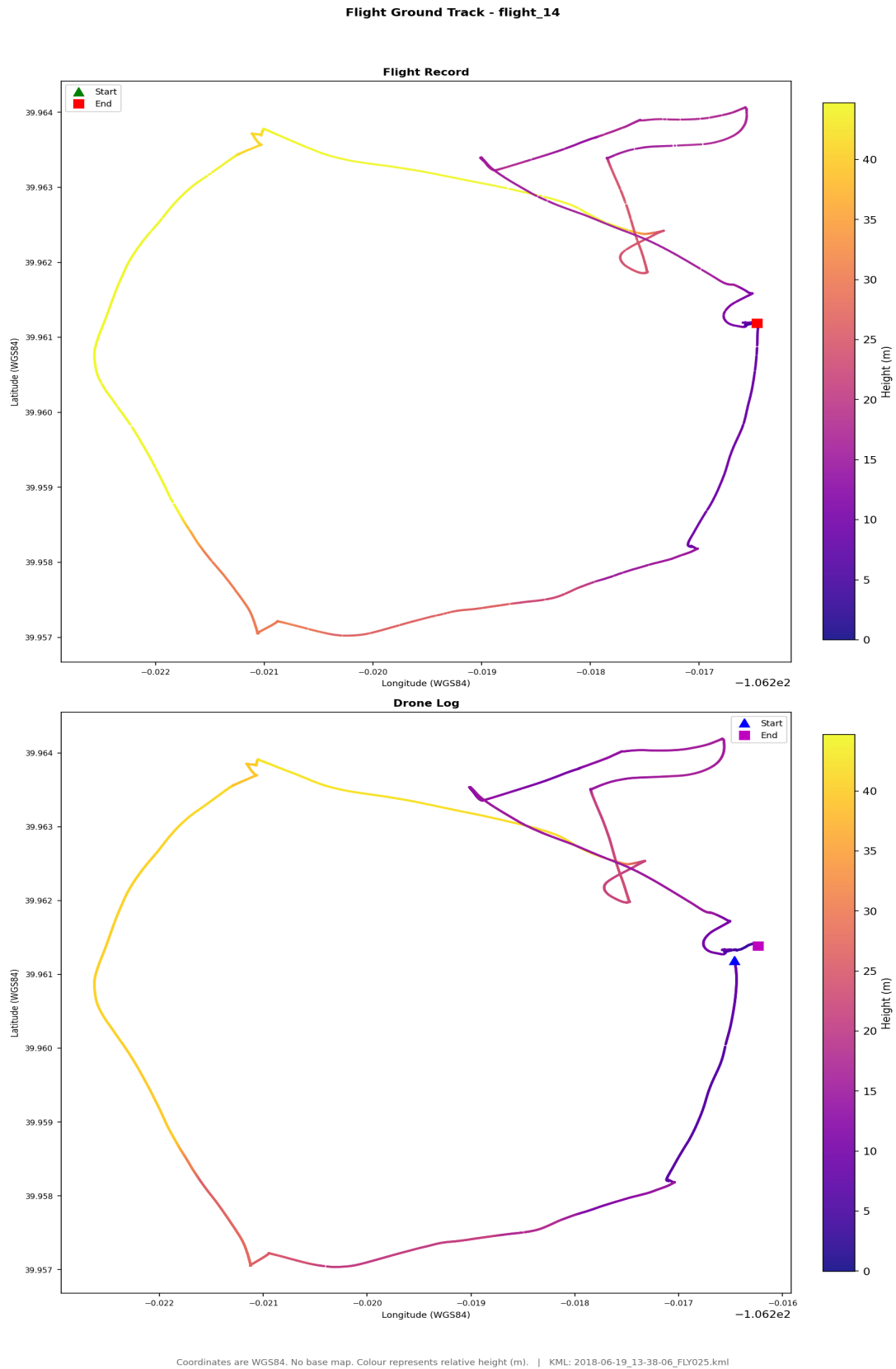
## Drone Log - Sensor Telemetry

### Drone Log Telemetry - flight\_14



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2018-06-19\_13-38-06\_FLY025.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 15

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** e4a06a59b0a376b6c009d70845d4a37ae8a40527c064788223a45f8476344da2  
**Drone Log Name:** FLY027.csv  
**Start:** 2018-06-20T20:15:05.429000+00:00  
**End:** 2018-06-20T20:16:12.946000+00:00  
**Duration:** 67.5 s

### Flight Metrics

**Peak Height:** 0.0031376458781003508 m  
**Distance Travelled:** 1.6417395072780427 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

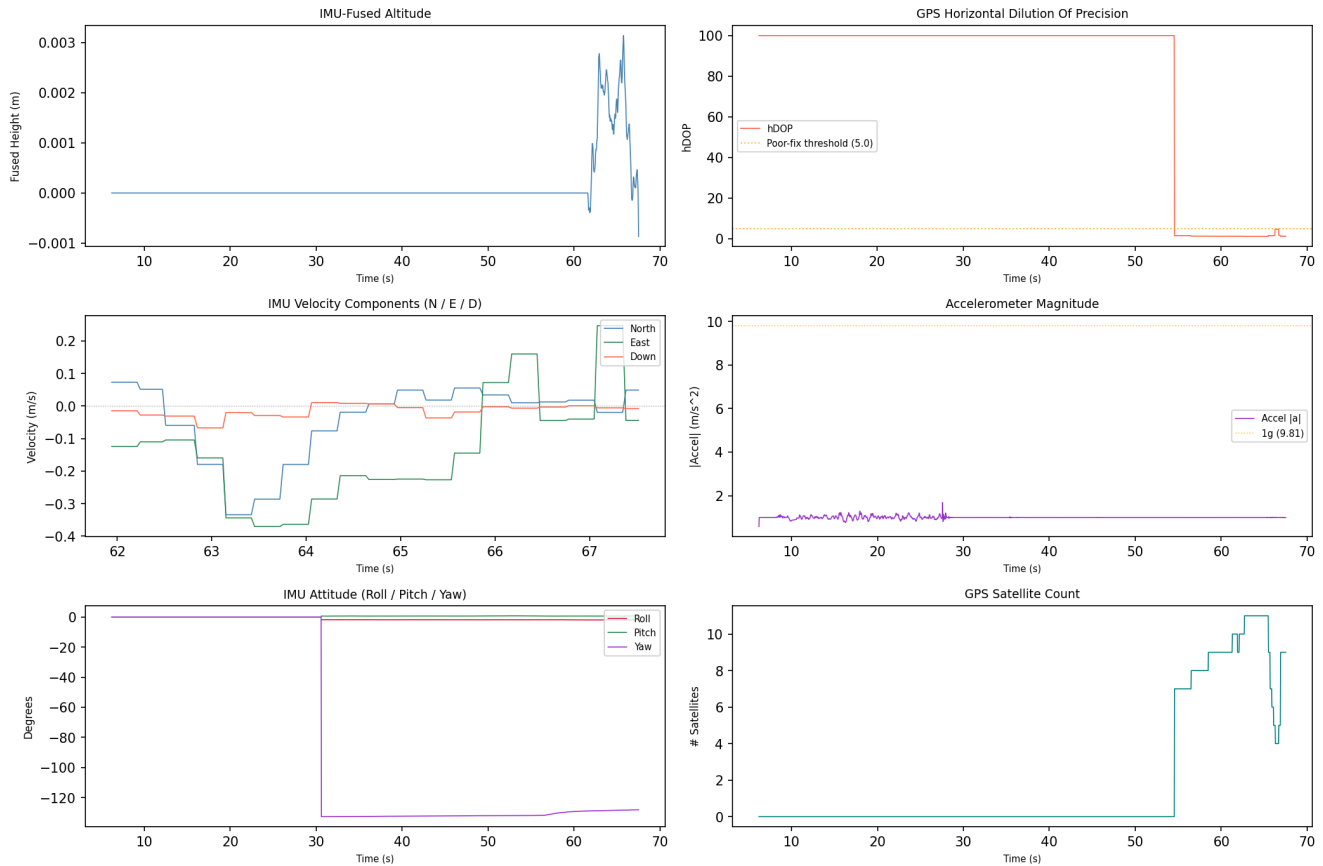
1. Manual
2. Atti
3. GPS\_Atti

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-20 20:16:00 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-20 20:16:11 | drone_flight_storage:dl | Reached peak height | 0.0 m   |
| 2018-06-20 20:16:13 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

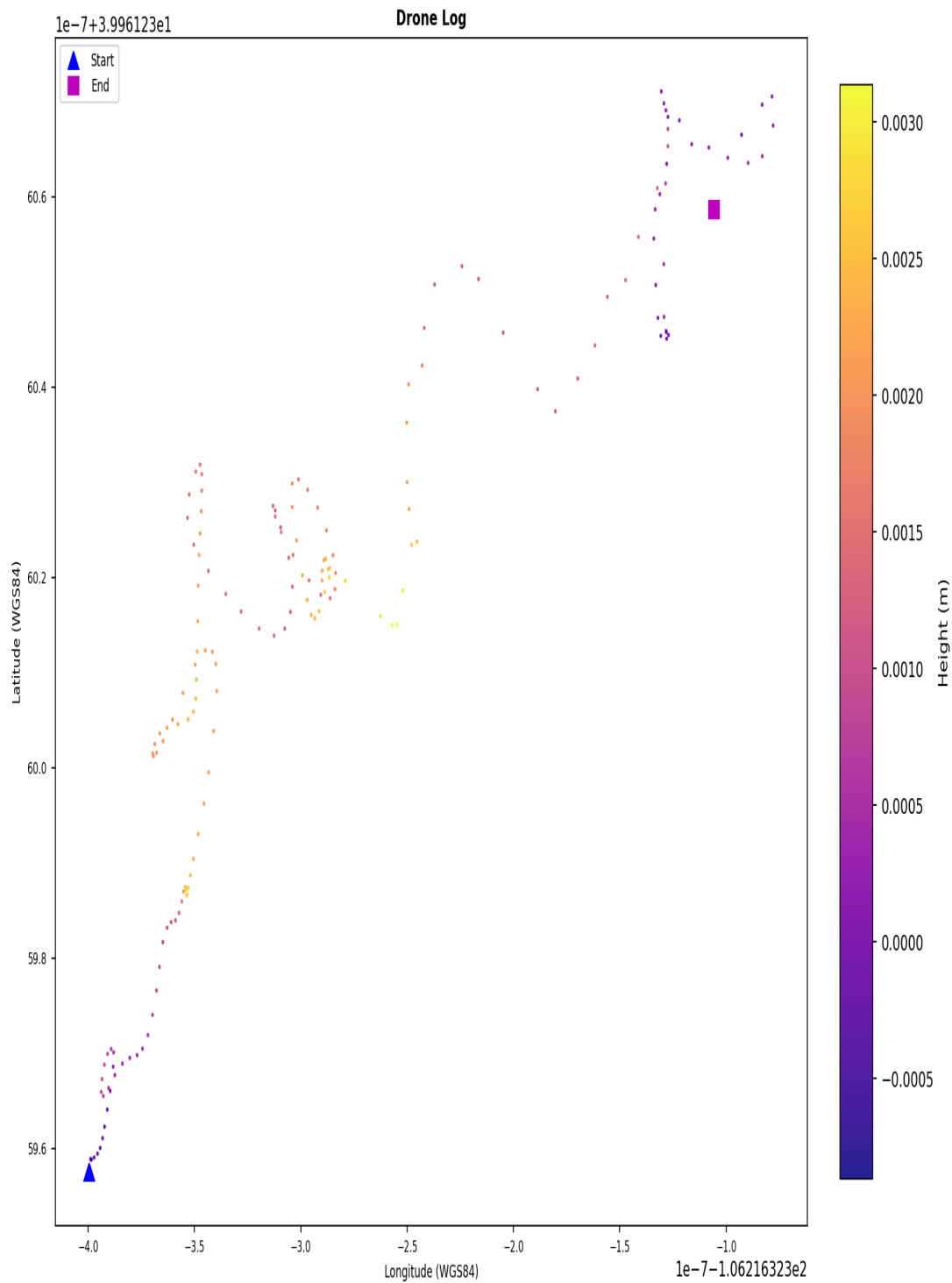
### Drone Log Telemetry - flight\_15



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_15



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY027.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY027.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 16

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 73d030763c62c201e442aed539b0f08d1f12e08ceb3620ca09ac8afed1091797  
**Drone Log Name:** FLY028.csv  
**Start:** 2018-06-20T20:17:01.341000+00:00  
**End:** 2018-06-20T20:32:00.175000+00:00  
**Duration:** 898.8 s

### Flight Metrics

**Peak Height:** 36.245934634710785 m  
**Distance Travelled:** 2476.023038884696 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AutoTakeoff
6. GoHome
7. AutoLanding

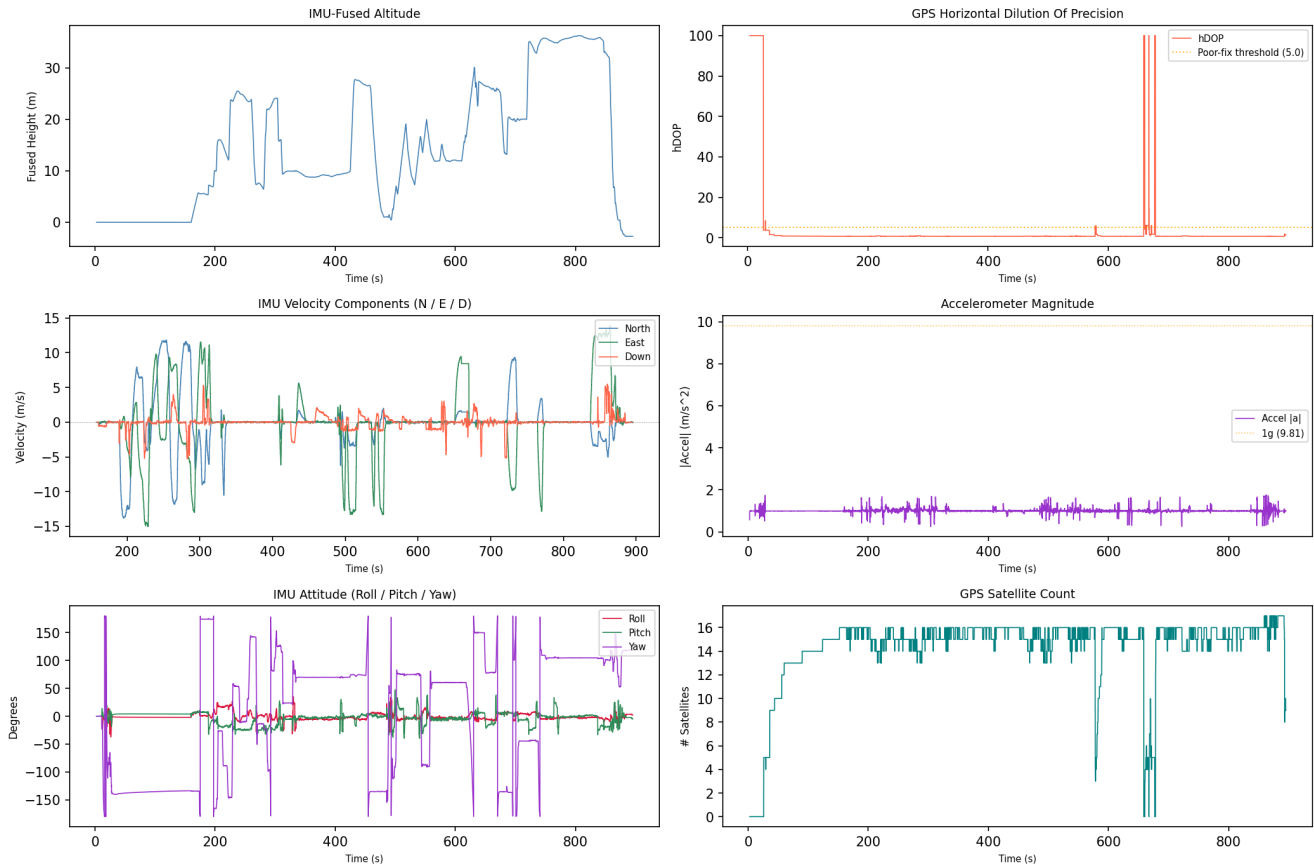
### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-20 20:17:31 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-20 20:19:41 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-20 20:30:30 | drone_flight_storage:dl | Reached peak height | 36.2 m  |
| 2018-06-20 20:31:48 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-20 20:31:59 | drone_flight_storage:dl | Log ended           |         |



## Drone Log - Sensor Telemetry

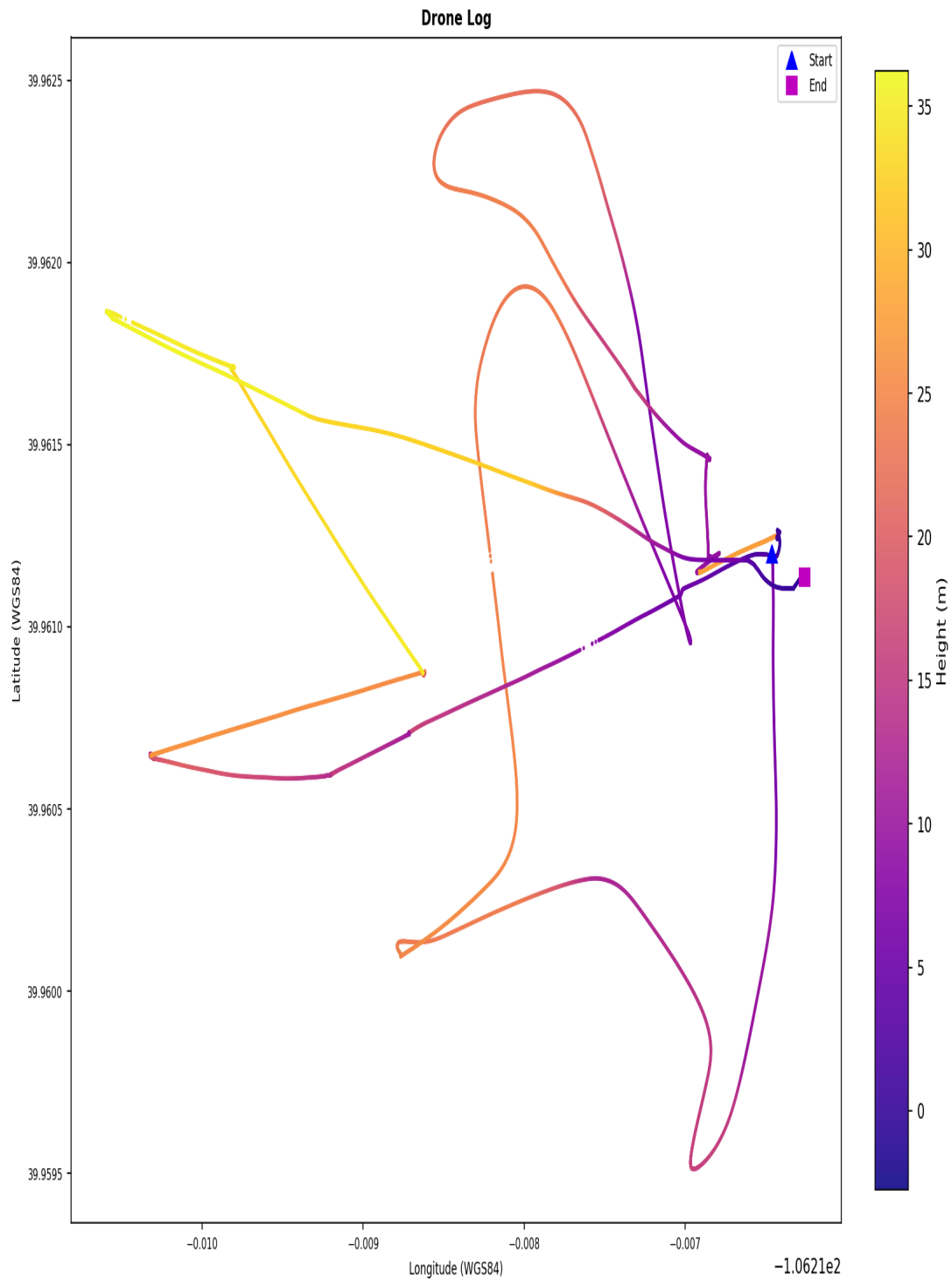
### Drone Log Telemetry - flight\_16



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_16



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY028.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY028.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 17

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 399974da349aa149f5d0bd2860c4afd58d22b61a2f1389281b46638acf7ee2ea  
**Drone Log Name:** FLY029.csv  
**Start:** 2018-06-20T22:43:55.954000+00:00  
**End:** 2018-06-20T22:58:28.848000+00:00  
**Duration:** 872.9 s

### Flight Metrics

**Peak Height:** 14.612969929858322 m  
**Distance Travelled:** 477.46596085850445 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

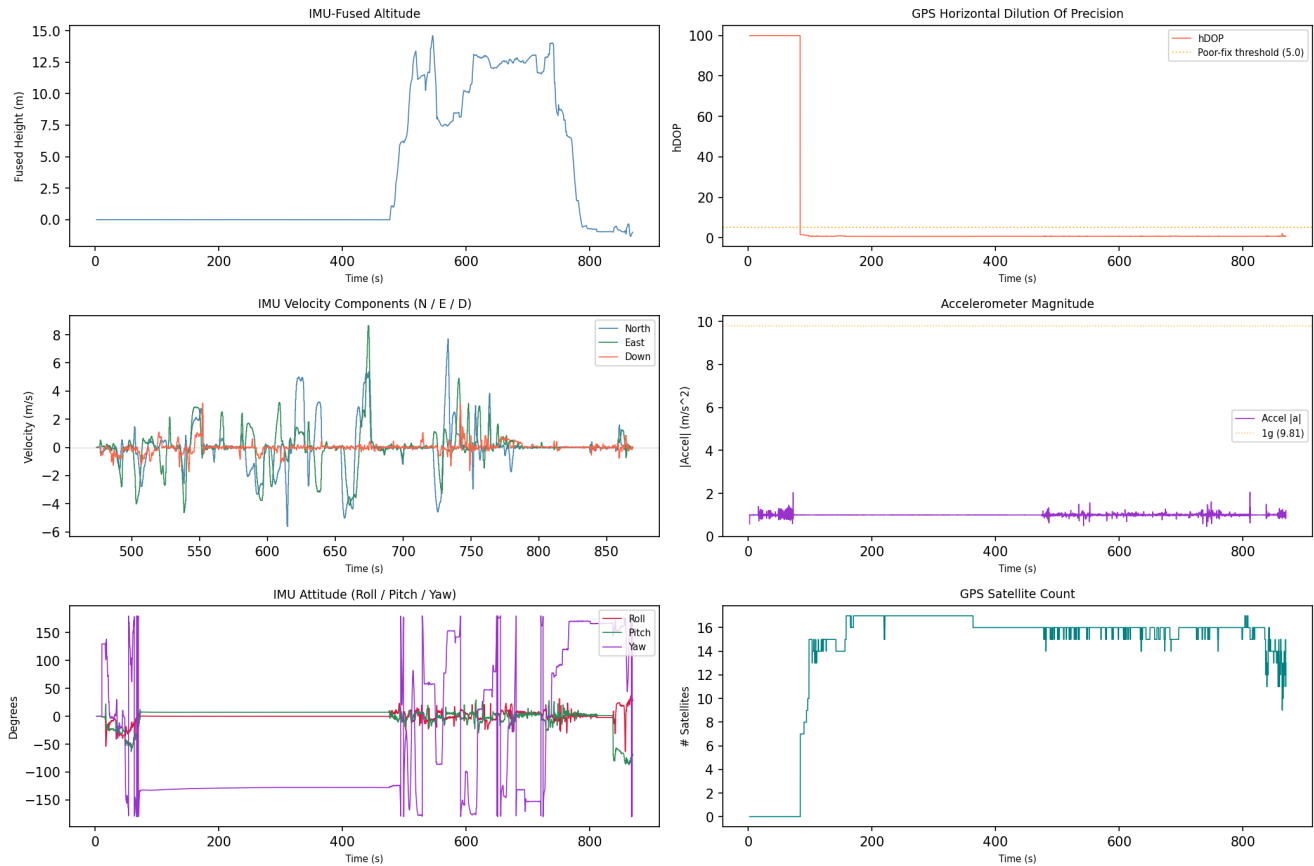
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AutoTakeoff
6. AutoLanding
7. FORCE\_LANDING

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-20 22:45:24 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-20 22:51:53 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-20 22:53:05 | drone_flight_storage:dl | Reached peak height | 14.6 m  |
| 2018-06-20 22:57:32 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-20 22:58:28 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

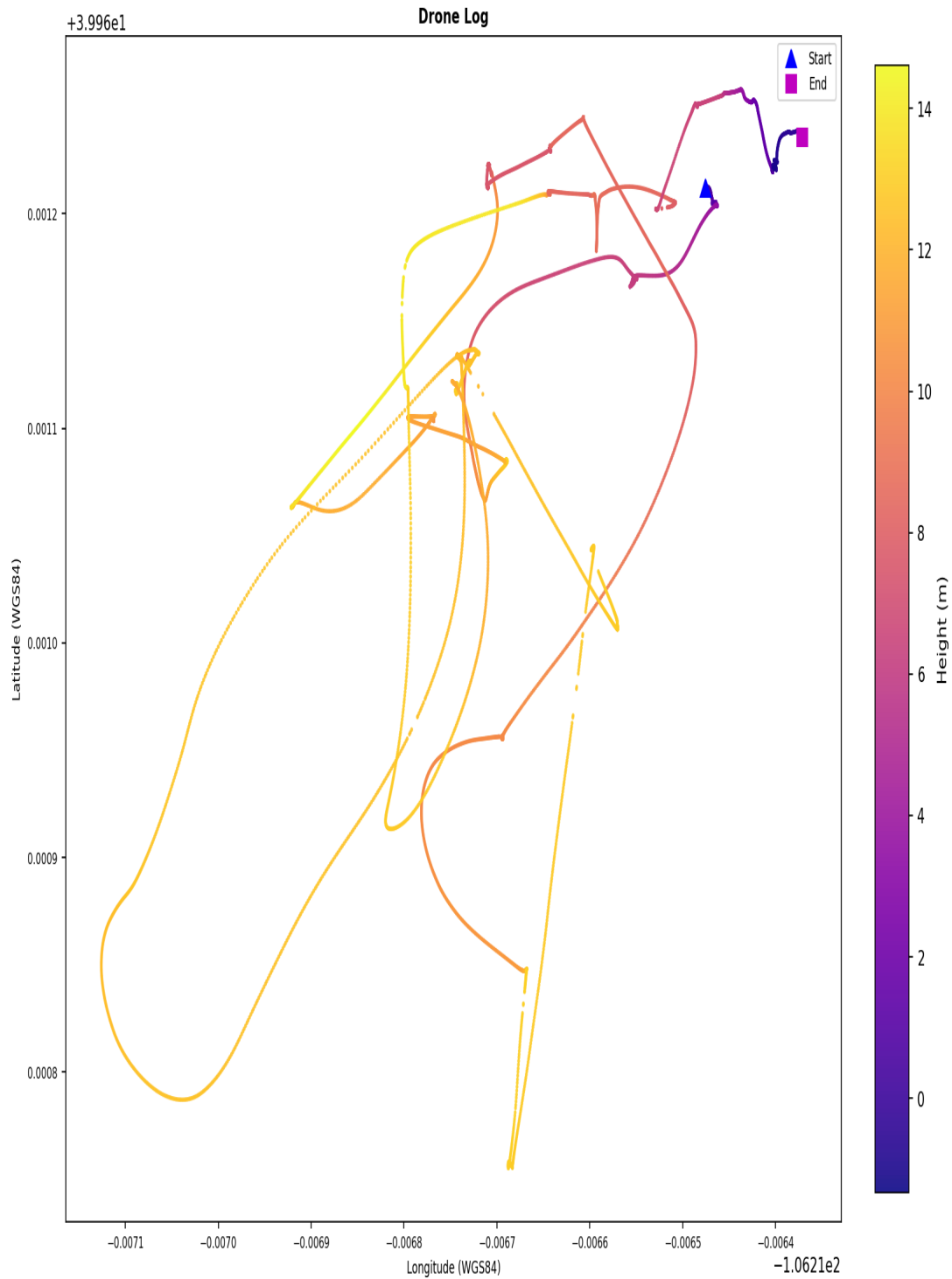
### Drone Log Telemetry - flight\_17



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_17



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY029.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY029.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 18

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 5db2d6ce42945d1d3a81de58789cff1386e32352f78a0b2eab0da289ed03ca3  
**Drone Log Name:** FLY030.csv  
**Start:** 2018-06-20T22:59:37.018000+00:00  
**End:** 2018-06-20T23:15:38.755000+00:00  
**Duration:** 961.7 s

### Flight Metrics

**Peak Height:** 12.83920897341924 m  
**Distance Travelled:** 2059.404824217 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

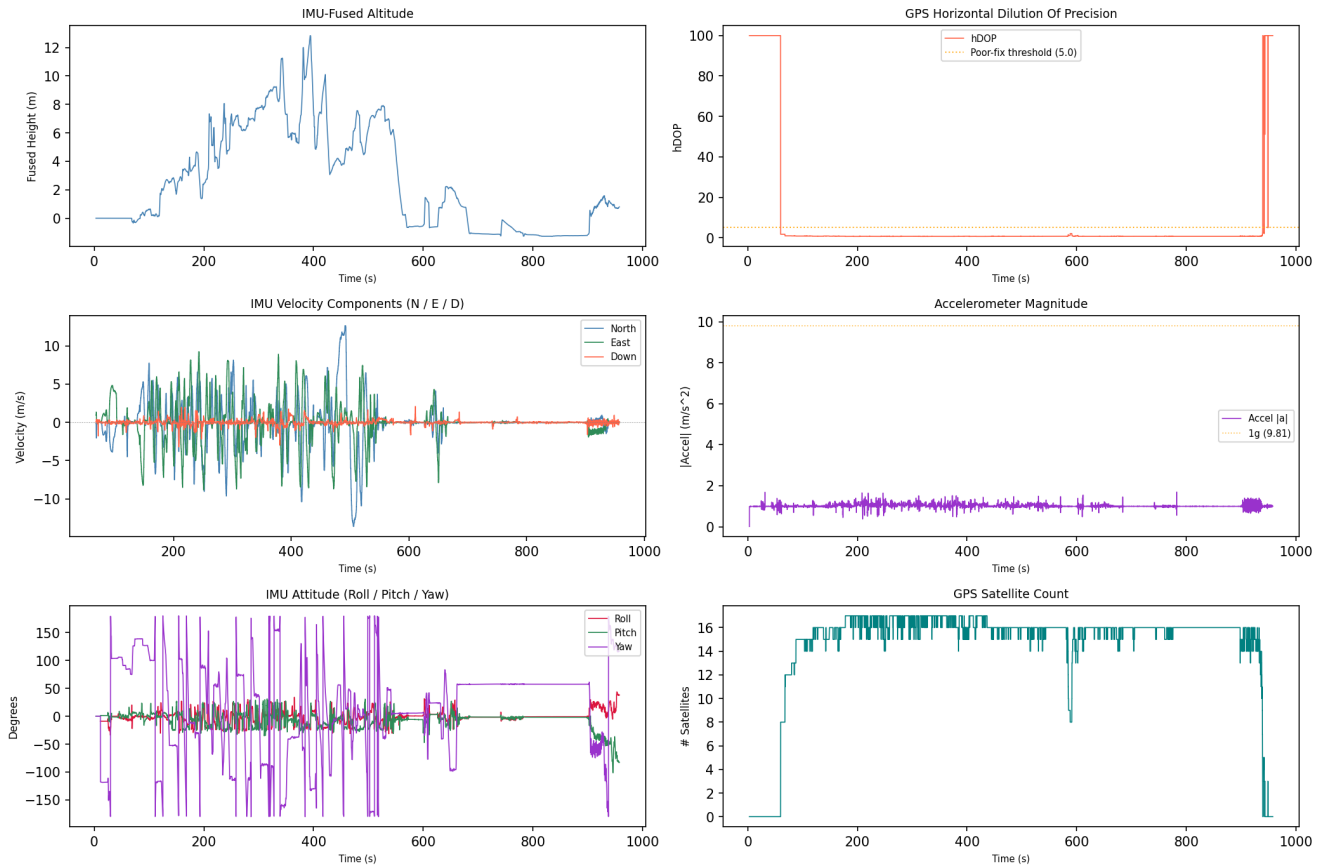
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AutoTakeoff
6. AutoLanding
7. FORCE\_LANDING

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| -                   | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-20 23:00:41 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-20 23:06:14 | drone_flight_storage:dl | Reached peak height | 12.8 m  |
| 2018-06-20 23:09:12 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-20 23:10:03 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-20 23:11:05 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-20 23:11:58 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-20 23:12:43 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-20 23:15:37 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

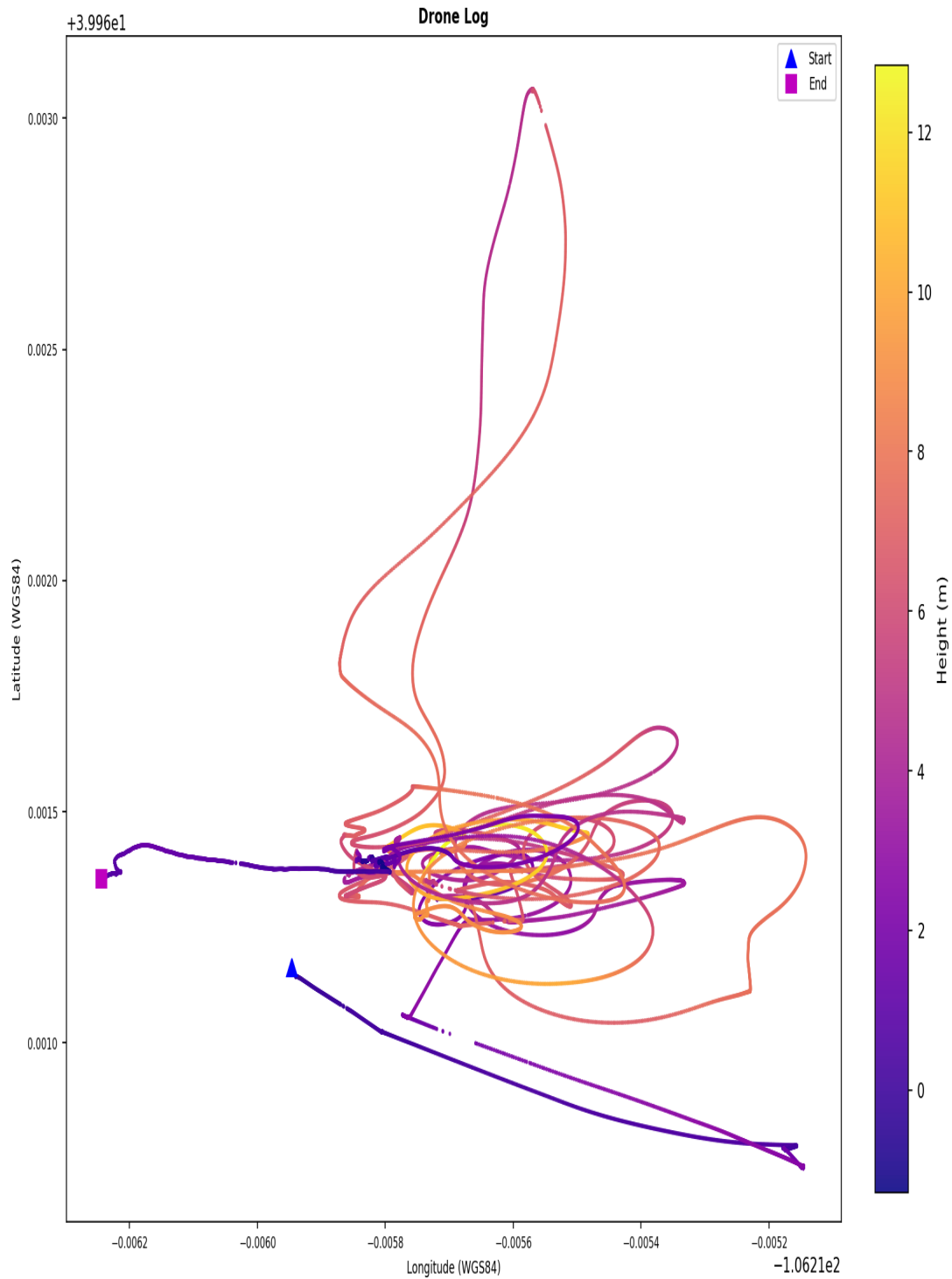
### Drone Log Telemetry - flight\_18



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_18



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY030.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY030.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.



## 4. Flight Analysis - Flight 19

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 81fb20e5664545368d149a50579362800c943bdc816d1c9478bbe65f1759f5b9  
**Drone Log Name:** FLY031.csv  
**Start:** 2018-06-20T23:55:40.483000+00:00  
**End:** 2018-06-21T00:04:30.655000+00:00  
**Duration:** 530.2 s

### Flight Metrics

**Peak Height:** 28.246418126023173 m  
**Distance Travelled:** 1596.0892646311215 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

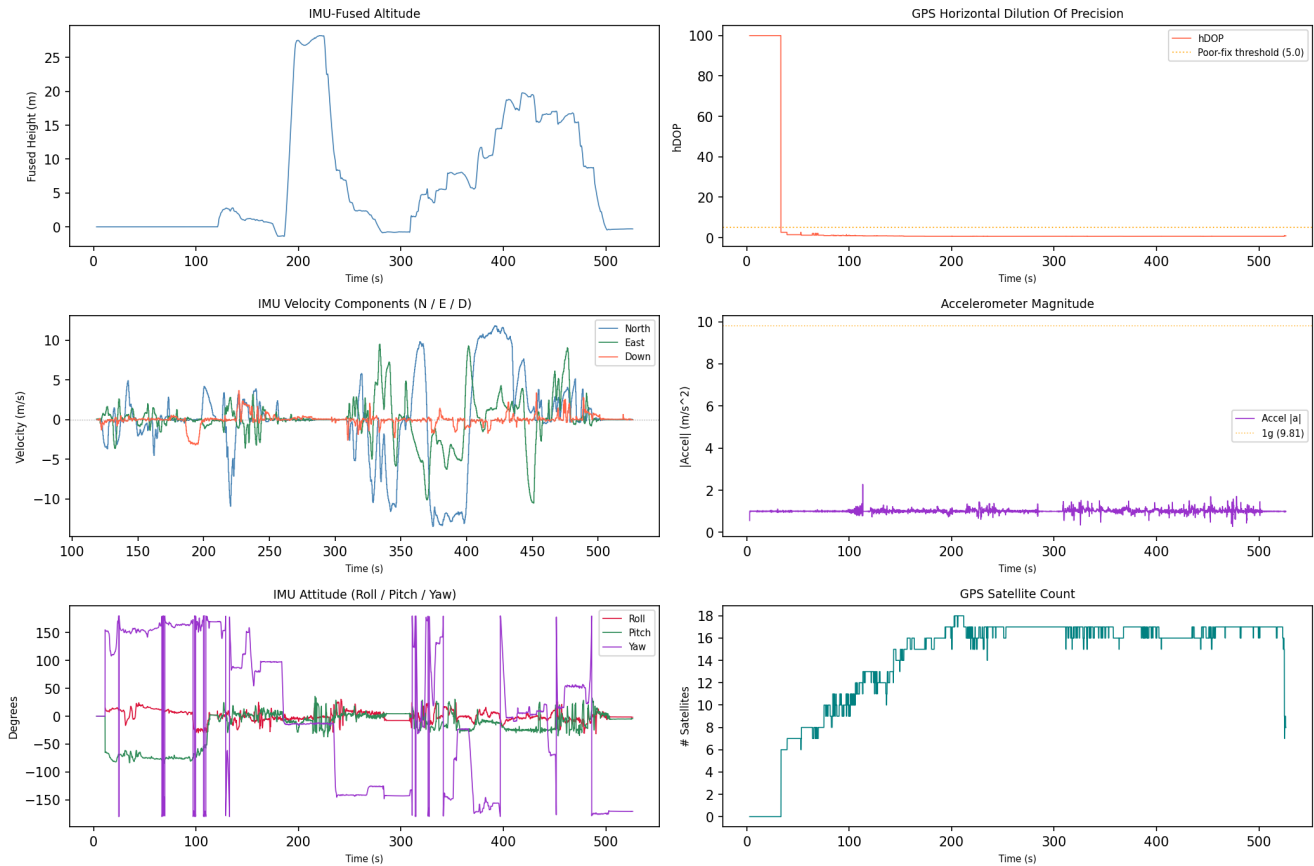
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AutoTakeoff
6. GoHome
7. AssitedTakeoff
8. AutoLanding

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-20 23:56:18 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-20 23:57:42 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-20 23:59:25 | drone_flight_storage:dl | Reached peak height | 28.2 m  |
| 2018-06-21 00:00:41 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 00:00:47 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 00:04:23 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 00:04:30 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

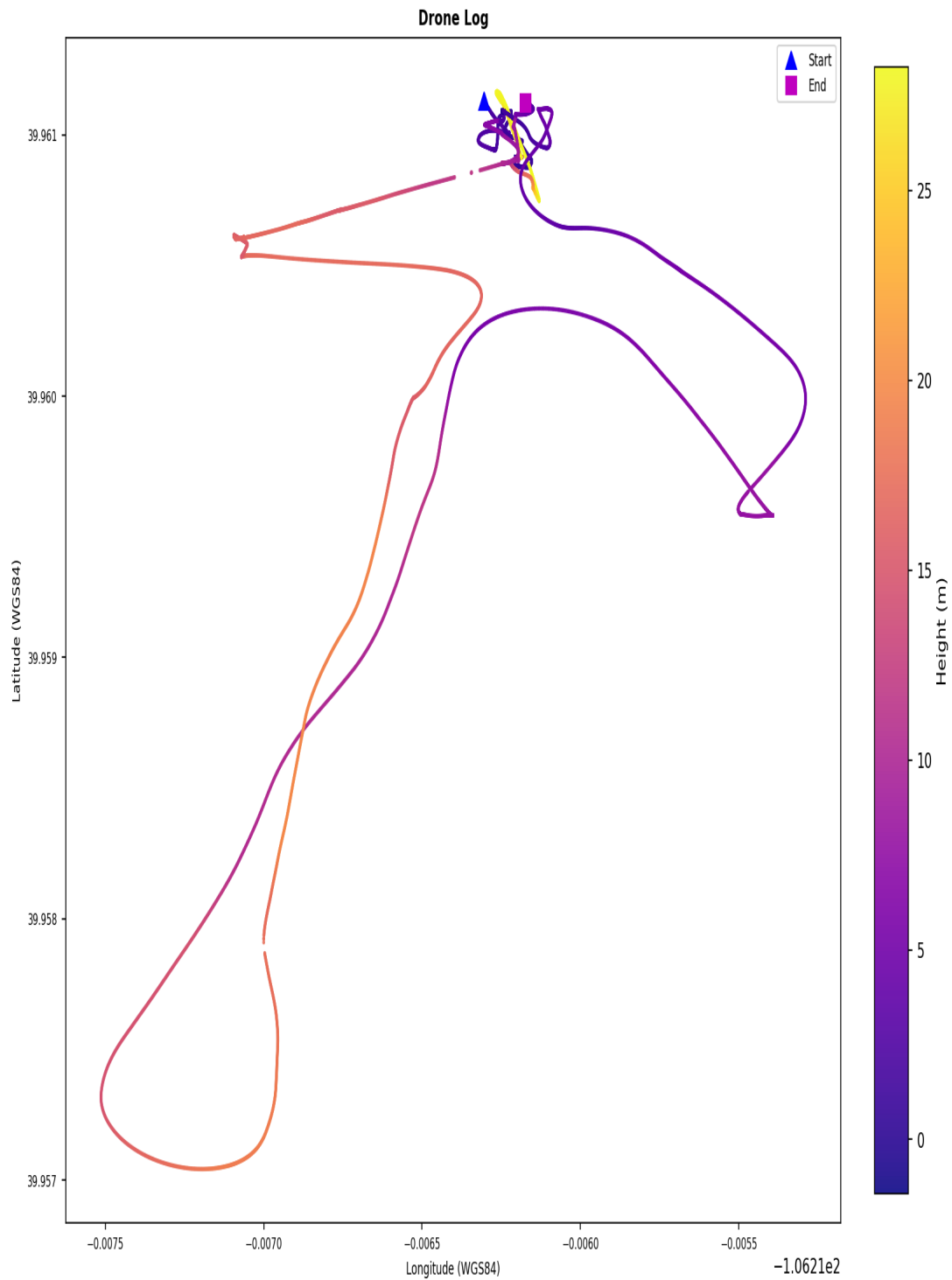
### Drone Log Telemetry - flight\_19



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_19



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY031.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 20

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** fc13804389738d932de721ea0e897c4c30f0185fadc89fdb0493990c897e660d  
**Drone Log Name:** FLY033.csv  
**Start:** 2018-06-21T15:18:02.111000+00:00  
**End:** 2018-06-21T15:35:06.869000+00:00  
**Duration:** 1024.8 s

### Flight Metrics

**Peak Height:** 24.990842999608596 m  
**Distance Travelled:** 344.48859988762723 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

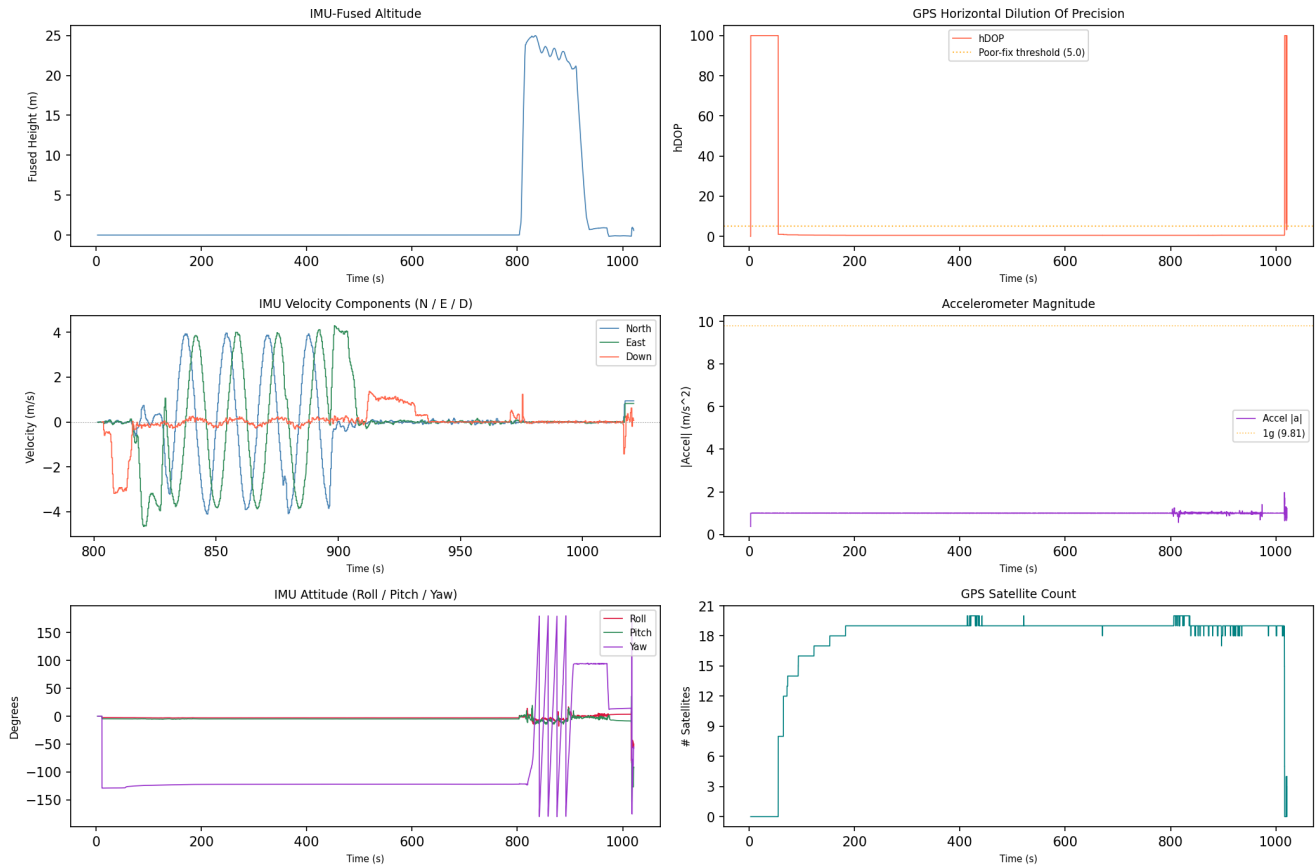
1. Atti
2. GPS\_Atti
3. Manual
4. ASST\_TAKEOFF
5. AutoTakeoff
6. NaviGo
7. AutoLanding
8. FORCE\_LANDING

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| 0002-11-30 00:00:00 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 15:19:01 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-21 15:31:26 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 15:31:59 | drone_flight_storage:dl | Reached peak height | 25.0 m  |
| 2018-06-21 15:34:20 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 15:35:05 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

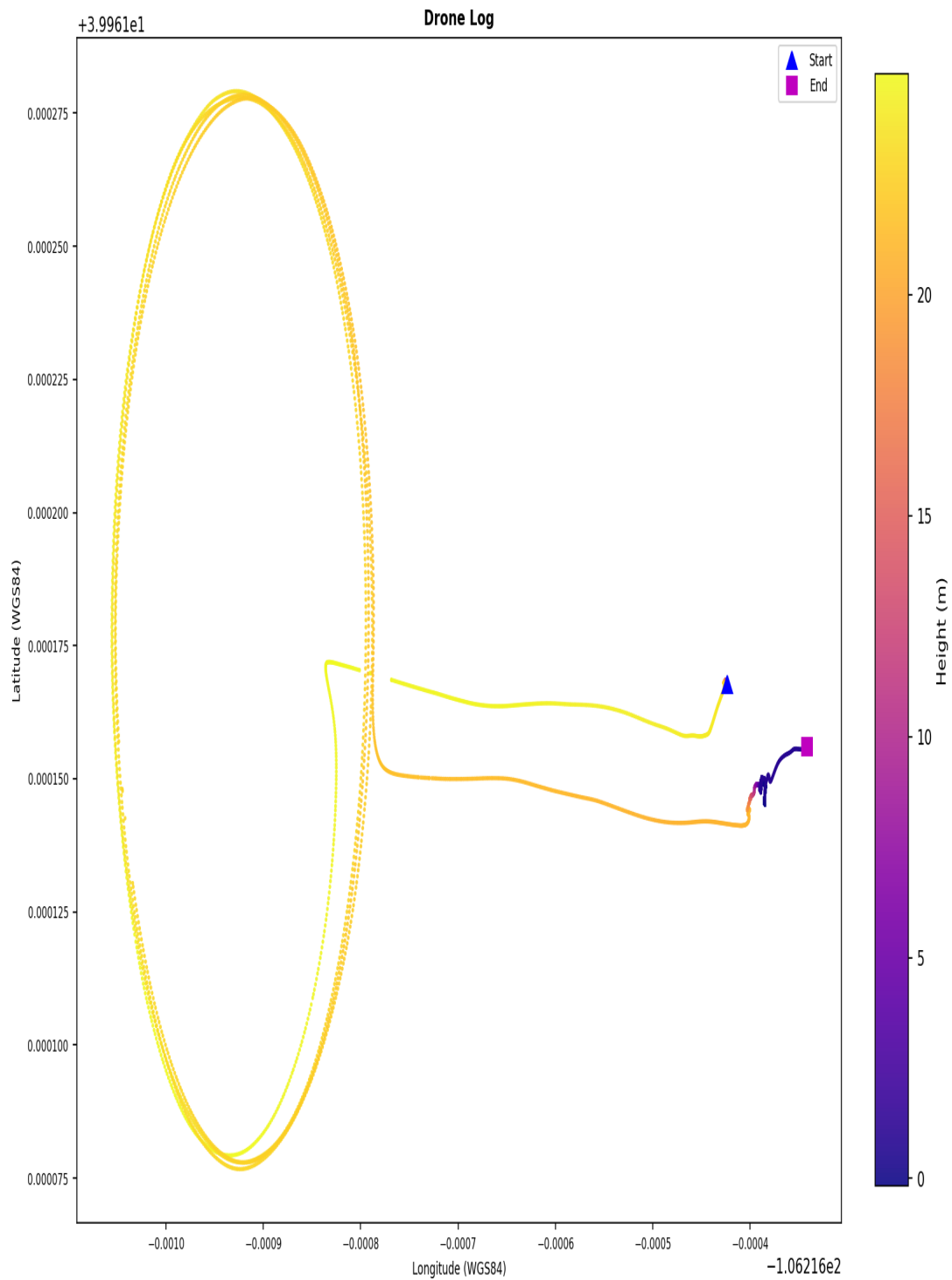
### Drone Log Telemetry - flight\_20



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_20



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY033.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY033.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 21

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 92ec5e5b4476048ef6acba5fedbf70f9e38c8d9c5ca0f993d237f195b6f3323e  
**Drone Log Name:** FLY035.csv  
**Start:** 2018-06-21T16:45:22.479000+00:00  
**End:** 2018-06-21T17:13:00.141000+00:00  
**Duration:** 1657.7 s

### Flight Metrics

**Peak Height:** 5.0009915162911565 m  
**Distance Travelled:** 211.40547797816185 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

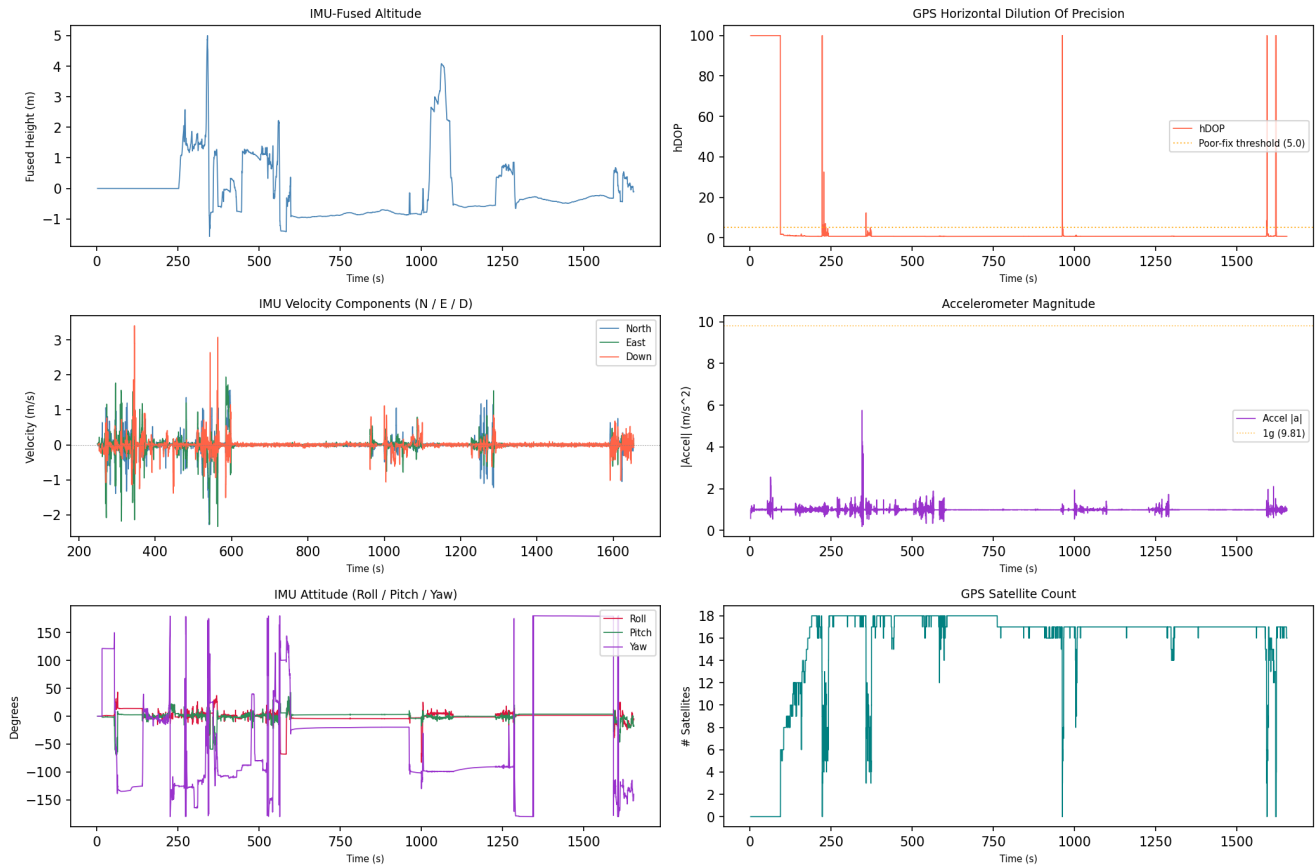
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AssitedTakeoff

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 16:47:01 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-21 16:49:34 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 16:51:06 | drone_flight_storage:dl | Reached peak height | 5.0 m   |
| 2018-06-21 16:51:16 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 16:51:42 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 16:52:38 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 16:52:49 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 16:54:52 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:01:45 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 17:02:05 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:02:17 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 17:03:44 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:05:45 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 17:06:56 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:12:59 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

### Drone Log Telemetry - flight\_21

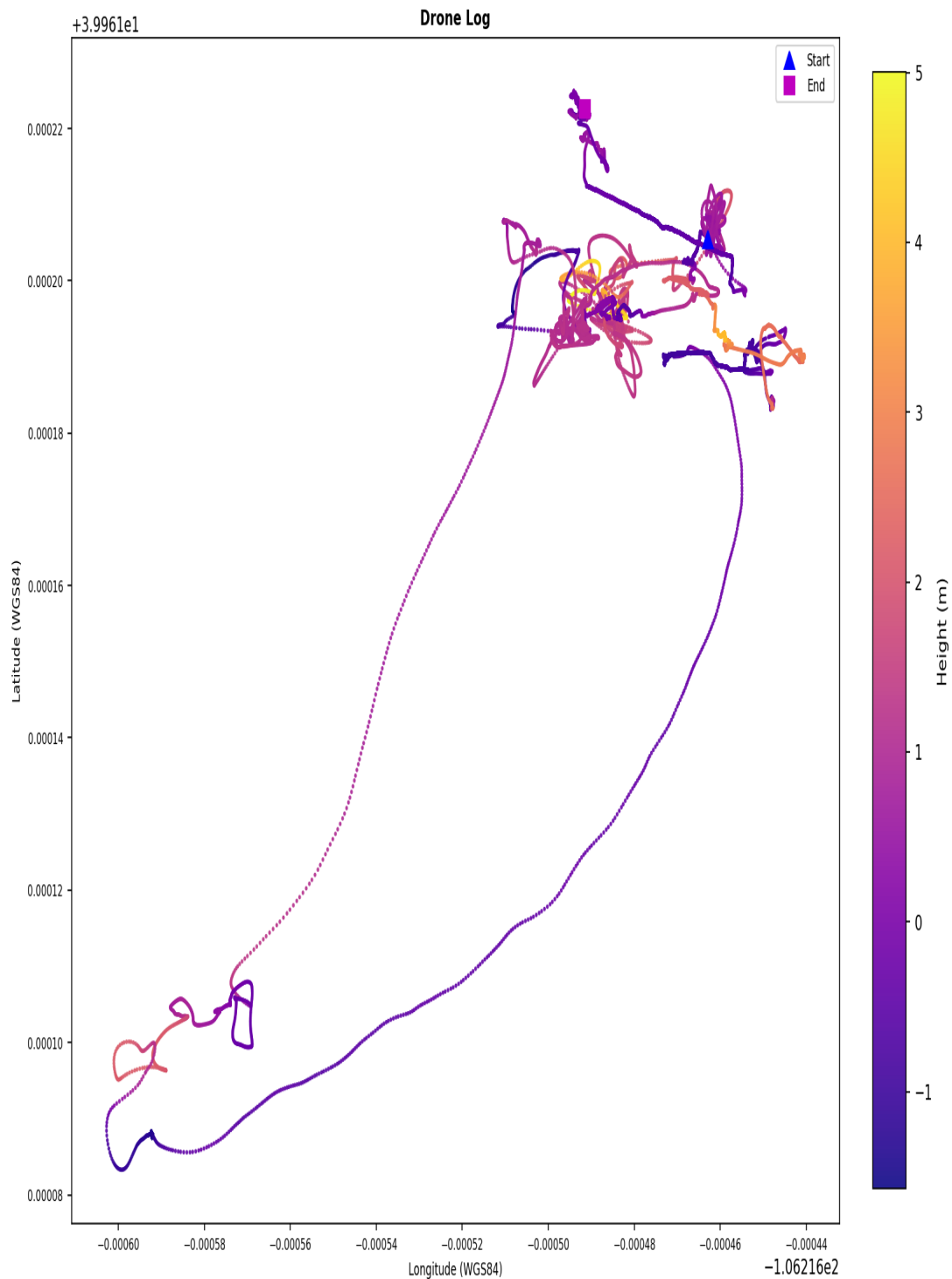


GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.



## Flight Ground Track

### Flight Ground Track - flight\_21



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY035.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY035.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 22

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** e0c86ecde7e6e6ab3dd85c8ddef5fb211af349213a43c9661700d06ba4a73b1e  
**Drone Log Name:** FLY036.csv  
**Start:** 2018-06-21T17:01:15.348000+00:00  
**End:** 2018-06-21T17:19:46.202000+00:00  
**Duration:** 1110.9 s

### Flight Metrics

**Peak Height:** 7.117498453557568 m  
**Distance Travelled:** 65.85851268761083 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

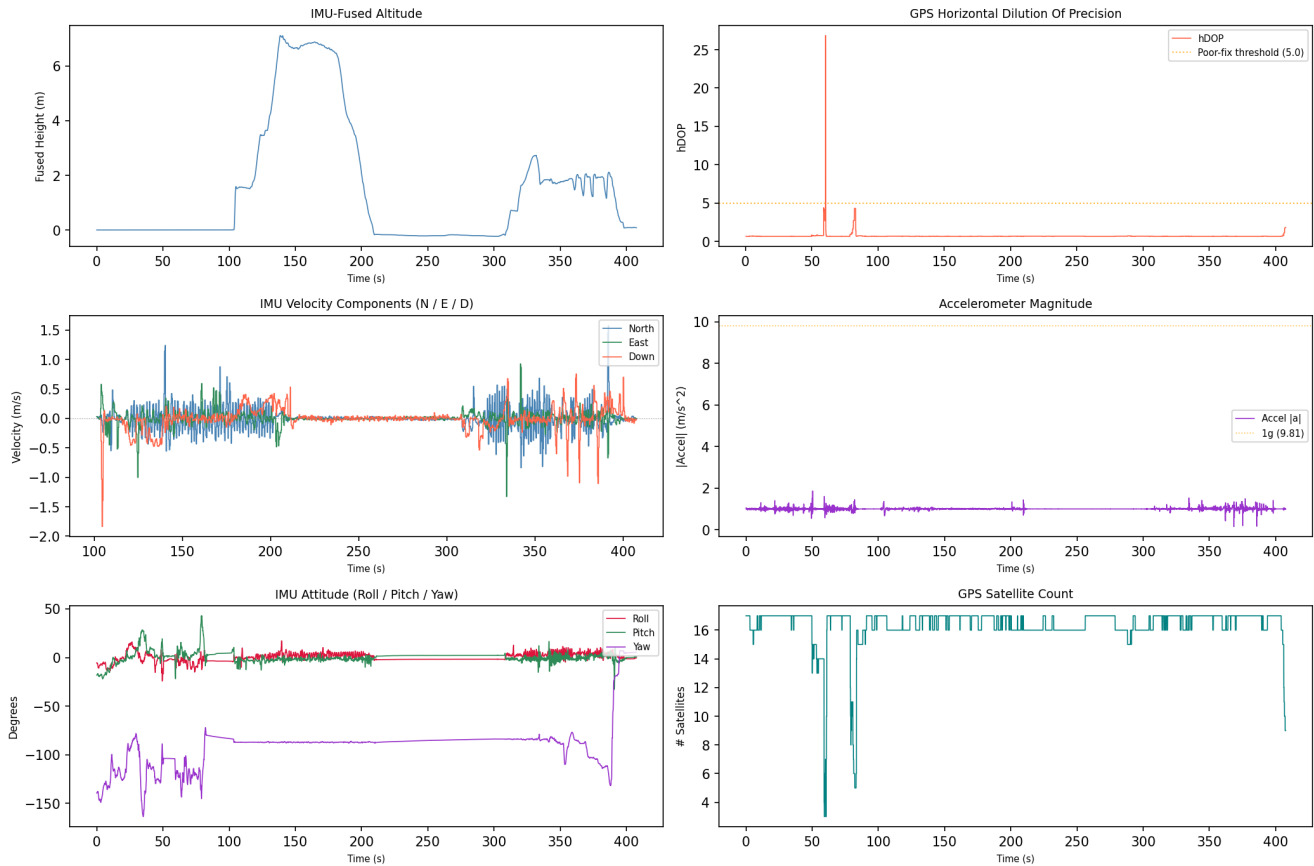
1. GPS\_Atti
2. ASST\_TAKEOFF
3. AssitedTakeoff
4. AutoLanding
5. FORCE\_LANDING

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:12:59 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-21 17:14:40 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 17:15:17 | drone_flight_storage:dl | Reached peak height | 7.1 m   |
| 2018-06-21 17:16:30 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:18:00 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 17:19:39 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:19:46 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

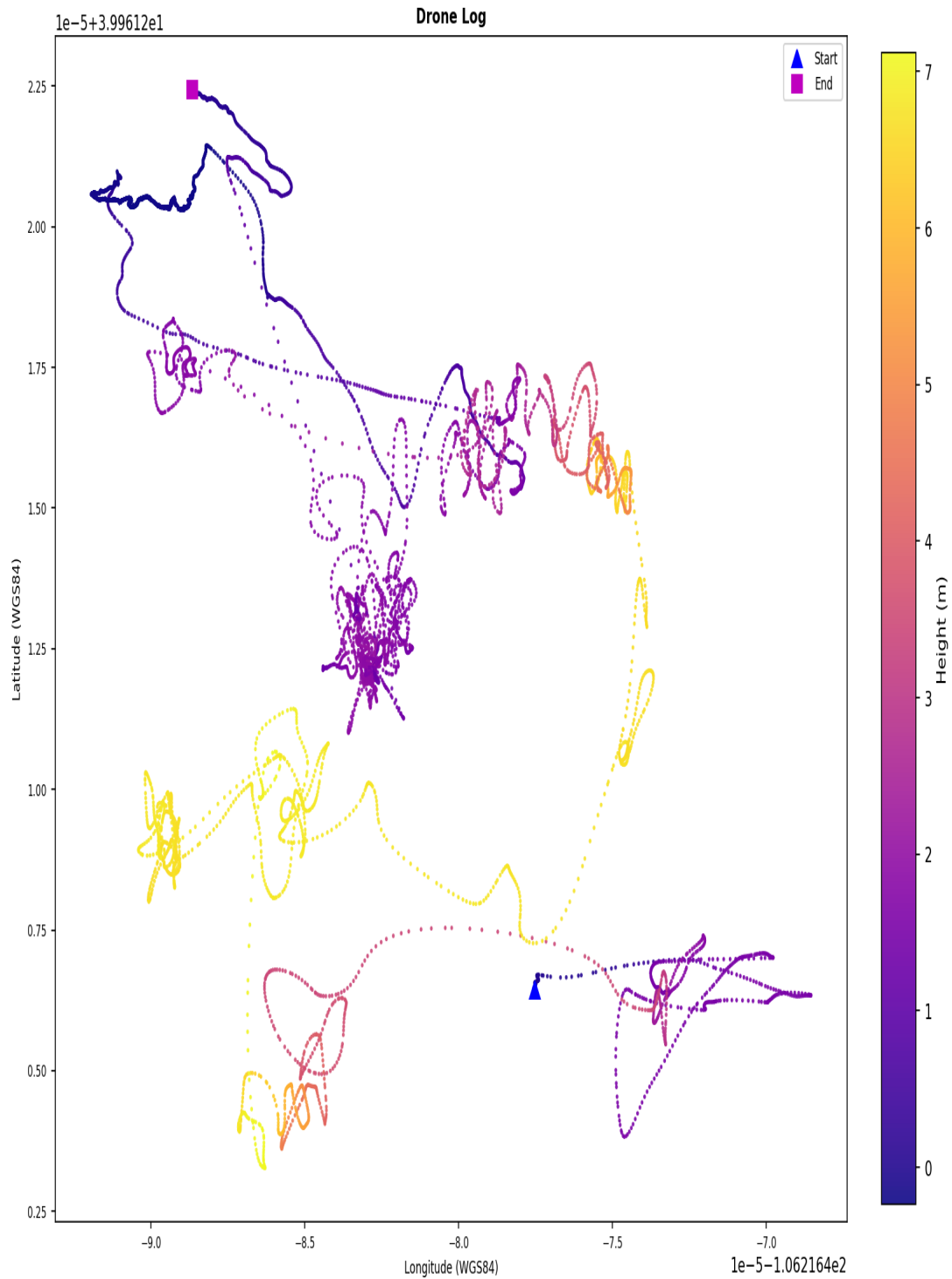
### Drone Log Telemetry - flight\_22



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_22



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY036.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 4. Flight Analysis - Flight 23

### Correlation and Flight Window

**Status:** Unmatched (no drone log corroboration)  
**Drone Log SHA-256:** 7125863c75057341cc39317b0d2c30cc95f613cb805a2bfff2507aa9fb5e41216  
**Drone Log Name:** FLY037.csv  
**Start:** 2018-06-21T17:20:35.762000+00:00  
**End:** 2018-06-21T17:36:54.614000+00:00  
**Duration:** 978.9 s

### Flight Metrics

**Peak Height:** 7.798884657897458 m  
**Distance Travelled:** 97.33082812182377 m  
**Video Recording:** No  
**Photo Capture:** No

### Fly Modes Observed (Chronological)

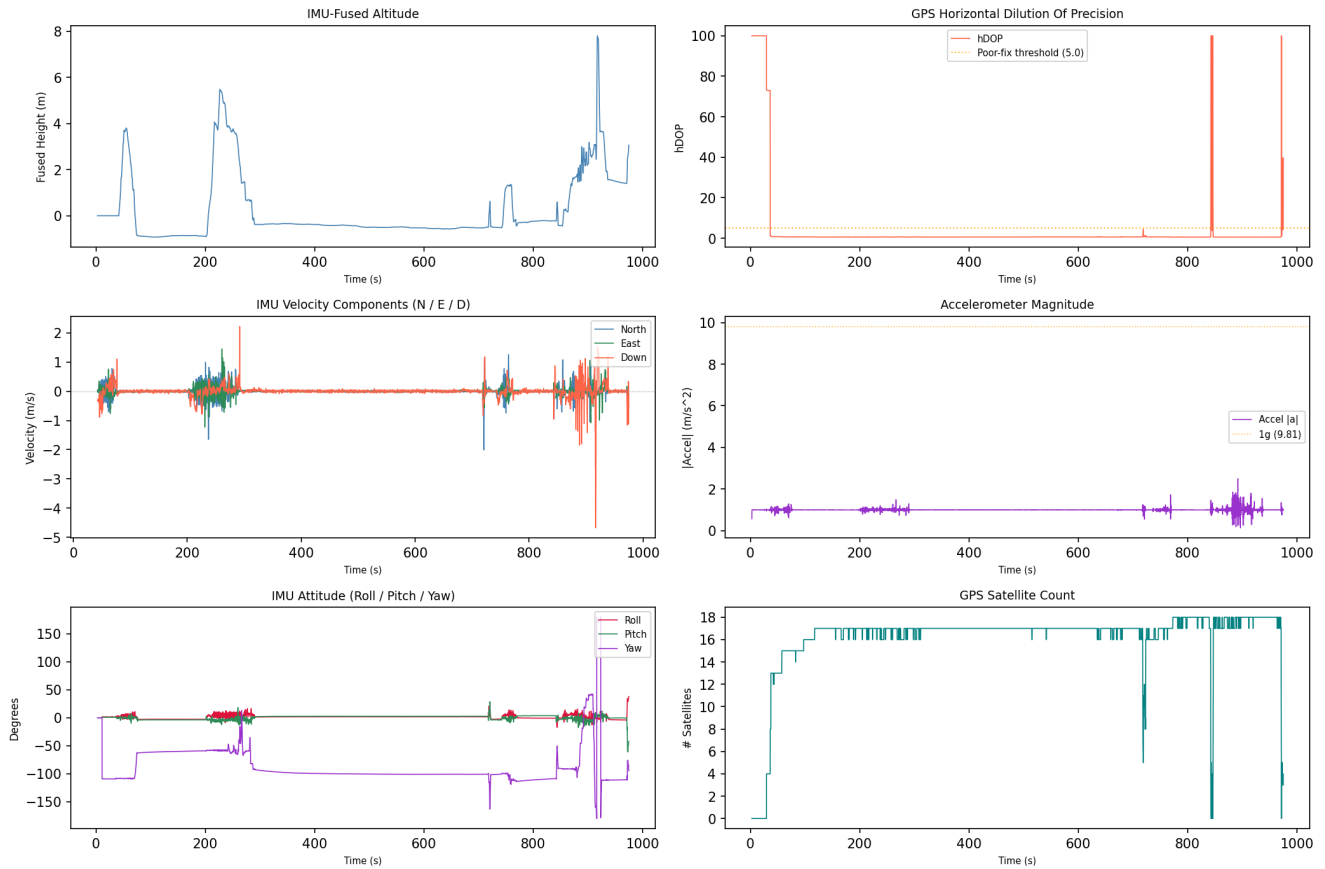
1. Manual
2. Atti
3. GPS\_Atti
4. ASST\_TAKEOFF
5. AssitedTakeoff

### Key Events

| Timestamp (UTC)     | Source                  | Event               | Details |
|---------------------|-------------------------|---------------------|---------|
| -                   | drone_flight_storage:dl | Motor turned off    | off     |
| -                   | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 17:21:16 | drone_flight_storage:dl | Log started         | P4P     |
| 2018-06-21 17:21:56 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:23:57 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 17:25:31 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:32:56 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 17:33:30 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:34:51 | drone_flight_storage:dl | Motor turned on     | on      |
| 2018-06-21 17:35:57 | drone_flight_storage:dl | Reached peak height | 7.8 m   |
| 2018-06-21 17:36:18 | drone_flight_storage:dl | Motor turned off    | off     |
| 2018-06-21 17:36:55 | drone_flight_storage:dl | Log ended           |         |

## Drone Log - Sensor Telemetry

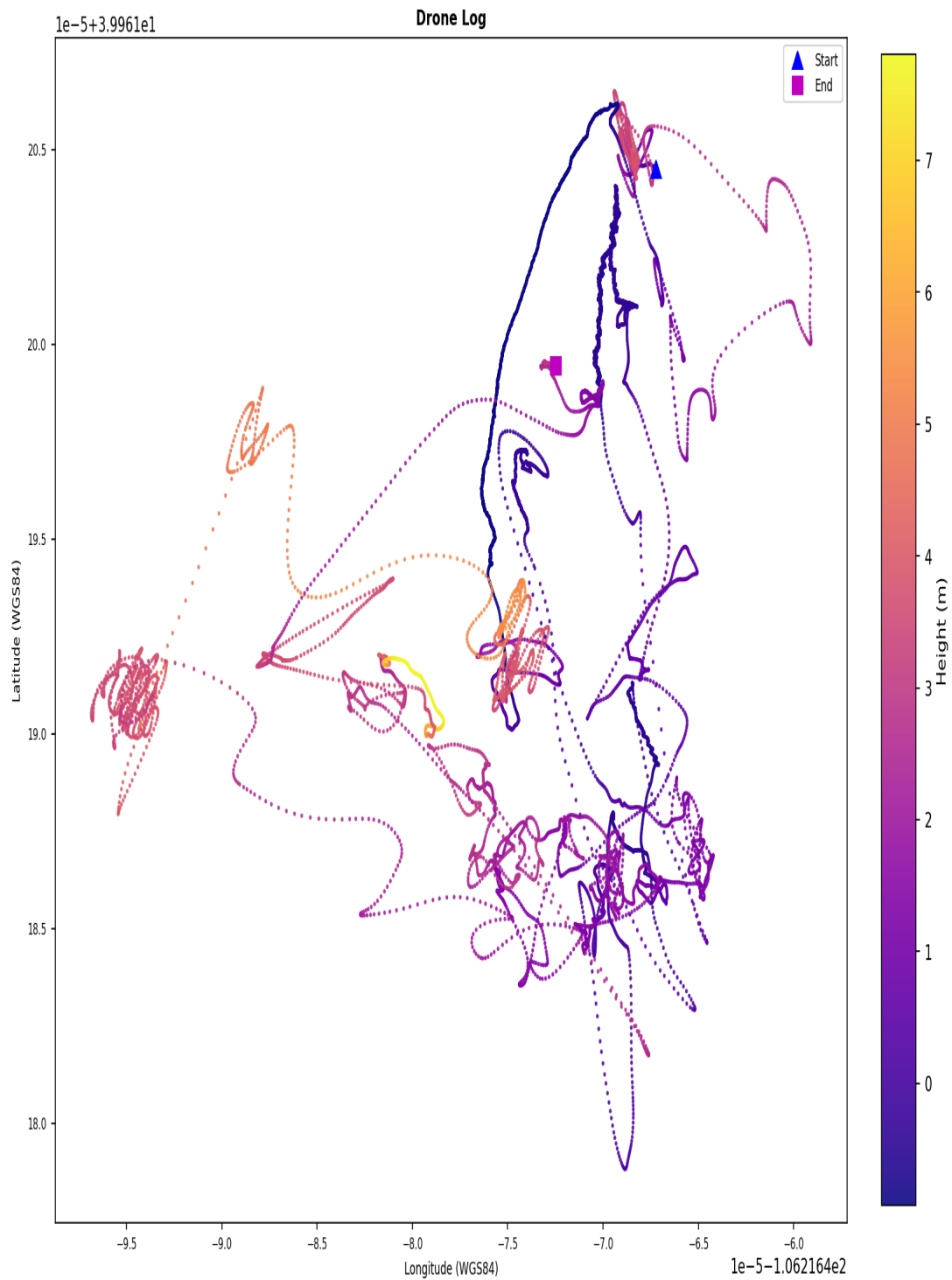
Drone Log Telemetry - flight\_23



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of  $\pm 1.5$  m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

## Flight Ground Track

### Flight Ground Track - flight\_23



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY037.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY037.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

## 5. Artefact Coverage

### Source Coverage

| Source               | Detected |
|----------------------|----------|
| controller_android   | No       |
| controller_ios       | Yes      |
| drone_sd             | No       |
| drone_flight_storage | Yes      |

### Artefact Counts per Source and Category

| Source               | Account Data | Databases | Drone Logs | Flight Logs | Flight Records | Images | Videos |
|----------------------|--------------|-----------|------------|-------------|----------------|--------|--------|
| controller_ios       | 2            | 15        | 1          | 5           | 1              | 0      | 0      |
| drone_flight_storage | 0            | 0         | 40         | 0           | 0              | 0      | 0      |

### Data Quality Notes per Category

#### Databases (15 files):

- 9 database(s) empty (controller\_ios)

#### Drone Logs (41 files):

- 41 file(s) with row anomalies
- 41 file(s) with empty columns
- 41 file(s) with constant-value columns

#### Flight Logs (5 files):

- formats: 4 bracketed\_log, 1 json\_array\_log

#### Flight Records (1 files):

- 1 file(s) with empty columns
- 1 file(s) with constant-value columns



## 6. Data Quality and Anomalies

### Data Quality Observations

- Drone\_logs: 41 file(s) with row anomalies detected in telemetry data.
- Drone\_logs: 41 file(s) with empty columns - some telemetry channels contain no data.
- Drone\_logs: 41 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight\_logs: log files present in the following format(s): 4 bracketed\_log, 1 json\_array\_log.
- Flight\_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight\_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

### Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - images]: no artefacts found
4. [p3 - controller ios - videos]: no artefacts found
5. [p3 - drone sd]: source evidence not found
6. [p4 - controller android]: source evidence not found
7. [p4 - drone sd]: source evidence not found
8. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY000.csv
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY001.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY002.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY003.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY014.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY015.csv
14. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY016.csv
15. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY017.csv
16. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY018.csv
17. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY019.csv
18. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY020.csv
19. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY021.csv
20. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY022.csv
21. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY023.csv
22. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY024.csv
23. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY034.csv
24. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY038.csv
25. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm\_FLY039.csv

### Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

#### Databases

5255DF85724869CD36D4AFCD12AE092D.db

- database\_empty: all tables empty

collectionList.db

- database\_empty: all tables empty

djiFMDB.db

- database\_empty: all tables empty

djiFMDB\_1.db

- database\_empty: all tables empty

dji\_fs\_upgrade\_local\_cache.db

- database\_empty: all tables empty

flysafe\_app\_dynamic\_areas.db

- database\_empty: all tables empty

**flysafe\_app\_dynamic\_areas\_1.db**

- database\_empty: all tables empty

**flysafe\_dji\_flight\_dynamic\_areas.db**

- database\_empty: all tables empty

**flysafe\_dji\_flight\_dynamic\_areas\_1.db**

- database\_empty: all tables empty

**Drone Logs****norm\_2018-06-19\_13-38-06\_FLY025.csv**

- timestamp\_gap: 1 row(s) affected
- missing\_gps: 8 row(s) affected
- motor\_airborne\_off: 1399 row(s) affected
- gps\_poor\_fix: 464 row(s) affected
- contains\_no\_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C
- contains\_constant\_value: 41 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, BatteryInfo:dyna\_cnt:D, BatteryInfo:f\_cap:D, BatteryInfo:out\_ctl:D (+36 more)
- gps\_accuracy\_poor: 460 row(s) affected

**norm\_FLY000.csv**

- timestamp\_gap: 1 row(s) affected
- missing\_gps: 4 row(s) affected
- zero\_coordinate: 819 row(s) affected
- gps\_poor\_fix: 819 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 123 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+118 more)
- gps\_accuracy\_poor: 819 row(s) affected
- attitude\_out\_of\_bounds: 2 row(s) affected

**norm\_FLY001.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 10598 row(s) affected
- gps\_poor\_fix: 10598 row(s) affected
- contains\_no\_value: 24 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+19 more)
- contains\_constant\_value: 101 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+96 more)
- gps\_accuracy\_poor: 10598 row(s) affected
- quaternion\_invalid: 227 row(s) affected
- attitude\_out\_of\_bounds: 118 row(s) affected

**norm\_FLY002.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 3644 row(s) affected
- gps\_poor\_fix: 6443 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 127 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+122 more)
- gps\_accuracy\_poor: 6443 row(s) affected
- quaternion\_invalid: 80 row(s) affected

**norm\_FLY003.csv**

- missing\_gps: 7 row(s) affected
- zero\_coordinate: 22922 row(s) affected
- gps\_poor\_fix: 22922 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 113 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+108 more)
- gps\_accuracy\_poor: 22922 row(s) affected
- quaternion\_invalid: 245 row(s) affected

**norm\_FLY004.csv**

- missing\_gps: 7 row(s) affected
- zero\_coordinate: 557 row(s) affected
- motor\_airborne\_off: 463 row(s) affected
- gps\_poor\_fix: 945 row(s) affected
- contains\_no\_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C
- contains\_constant\_value: 54 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryInfo:out\_ctl:D (+49 more)
- gps\_accuracy\_poor: 995 row(s) affected
- quaternion\_invalid: 249 row(s) affected

**norm\_FLY005.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 614 row(s) affected
- altitude\_negative: 706 row(s) affected
- gps\_poor\_fix: 734 row(s) affected
- contains\_no\_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, osd\_data:lowVoltage
- contains\_constant\_value: 58 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryInfo:out\_ctl:D (+53 more)
- gps\_accuracy\_poor: 767 row(s) affected
- quaternion\_invalid: 238 row(s) affected
- attitude\_out\_of\_bounds: 93 row(s) affected

**norm\_FLY006.csv**

- missing\_gps: 13 row(s) affected
- zero\_coordinate: 662 row(s) affected
- altitude\_negative: 1686 row(s) affected
- motor\_airborne\_off: 1219 row(s) affected
- gps\_poor\_fix: 1279 row(s) affected
- contains\_no\_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:dataLost (+1 more)
- contains\_constant\_value: 85 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+80 more)
- gps\_accuracy\_poor: 1214 row(s) affected
- quaternion\_invalid: 247 row(s) affected
- attitude\_out\_of\_bounds: 201 row(s) affected

**norm\_FLY007.csv**

- missing\_gps: 7 row(s) affected
- zero\_coordinate: 190 row(s) affected
- motor\_airborne\_off: 409 row(s) affected
- gps\_poor\_fix: 803 row(s) affected
- contains\_no\_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:dataLost (+1 more)
- contains\_constant\_value: 72 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+67 more)
- gps\_accuracy\_poor: 803 row(s) affected
- quaternion\_invalid: 225 row(s) affected

**norm\_FLY008.csv**

- missing\_gps: 6 row(s) affected
- zero\_coordinate: 704 row(s) affected
- motor\_airborne\_off: 6839 row(s) affected
- gps\_poor\_fix: 725 row(s) affected
- contains\_no\_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C
- contains\_constant\_value: 52 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, BatteryInfo:dyna\_cnt:D, BatteryInfo:out\_ctl:D, BatteryInfo:out\_ctl\_f:D (+47 more)
- gps\_accuracy\_poor: 761 row(s) affected
- quaternion\_invalid: 238 row(s) affected

**norm\_FLY009.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 500 row(s) affected
- gps\_poor\_fix: 500 row(s) affected
- contains\_no\_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, IMU\_ATT(0):relativeHeight:C (+1 more)
- contains\_constant\_value: 91 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+86 more)
- gps\_accuracy\_poor: 506 row(s) affected
- quaternion\_invalid: 226 row(s) affected

**norm\_FLY010.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 857 row(s) affected
- altitude\_negative: 37 row(s) affected
- motor\_airborne\_off: 130 row(s) affected
- gps\_poor\_fix: 1193 row(s) affected
- contains\_no\_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:dataLost (+1 more)
- contains\_constant\_value: 72 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+67 more)
- gps\_accuracy\_poor: 1165 row(s) affected
- quaternion\_invalid: 244 row(s) affected

**norm\_FLY011.csv**

- missing\_gps: 6 row(s) affected
- zero\_coordinate: 370 row(s) affected
- altitude\_negative: 2521 row(s) affected
- motor\_airborne\_off: 1002 row(s) affected
- gps\_poor\_fix: 852 row(s) affected
- contains\_no\_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:dataLost
- contains\_constant\_value: 47 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryInfo:right:D (+42 more)
- gps\_accuracy\_poor: 912 row(s) affected
- quaternion\_invalid: 242 row(s) affected

**norm\_FLY012.csv**

- missing\_gps: 7 row(s) affected
- zero\_coordinate: 710 row(s) affected
- altitude\_negative: 610 row(s) affected
- gps\_poor\_fix: 787 row(s) affected
- contains\_no\_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:dataLost
- contains\_constant\_value: 54 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryInfo:right:D (+49 more)
- gps\_accuracy\_poor: 776 row(s) affected
- quaternion\_invalid: 245 row(s) affected

**norm\_FLY013.csv**

- missing\_gps: 12 row(s) affected
- zero\_coordinate: 428 row(s) affected
- motor\_airborne\_off: 10822 row(s) affected
- gps\_poor\_fix: 748 row(s) affected
- contains\_no\_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C
- contains\_constant\_value: 52 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryInfo:right:D (+47 more)
- gps\_accuracy\_poor: 1221 row(s) affected
- quaternion\_invalid: 237 row(s) affected
- attitude\_out\_of\_bounds: 344 row(s) affected

**norm\_FLY014.csv**

- missing\_gps: 12 row(s) affected
- zero\_coordinate: 1340 row(s) affected
- gps\_poor\_fix: 1340 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 113 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+108 more)
- gps\_accuracy\_poor: 1340 row(s) affected
- quaternion\_invalid: 246 row(s) affected

**norm\_FLY015.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 5427 row(s) affected
- gps\_poor\_fix: 5427 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 115 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+110 more)
- gps\_accuracy\_poor: 5427 row(s) affected
- quaternion\_invalid: 254 row(s) affected

**norm\_FLY016.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 471 row(s) affected
- gps\_poor\_fix: 471 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 121 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+116 more)
- gps\_accuracy\_poor: 471 row(s) affected
- quaternion\_invalid: 258 row(s) affected

**norm\_FLY017.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 732 row(s) affected
- gps\_poor\_fix: 732 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 117 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+112 more)
- gps\_accuracy\_poor: 732 row(s) affected
- quaternion\_invalid: 258 row(s) affected

**norm\_FLY018.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 12495 row(s) affected
- gps\_poor\_fix: 12495 row(s) affected
- contains\_no\_value: 24 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+19 more)
- contains\_constant\_value: 108 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+103 more)
- gps\_accuracy\_poor: 12495 row(s) affected
- quaternion\_invalid: 258 row(s) affected
- attitude\_out\_of\_bounds: 128 row(s) affected

**norm\_FLY019.csv**

- missing\_gps: 351 row(s) affected
- contains\_no\_value: 23 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+18 more)
- contains\_constant\_value: 41 column(s) - IMUCalcs(0):PosD:C, IMUCalcs(0):PosE:C, IMUCalcs(0):PosN:C, IMUCalcs(0):PositionError:C, IMUCalcs(0):VelocityError:C (+36 more)
- quaternion\_invalid: 31 row(s) affected

**norm\_FLY020.csv**

- missing\_gps: 4 row(s) affected

- zero\_coordinate: 6352 row(s) affected
- gps\_poor\_fix: 6352 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 138 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+133 more)
- gps\_accuracy\_poor: 6352 row(s) affected
- quaternion\_invalid: 79 row(s) affected

**norm\_FLY021.csv**

- timestamp\_gap: 2 row(s) affected
- missing\_gps: 11 row(s) affected
- zero\_coordinate: 30443 row(s) affected
- gps\_poor\_fix: 30443 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 118 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+113 more)
- gps\_accuracy\_poor: 30443 row(s) affected
- quaternion\_invalid: 226 row(s) affected

**norm\_FLY022.csv**

- timestamp\_gap: 1 row(s) affected
- missing\_gps: 9 row(s) affected
- zero\_coordinate: 94 row(s) affected
- gps\_poor\_fix: 94 row(s) affected
- contains\_no\_value: 30 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+25 more)
- contains\_constant\_value: 111 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+106 more)
- gps\_accuracy\_poor: 94 row(s) affected
- quaternion\_invalid: 95 row(s) affected
- attitude\_out\_of\_bounds: 2 row(s) affected

**norm\_FLY023.csv**

- missing\_gps: 7 row(s) affected
- zero\_coordinate: 1444 row(s) affected
- gps\_poor\_fix: 1444 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 120 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+115 more)
- gps\_accuracy\_poor: 1444 row(s) affected
- quaternion\_invalid: 243 row(s) affected

**norm\_FLY024.csv**

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 1102 row(s) affected
- gps\_poor\_fix: 1102 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 120 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+115 more)
- gps\_accuracy\_poor: 1102 row(s) affected
- quaternion\_invalid: 219 row(s) affected
- attitude\_out\_of\_bounds: 74 row(s) affected

**norm\_FLY025.csv**

- missing\_gps: 8 row(s) affected
- zero\_coordinate: 738 row(s) affected
- gps\_poor\_fix: 822 row(s) affected
- contains\_no\_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:dataLost (+1 more)
- contains\_constant\_value: 92 column(s) - ATTI\_MINIO:s\_rsv00, ATTI\_MINIO:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+87 more)
- gps\_accuracy\_poor: 822 row(s) affected

- quaternion\_invalid: 217 row(s) affected

#### norm\_FLY026.csv

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 682 row(s) affected
- motor\_airborne\_off: 2083 row(s) affected
- gps\_poor\_fix: 4879 row(s) affected
- contains\_no\_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, osd\_data:lowVoltage
- contains\_constant\_value: 59 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, BatteryInfo:dyna\_cnt:D, BatteryInfo:out\_ctl:D, BatteryInfo:out\_ctl\_f:D (+54 more)
- gps\_accuracy\_poor: 4867 row(s) affected
- quaternion\_invalid: 220 row(s) affected

#### norm\_FLY027.csv

- missing\_gps: 7 row(s) affected
- zero\_coordinate: 1365 row(s) affected
- gps\_poor\_fix: 1403 row(s) affected
- contains\_no\_value: 7 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, IMU\_ATT(0):relativeHeight:C (+2 more)
- contains\_constant\_value: 98 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+93 more)
- gps\_accuracy\_poor: 1403 row(s) affected
- quaternion\_invalid: 696 row(s) affected

#### norm\_FLY028.csv

- missing\_gps: 16 row(s) affected
- zero\_coordinate: 653 row(s) affected
- altitude\_negative: 620 row(s) affected
- gps\_poor\_fix: 785 row(s) affected
- contains\_no\_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, osd\_data:lowVoltage
- contains\_constant\_value: 59 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, BatteryInfo:dyna\_cnt:D, BatteryInfo:out\_ctl:D, BatteryInfo:out\_ctl\_f:D (+54 more)
- gps\_accuracy\_poor: 982 row(s) affected
- quaternion\_invalid: 222 row(s) affected

#### norm\_FLY029.csv

- missing\_gps: 10 row(s) affected
- zero\_coordinate: 1707 row(s) affected
- altitude\_negative: 2460 row(s) affected
- gps\_poor\_fix: 2387 row(s) affected
- contains\_no\_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C
- contains\_constant\_value: 55 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryStatus:capLevel2:D (+50 more)
- gps\_accuracy\_poor: 2387 row(s) affected
- quaternion\_invalid: 230 row(s) affected

#### norm\_FLY030.csv

- missing\_gps: 7 row(s) affected
- zero\_coordinate: 458 row(s) affected
- altitude\_negative: 8433 row(s) affected
- altitude\_spike: 1 row(s) affected
- motor\_airborne\_off: 850 row(s) affected
- gps\_poor\_fix: 2110 row(s) affected
- contains\_no\_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, osd\_data:lowVoltage
- contains\_constant\_value: 59 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, BatteryInfo:dyna\_cnt:D, BatteryInfo:out\_ctl:D, BatteryInfo:out\_ctl\_f:D (+54 more)
- gps\_accuracy\_poor: 2088 row(s) affected
- quaternion\_invalid: 218 row(s) affected

- attitude\_out\_of\_bounds: 33 row(s) affected

#### norm\_FLY031.csv

- missing\_gps: 8 row(s) affected
- zero\_coordinate: 702 row(s) affected
- altitude\_negative: 2018 row(s) affected
- gps\_poor\_fix: 883 row(s) affected
- contains\_no\_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:dataLost
- contains\_constant\_value: 54 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, BatteryInfo:dyna\_cnt:D, BatteryInfo:right:D, BatteryStatus:auth\_fail (+49 more)
- gps\_accuracy\_poor: 883 row(s) affected
- quaternion\_invalid: 223 row(s) affected

#### norm\_FLY032.csv

- missing\_gps: 6 row(s) affected
- zero\_coordinate: 449 row(s) affected
- gps\_poor\_fix: 449 row(s) affected
- contains\_no\_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, IMU\_ATT(0):relativeHeight:C (+1 more)
- contains\_constant\_value: 97 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+92 more)
- gps\_accuracy\_poor: 449 row(s) affected
- quaternion\_invalid: 221 row(s) affected

#### norm\_FLY033.csv

- missing\_gps: 6 row(s) affected
- zero\_coordinate: 1525 row(s) affected
- altitude\_negative: 744 row(s) affected
- gps\_poor\_fix: 1633 row(s) affected
- contains\_no\_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, osd\_data:lowVoltage
- contains\_constant\_value: 59 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryInfo:out\_ctl:D (+54 more)
- gps\_accuracy\_poor: 1643 row(s) affected
- quaternion\_invalid: 224 row(s) affected
- attitude\_out\_of\_bounds: 117 row(s) affected

#### norm\_FLY034.csv

- missing\_gps: 9 row(s) affected
- zero\_coordinate: 1749 row(s) affected
- gps\_poor\_fix: 10701 row(s) affected
- contains\_no\_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains\_constant\_value: 108 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+103 more)
- gps\_accuracy\_poor: 10701 row(s) affected
- quaternion\_invalid: 227 row(s) affected

#### norm\_FLY035.csv

- missing\_gps: 11 row(s) affected
- zero\_coordinate: 2678 row(s) affected
- altitude\_negative: 28614 row(s) affected
- motor\_airborne\_off: 283 row(s) affected
- gps\_poor\_fix: 2818 row(s) affected
- contains\_no\_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:dataLost
- contains\_constant\_value: 59 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryInfo:right:D (+54 more)
- gps\_accuracy\_poor: 2842 row(s) affected
- quaternion\_invalid: 384 row(s) affected



**norm\_FLY036.csv**

- timestamp\_gap: 1 row(s) affected
- missing\_gps: 6 row(s) affected
- altitude\_negative: 2906 row(s) affected
- gps\_poor\_fix: 19 row(s) affected
- contains\_no\_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:appLost (+1 more)
- contains\_constant\_value: 68 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryInfo:f\_cap:D (+63 more)
- gps\_accuracy\_poor: 7 row(s) affected

**norm\_FLY037.csv**

- missing\_gps: 7 row(s) affected
- zero\_coordinate: 761 row(s) affected
- altitude\_negative: 20062 row(s) affected
- motor\_airborne\_off: 1097 row(s) affected
- gps\_poor\_fix: 869 row(s) affected
- contains\_no\_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU\_ATT(0):absoluteHeight:C, IMU\_ATT(0):distanceHP:C, RC\_Info:dataLost (+1 more)
- contains\_constant\_value: 60 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:vel\_level:D, BatteryInfo:dyna\_cnt:D, BatteryInfo:out\_ctl:D (+55 more)
- gps\_accuracy\_poor: 1131 row(s) affected
- quaternion\_invalid: 218 row(s) affected

**norm\_FLY038.csv**

- missing\_gps: 9 row(s) affected
- zero\_coordinate: 3907 row(s) affected
- gps\_poor\_fix: 3907 row(s) affected
- contains\_no\_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains\_constant\_value: 117 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+112 more)
- gps\_accuracy\_poor: 3907 row(s) affected
- quaternion\_invalid: 232 row(s) affected

**norm\_FLY039.csv**

- missing\_gps: 9 row(s) affected
- zero\_coordinate: 16091 row(s) affected
- gps\_poor\_fix: 16091 row(s) affected
- contains\_no\_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains\_constant\_value: 116 column(s) - ATTI\_MINI0:s\_rsv00, ATTI\_MINI0:s\_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air\_vbx:D, AirSpeed:air\_vby:D (+111 more)
- gps\_accuracy\_poor: 16091 row(s) affected
- quaternion\_invalid: 220 row(s) affected

**Flight Logs****2018-06-19\_13\_39\_46-07JDE5300200GN.dat**

- format: json\_array\_log | entries: 6

**upgradeLog\_0.txt**

- format: bracketed\_log | entries: 10

**upgradeLog\_0\_1.txt**

- format: bracketed\_log | entries: 11

**upgradeLog\_1.txt**

- format: bracketed\_log | entries: 4

**upgradeLog\_1\_1.txt**

- format: bracketed\_log | entries: 4

**Flight Records****norm\_DJIFlightRecord\_2018-06-19\_13-39-43\_.csv**

- contains\_no\_value: 27 column(s) - APP\_GPS.accuracy, APP\_GPS.latitude, APP\_GPS.longitude, APP\_SER\_WARN.warn, CENTER\_BATTERY.isBatteryOnCharge (+22 more)
- contains\_constant\_value: 170 column(s) - APP\_TIP.tip, CAMERA\_INFO.cameraType, CAMERA\_INFO.connectState, CAMERA\_INFO.encryptStatus, CAMERA\_INFO.firmErrorType (+165 more)

## 7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Drone Model inconsistency detected - conflicting values observed: P4 Professional, P4P. Manual verification required.
2. DJI Application Version inconsistency detected - conflicting values observed: 3.1.38, 4.2.20. Manual verification required.
3. Product Type inconsistency detected - conflicting values observed: 24, 255. Manual verification required.
4. Note: no firmware version found - Device and account information.
5. Last App Launch inconsistency detected - conflicting values observed: 2018-06-19T18:57:41Z, 2018-06-19T19:35:26Z. Manual verification required.
6. 22 flight record(s) could not be correlated to a drone log (flight\_01, flight\_02, flight\_03, flight\_04, flight\_05, flight\_06, flight\_07, flight\_08, flight\_09, flight\_10, flight\_11, flight\_12, flight\_13, flight\_15, flight\_16, flight\_17, flight\_18, flight\_19, flight\_20, flight\_21, flight\_22, flight\_23). Independent corroboration is absent for these flight(s).
7. Photo(s) were recorded during the flight. Copies may reside on both the controller and the drone SD card - both sources should be examined.
8. 6 database file(s) contain records and should be examined for relevant artefacts.

## Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

| com.dji.go.plist | Account Data                                                     | controller_ios | extract_logical |
|------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:         | 19619241cd0749ff506a7fe3abb28f542c304cd7834eec0d5eefbb7f27f484ba |                |                 |
| SHA-1:           | 7b8a5e7b38ca51c4d6a469e52092ddc6c5e76dba                         |                |                 |

| com.dji.pilot.plist | Account Data                                                     | controller_ios | extract_logical |
|---------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:            | 756d292792b4e41c748d5c136e23eb848388d4c6788ae8333ec6a42ff7eac64a |                |                 |
| SHA-1:              | 9ecb8037650e47b8a926b8e26ea2367f64d09833                         |                |                 |

| collectionList.db | Databases                                                        | controller_ios | extract_logical |
|-------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:          | 52ff80590e7a6e418b7aaeede8b04a62f661d77566de2e15827308fd1099cdb5 |                |                 |
| SHA-1:            | 6ed495c00b14c89a44297e47ca17df469c6bcc2b                         |                |                 |

| 5255DF85724869CD36D4AFCD12AE092D.db | Databases                                                        | controller_ios | extract_logical |
|-------------------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                            | aae50532bee18355508de941e178ea847cd725018bb03a93589944f7cad4898e |                |                 |
| SHA-1:                              | 7ecb7ed8a4b0ecf3028259a0eb0dcee998fcbc0c                         |                |                 |

| 538103E3AB6E8F987EC10913E64A3E24.db | Databases                                                        | controller_ios | extract_logical |
|-------------------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                            | 6df74165eb44a8570bdc89f0038e0831de6ba509aa2aa2bd15aa1a6a2786cea0 |                |                 |
| SHA-1:                              | f2d9e8167faa28bbb2685b330a7fabe2732943aa                         |                |                 |

| djiFMDB.db | Databases                                                        | controller_ios | extract_logical |
|------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:   | 5fdae73b94d5e7a078a8bfd1afda762eae9a5bff869139e9242dfecc749fd52a |                |                 |
| SHA-1:     | b1d200f2a2416e3c776ebc4e1779312198adf52b                         |                |                 |

| dji_fs_upgrade_local_cache.db | Databases                                                        | controller_ios | extract_logical |
|-------------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                      | 160afa85388b924b8dccb457db44c630a72905aff9e3d6ebcdf2eab6ded1ab49 |                |                 |
| SHA-1:                        | c9a66614704ea8d5b877afc7ddad8f99b4a2d056                         |                |                 |

| flysafe_app_dynamic_areas.db | Databases                                                        | controller_ios | extract_logical |
|------------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                     | 483da04ac8232310746927e831150673403cb79e12dbedf322869a1317fa7fcf |                |                 |
| SHA-1:                       | 4e8e284810efa83aa0861217939fff02c9d7ad12                         |                |                 |

| flysafe_areas_djigo_en.db | Databases                                                       | controller_ios | extract_logical |
|---------------------------|-----------------------------------------------------------------|----------------|-----------------|
| SHA-256:                  | 1a1c024072fba37bfbf1198f4227fcf2f4ffad347ad9a93b164a7077cd0b6f8 |                |                 |
| SHA-1:                    | 761bad6393f8f8a28ecea57069626b9a9eaa6de4                        |                |                 |

| flysafe_dji_flight_dynamic_areas.db | Databases                                                        | controller_ios | extract_logical |
|-------------------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                            | 6d434e165f689c505c6d051cc3ba0aa7dd26737023ff1aee562fd88f106a8953 |                |                 |
| SHA-1:                              | 80d3466fd6360f9edab821a05cfbf97ac0ce6804                         |                |                 |

| flysafe_license_unlock.db | Databases                                                        | controller_ios | extract_logical |
|---------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                  | 115ed6d632bdfbf27aad4f215055971b109fedc49cc4f35a6582fcb83fb6a91e |                |                 |
| SHA-1:                    | b738513795ac730bcb9fa3f713d4c1f45f1d7433                         |                |                 |

| flysafe_static_circle.db | Databases                                                        | controller_ios | extract_logical |
|--------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                 | 4787fc859c865dab6b3d0db8eb68a55c764ca5d13105b14429cc2d455708cfb4 |                |                 |
| SHA-1:                   | ccf2d5995e798abe1f1255c252c28ec8ae699dd4                         |                |                 |

| djiFMDB_1.db | Databases                                                       | controller_ios | extract_logical |
|--------------|-----------------------------------------------------------------|----------------|-----------------|
| SHA-256:     | 5fd4e73b94d5e7a078a8bfd1afda762eae9a5bff869139e9242dfec749fd52a |                |                 |
| SHA-1:       | b1d200f2a2416e3c776ebc4e1779312198adf52b                        |                |                 |

| flysafe_app_dynamic_areas_1.db | Databases                                                        | controller_ios | extract_logical |
|--------------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                       | ce02ae43e9061d526429d75f4aab106e9dc0133a90ac005df05e984ccf52e6fe |                |                 |
| SHA-1:                         | 5c25a94e52815ddf092eccb33456a159cc66f577                         |                |                 |

| flysafe_areas_djigo.db | Databases                                                        | controller_ios | extract_logical |
|------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:               | b5a6d15e6e04e9cbcc0902a392540a2b25d2a9722c6219d5cf34dfad90f838c8 |                |                 |
| SHA-1:                 | a2836e85087f06b331fbfea45025b05cf35f5942                         |                |                 |

| flysafe_dji_flight_dynamic_areas_1.db | Databases                                                       | controller_ios | extract_logical |
|---------------------------------------|-----------------------------------------------------------------|----------------|-----------------|
| SHA-256:                              | 56ef148210fcef9bca82dc86b201fbd340f32c9a247b9341ba85c54a21918c6 |                |                 |
| SHA-1:                                | 91649bae84edd1c230392932c2a5ad0e917b0f7c                        |                |                 |

| flysafe_static_circle_1.db | Databases                                                        | controller_ios | extract_logical |
|----------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                   | 4787fc859c865dab6b3d0db8eb68a55c764ca5d13105b14429cc2d455708cfb4 |                |                 |
| SHA-1:                     | ccf2d5995e798abe1f1255c252c28ec8ae699dd4                         |                |                 |

| 2018-06-19_13-38-06_FLY025.DAT | Drone Logs                                                       | controller_ios | extract_logical |
|--------------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                       | af7d8dbc8d9704865f58e1992f6a3281b8a7f1ac3abe99888748aad7a45429c7 |                |                 |
| SHA-1:                         | a0250e45fe4cd135f442a3984c150aba59f8fa68                         |                |                 |

| upgradeLog_0.txt | Flight Logs                                                      | controller_ios | extract_logical |
|------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:         | aa6b86f4a985918ed8f1db2ce4580059d64d1ec6785d49c29f2d44174deb9b85 |                |                 |
| SHA-1:           | 12c5c2b461de7f0956a475888064ab735adc245a                         |                |                 |

| upgradeLog_1.txt | Flight Logs                                                      | controller_ios | extract_logical |
|------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:         | 3c18245a5d67ba6cdc0ee896d196b67023a84ee68cc69add93d64d928232b402 |                |                 |
| SHA-1:           | b2ec714c31d55cb52b26694766928da9f59ea673                         |                |                 |

| 2018-06-19_13_39_46-07JDE5300200GN.dat | Flight Logs                                                     | controller_ios | extract_logical |
|----------------------------------------|-----------------------------------------------------------------|----------------|-----------------|
| SHA-256:                               | 9d3f351a10b80e9bb9fe724f7e0e91c07750719efa302e3743d9fdb2e6de33c |                |                 |
| SHA-1:                                 | 837c9b9d1cd952a9a9f9c19255ffc0b17c8c764a                        |                |                 |

| upgradeLog_0_1.txt | Flight Logs                                                      | controller_ios | extract_logical |
|--------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:           | c386b93daca3761522ef1e40eca44b3d86071737274b6c75381cdaf7f1077d84 |                |                 |
| SHA-1:             | 3892e404514e8ff649866a68b04f0815baf6fcf4                         |                |                 |

| upgradeLog_1_1.txt | Flight Logs                                                      | controller_ios | extract_logical |
|--------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:           | 9ecb4d3e10731143b8e66dce5c4026af364de93c48970da4d8cf5aeaa5fe07c5 |                |                 |
| SHA-1:             | 0ffe3130597fd0df8027b38b4b25e46af18b4f19                         |                |                 |

| DJIFlightRecord_2018-06-19_13-39-43_.txt | Flight Records                                                   | controller_ios | extract_logical |
|------------------------------------------|------------------------------------------------------------------|----------------|-----------------|
| SHA-256:                                 | b056e5524596a35c58a9edcbf1285f2042d3e92c734c41f7c495c1967a94a87a |                |                 |
| SHA-1:                                   | 5286a3e791e7230e7a76217e1b18bb8db16b80d2                         |                |                 |

| FLY000.DAT | Drone Logs                                                       | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|----------------------|-----------------|
| SHA-256:   | 86b80b703059f7c5bdf9d979ff4f826439ca01b127ec17653b11c56b0372176a |                      |                 |
| SHA-1:     | e2fd5aee6463552b0f72fca406f8d5072e2525c7                         |                      |                 |

| FLY001.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 2024320114e0f071397184ba83b1cf554d183fe9f76a9a9d9e52703bafac1653 |            |                      |                 |
| SHA-1:     | 40eccd821d64bce22f63f173607111b23c6be242                         |            |                      |                 |

| FLY002.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 49cd52dacc62885015578f4ae17a6d79a0968a49ad40b865778cc3683c121ade |            |                      |                 |
| SHA-1:     | 0130ad211e7764372c1d3917906436df2ae90d85                         |            |                      |                 |

| FLY003.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 97d09a055b2ef085a863b9abd179c5379de6472d65cdfec4ac8694c690dd7b8c |            |                      |                 |
| SHA-1:     | d50cbee00e267a85fe3f9341315d025d1ca73241                         |            |                      |                 |

| FLY004.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | a323d7c3c49c3bb4b88c99ff9a8d7aa52db135f2a74628b342f84a3999677643 |            |                      |                 |
| SHA-1:     | 53ad38848bfd6430434b400e63324e83738b6285                         |            |                      |                 |

| FLY005.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | ae190556d83ac550ad76cc8dfb33bbe72c7d202dbd2f69912c3484aca5f41e60 |            |                      |                 |
| SHA-1:     | e8b283fdb98a770accb5f18445354a61e55399f5                         |            |                      |                 |

| FLY006.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | f38541476f394f992068d6f9106749a05f6ff735b3267e82c80175743f2db17e |            |                      |                 |
| SHA-1:     | 065f9d3a37aa64d17175bc46ed2c9876a44f6616                         |            |                      |                 |

| FLY007.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | bfcf9ef316d741d55c62f4b4122ec8048f5961e389533792b08534cd5b934d07 |            |                      |                 |
| SHA-1:     | d48f695689b3281f697c56aa3906859a675b39ab                         |            |                      |                 |

| FLY008.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 50c485f60130b103ea00be88f76ad6d2faeac356100742f6943b0a048f61bf22 |            |                      |                 |
| SHA-1:     | 1987e3c5ec33b5702285e8dfabdb4d1d8c7f9784                         |            |                      |                 |

| FLY009.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | b27cc3a442ee4e3b2204684255df02ec83b5b7b8222afbd01415c464f95d635a |            |                      |                 |
| SHA-1:     | 078d3600e091478cd9d3deb54d8e4cd60d4aea04                         |            |                      |                 |

| FLY010.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 5135658a19aa96a1434bf045786928e77cb74af63d756bbd97cbbe0195ad8742 |            |                      |                 |
| SHA-1:     | 7994ce125b263d233c17563ab8f329dd2dc161e9                         |            |                      |                 |

| FLY011.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | d470a6f91bdc89ef0e96cb18b5114faa3fa72731355ef47db98bc46c6621d412 |            |                      |                 |
| SHA-1:     | 5bfd0adca5f33888761b214ffa74dc87e0132348                         |            |                      |                 |

| FLY012.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | b41d7078019b5cd722b9f6e34359b5c43a00e79bbab845cb43fa07582104c3ce |            |                      |                 |
| SHA-1:     | 29603497e1d9840d8b8aee7dc381cbe4d546d0c5                         |            |                      |                 |

| FLY013.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 12cad1f7ee43edf7a20ef389161d1501ec2e85b98002e7a13c4aced3abe40c82 |            |                      |                 |
| SHA-1:     | c3a2c793a6d597f3281af4985bcae10c7c5039b6                         |            |                      |                 |

| FLY014.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | a1132deaa1e0fe66c254d8016a888d2ce99707d39188718db99cdbfda3c0790c |            |                      |                 |
| SHA-1:     | e11d33b4f90cdd6a19b0bef61720d19872000ec0                         |            |                      |                 |

| FLY015.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | bfcfc2d7008ab241db5a8e7a2711b8dcf1ed3c3ebba1cac71b5de182b9dc78ec |            |                      |                 |
| SHA-1:     | 428ddf37dd84e9c77c16fe9ce1d1b26f03c8994c                         |            |                      |                 |

| FLY016.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | f07b2d87707048f708432904c0bdcdb4b2d6d86f2730bfb5b62528cf0e87cd42 |            |                      |                 |
| SHA-1:     | c8c8511348ba88a90de34eb17bef24380239e8a2                         |            |                      |                 |

| FLY017.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | d0bae4f14cd607bdf8de7e4f28494270f32a58b51ea5faa119b9ea7361304d49 |            |                      |                 |
| SHA-1:     | 8176cb8778b3ea9ec71e70e7d3c59f51dcf2b92a                         |            |                      |                 |

| FLY018.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 3e95fd14412c701336ff7d8d84af984b834e80191f10ac8aa14471d13e15365e |            |                      |                 |
| SHA-1:     | de86eefb3aa83ba4d5f9f628e5771726c006e279                         |            |                      |                 |

| FLY019.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | bd19b8e0b2116fc6c34ab31396acc6ee383fc1dd8b13c7994a00e285655a8581 |            |                      |                 |
| SHA-1:     | a3d0d31af00e8b343e990869e499934e22e71a78                         |            |                      |                 |

| FLY020.DAT |                                                                   | Drone Logs | drone_flight_storage | extract_logical |
|------------|-------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 75ab7f356dd1cebbcb02766c21f4afd721e18700973198feb3e5c4db9b0048098 |            |                      |                 |
| SHA-1:     | a873a6438beb295140069c6112b2c9c9a9d056a6                          |            |                      |                 |

| FLY021.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 5d32e57f2a77cf6ab15ecd43bc56ac3570aa9c3d2ed082d4f554c28cbb71ad6a |            |                      |                 |
| SHA-1:     | 37f229add49a021354f47dd2947b50d9305535a7                         |            |                      |                 |

| FLY022.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 487de1d89034fabd0e948a833caf2c14621d6fe2584a57dc75c4307f68052891 |            |                      |                 |
| SHA-1:     | da26d83b24684914a87ee2e07e34f2c1feecd7a3                         |            |                      |                 |

| FLY023.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 403845479d68aaa261898b17cb812677aa55e7ff41790c027b9a8919545d828d |            |                      |                 |
| SHA-1:     | d8291ebba2fa44c27ac43f5b4a63204e117f851f                         |            |                      |                 |

| FLY024.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | a3a26fd56cc5d07ceebb4b88ba5e882a16af4d9d055bdc8c8631c0db9969b0e7 |            |                      |                 |
| SHA-1:     | db9c3763be2d7052d164a3689157356003afa363                         |            |                      |                 |

| FLY025.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 0262a1b25fa6837a1c1504b642843729068d4d14820bea573dd658b663c4da77 |            |                      |                 |
| SHA-1:     | 6a8995df18b83036170c4585caf52c746311dde6                         |            |                      |                 |

| FLY026.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | d5a731ec334080b87170e8236d65f2f585773986a6d1d00edeeb2ee2b66cf079 |            |                      |                 |
| SHA-1:     | a7600d7e5e9514829fdc900c6c1dfcac8fc78f88                         |            |                      |                 |

| FLY027.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | fa4cfb51e6bca5e687c4787d5dddbd2c6ac2fde8ee81002606c5f13abc4fa700 |            |                      |                 |
| SHA-1:     | 548e2180d36de735b31547640ec09a01cb915837                         |            |                      |                 |

| FLY028.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | f2a433850d4f0b037348c705e1e1cf9209748262b6c16ee7b7f8008bda75efe0 |            |                      |                 |
| SHA-1:     | 25adf37848ee1ac87070698c5d7ef0dd7b6e97e0                         |            |                      |                 |

| FLY029.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 0f0b9a670d05011a5b751503d39e0657044de0013d7bda6e085a06da0cb705ce |            |                      |                 |
| SHA-1:     | bc25ebf980cbbb597793acb1bd9cecb32ce55cf3                         |            |                      |                 |

| FLY030.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 8c7771c2439a20a103411cbbdf0e4443f03363c47afb3141dc4887d2e4193ac3 |            |                      |                 |
| SHA-1:     | 3d655d73ca03894128711724e014c83b78c934dd                         |            |                      |                 |

| FLY031.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 2c127ce8ec2f96f8e52fde92e66d568f5e5137736e31d70b01e93cd5157a4f42 |            |                      |                 |
| SHA-1:     | 1dff8e00e2770ef862e663dc8c403a1e731e3738                         |            |                      |                 |

| FLY032.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | d6df6e40bdbd2b9c352f4de8d1914e26aa37ca91ced930fb113f350d5f265164 |            |                      |                 |
| SHA-1:     | 734ac27513f329462d98dda5a40d7bd3e4a37d4e                         |            |                      |                 |

| FLY033.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 79ab7320cc7a62a09445b8f9d9f568dfe9ed0150df4de9ff0656994234fe6f17 |            |                      |                 |
| SHA-1:     | 2741ce675a5e3466a5af5c4906daefce457045c2                         |            |                      |                 |

| FLY034.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 240a8c913b680e25d04fa06afb8a900b90ce5e9e88f91c166bd9b9c4ef95862c |            |                      |                 |
| SHA-1:     | 7169c6fcf1c271055ac2b8dcb23b08928b9009ed                         |            |                      |                 |

| FLY035.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | e3d7da6668717c35cc76dfe7a195365bc91d62a5959c81686338e5a02b53c56c |            |                      |                 |
| SHA-1:     | 4a5c7e7f39363a613f5eafe11b339c472c93b5d4                         |            |                      |                 |

| FLY036.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | c027c063a45dbc5b5a133feef886435e32af1b2f5804b86c577236119e006ecb |            |                      |                 |
| SHA-1:     | 81278e63b6a65639f0a8281a69d0da62164c1e09                         |            |                      |                 |

| FLY037.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 9f8c187a3708f14f0ad74868f0c4a6b53dbca1581df17c55b4c3edf9ddd310db |            |                      |                 |
| SHA-1:     | e9eaa83b74c057123050a069924069c4e8585261                         |            |                      |                 |

| FLY038.DAT |                                                                | Drone Logs | drone_flight_storage | extract_logical |
|------------|----------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 96c6ce796eaa493efa2141aea12722b6ce67305fab2a8c37e97c0909b3626b |            |                      |                 |
| SHA-1:     | dbb965c92216089ec11f4f985fc835a88fa639c0                       |            |                      |                 |

| FLY039.DAT |                                                                  | Drone Logs | drone_flight_storage | extract_logical |
|------------|------------------------------------------------------------------|------------|----------------------|-----------------|
| SHA-256:   | 82f853275ef320a1044603b6dbb47007bfe992cb5083e5183db490f0ed06e768 |            |                      |                 |
| SHA-1:     | 90db78c74d7700cce3f1780e5df23bab47e34c25                         |            |                      |                 |



## Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

### Controller Ios

| Category       | P3 Extracted | P4 Decoded | P5 Normalised |
|----------------|--------------|------------|---------------|
| Account Data   | 2            | 2          | 0             |
| Databases      | 15           | 0          | 0             |
| Drone Logs     | 1            | 2          | 1             |
| Flight Logs    | 5            | 0          | 0             |
| Flight Records | 1            | 1          | 1             |

### Drone Flight Storage

| Category   | P3 Extracted | P4 Decoded | P5 Normalised |
|------------|--------------|------------|---------------|
| Drone Logs | 40           | 80         | 40            |

## Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

**2026-06-15 14:24:31 parser\_ios v- rc=n/a**

cmd: ios\_backup.zip C:\Users\Floris\Documents\dfdof\_output\CASE-df005-2018-ios\p2\_image\_parsing\controller\_ios\_parsed  
out: controller\_ios\_parsed

**2026-06-15 14:36:04 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe 2018-06-19\_13-38-06\_FLY025.DAT  
out: 2018-06-19\_13-38-06\_FLY025.csv, 2018-06-19\_13-38-06\_FLY025.kml

**2026-06-15 14:36:05 txtlogtocsvtool v2018-06-11 rc=0**

cmd: TXTlogToCSVtool.exe DJIFlightRecord\_2018-06-19\_\_13-39-43\_.txt DJIFlightRecord\_2018-06-19\_\_13-39-43\_.csv  
out: DJIFlightRecord\_2018-06-19\_\_13-39-43\_.csv

**2026-06-15 14:37:51 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY000.DAT  
out: FLY000.csv, FLY000.kml

**2026-06-15 14:37:52 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY001.DAT  
out: FLY001.csv, FLY001.kml

**2026-06-15 14:37:52 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY002.DAT  
out: FLY002.csv, FLY002.kml

**2026-06-15 14:37:53 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY003.DAT  
out: FLY003.csv, FLY003.kml

**2026-06-15 14:37:54 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY004.DAT  
out: FLY004.csv, FLY004.kml

**2026-06-15 14:37:54 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY005.DAT  
out: FLY005.csv, FLY005.kml

**2026-06-15 14:37:55 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY006.DAT  
out: FLY006.csv, FLY006.kml

**2026-06-15 14:37:56 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY007.DAT  
out: FLY007.csv, FLY007.kml

**2026-06-15 14:37:57 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY008.DAT  
out: FLY008.csv, FLY008.kml

**2026-06-15 14:37:58 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY009.DAT  
out: FLY009.csv, FLY009.kml

**2026-06-15 14:37:59 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY010.DAT  
out: FLY010.csv, FLY010.kml

**2026-06-15 14:38:00 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY011.DAT

out: FLY011.csv, FLY011.kml

**2026-06-15 14:38:00 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY012.DAT

out: FLY012.csv, FLY012.kml

**2026-06-15 14:38:01 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY013.DAT

out: FLY013.csv, FLY013.kml

**2026-06-15 14:38:02 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY014.DAT

out: FLY014.csv, FLY014.kml

**2026-06-15 14:38:04 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY015.DAT

out: FLY015.csv, FLY015.kml

**2026-06-15 14:38:05 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY016.DAT

out: FLY016.csv, FLY016.kml

**2026-06-15 14:38:06 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY017.DAT

out: FLY017.csv, FLY017.kml

**2026-06-15 14:38:07 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY018.DAT

out: FLY018.csv, FLY018.kml

**2026-06-15 14:38:08 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY019.DAT

out: FLY019.csv, FLY019.kml

**2026-06-15 14:38:09 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY020.DAT

out: FLY020.csv, FLY020.kml

**2026-06-15 14:38:10 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY021.DAT

out: FLY021.csv, FLY021.kml

**2026-06-15 14:38:11 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY022.DAT

out: FLY022.csv, FLY022.kml

**2026-06-15 14:38:11 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY023.DAT

out: FLY023.csv, FLY023.kml

**2026-06-15 14:38:12 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY024.DAT

out: FLY024.csv, FLY024.kml

**2026-06-15 14:38:13 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY025.DAT

out: FLY025.csv, FLY025.kml

**2026-06-15 14:38:14 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY026.DAT

out: FLY026.csv, FLY026.kml

**2026-06-15 14:38:15 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY027.DAT

out: FLY027.csv, FLY027.kml

**2026-06-15 14:38:16 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY028.DAT

out: FLY028.csv, FLY028.kml

**2026-06-15 14:38:19 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY029.DAT

out: FLY029.csv, FLY029.kml

**2026-06-15 14:38:20 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY030.DAT

out: FLY030.csv, FLY030.kml

**2026-06-15 14:38:21 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY031.DAT

out: FLY031.csv, FLY031.kml

**2026-06-15 14:38:22 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY032.DAT

out: FLY032.csv, FLY032.kml

**2026-06-15 14:38:22 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY033.DAT

out: FLY033.csv, FLY033.kml

**2026-06-15 14:38:23 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY034.DAT

out: FLY034.csv, FLY034.kml

**2026-06-15 14:38:24 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY035.DAT

out: FLY035.csv, FLY035.kml

**2026-06-15 14:38:25 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY036.DAT

out: FLY036.csv, FLY036.kml

**2026-06-15 14:38:27 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY037.DAT

out: FLY037.csv, FLY037.kml

**2026-06-15 14:38:28 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY038.DAT

out: FLY038.csv, FLY038.kml

**2026-06-15 14:38:29 datcon v4.3.0 rc=n/a**

cmd: DatCon.4.3.0.exe FLY039.DAT

out: FLY039.csv, FLY039.kml

## Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

### P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

*Input: evidence directory. Output: classified evidence objects with hashes.*

### P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup\_info.json with device identifiers and installed DJI applications.

*Input: raw evidence files. Output: parsed directory structures, backup\_info.json.*

### P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

*Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).*

### P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

*Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.*

### P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

*Input: decoded artefacts. Output: normalised CSVs, anomaly observations.*

### P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap  $\geq 60$  s AND  $\geq 90\%$  of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs  $\leq 25$  m with  $\geq 5$  matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a  $\pm 2$  s window.

*Input: normalised CSVs. Output: timeline\_flightXX.json per flight.*

### P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further\_investigation section.

*Input: all phase outputs. Output: analysis dict with conclusions in state.json.*

### P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

*Input: state.json (all phase outputs). Output: PDF report + PNG plot files.*

**Disclaimer**

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.